Libertinus and diacritics

To do

Audit mark anchors to fit roman and italic anchor models; patch as necessary.

Add alt cap dot above for italic and semibold italic.

Look at the full-descender line for tall below marks. At $Y = -319$, 2–7 of the 32 below marks overlap almost every descending letter tested (p, q, g, ʒ; y is clear): mostly U+0318, U+0319, U+031C, U+031F and U+0339. The fault is not in any one letter, so the fix would be in the anchors of those marks or in the reference line itself.

Check the below anchor of J by eye in all styles. The designer's $Y = -319$ is deleted because J is not a full descender, and J now has the candidate (regular 120, -257; italic 146, -285; semibold 131, -267; semibold italic 187, -249). The best X is open. Y may need a curated anchor with the bowl correction of 10 instead of 2 (semibold 109, -259 looked right; regular had 109, -254).

Review the designer anchors more than 5 from a Y reference, which stage 5 lists but does not move (they are in the output of `run_patch_font.py`). Many are legitimate, such as stacked marks on precomposed glyphs. Likely errors include: above and below swapped on ɥ, ɥ and ƪ; k -214 below; ɣ -277 below, not flagged by the audit; and capitals near 805 in the other styles, the semibold problem again: Ɲ, Ƣ, Ɓ, Ƅ, Ƙ, Ƶ and others.

A side finding about the patched fonts. After Ç, the patched fonts render the plain acute (acutecomb), not acute.cap as the originals do. So the rebuilt ccmp seems to have dropped Ç, and possibly other glyphs, from the capital list. I haven't checked how many.

A side finding: semibold U+0351 and U+0357 have mark-anchor X = +100, while regular has −171. That looks wrong, and may come from the copies the original font has in U+0358–U+0363.

Anchor model

Libertinus has an anchor model where the base anchors are set to locations where the marks would be placed relative to the base. See the blue dots in the left panel of Figure 1; from top to bottom, the above-anchor is the blue dot above the base glyph, the overlay-anchor is blue dot inside the base glyph outline, and the below-anchor is the blue dot below the base glyph. The mark anchor is generally near the optical center of the mark glyph outline. See the red square in the middle panel of Figure 1; the mark's below-anchor is at the optical center of the mark. The base and mark are combined in the right panel of Figure 1 by superimposing their below-anchors.

Every font has five vertical reference lines (with respective Y coordinate values) – ascender line, capline, x-height or meanline, baseline, descender line – within which each glyph lives. These vertical reference lines are the gray vertical lines in Figures 1, 2, 3 and 4. Each glyph has a (tight) bounding box for real minimum and maximum X and Y coordinate values, as well as a so-called ideal bounding box where the glyph seems to live, but accounting for overshoots of bowls and serifs. The entire design goal is that the top and bottom of the ideal bounding box are two vertical reference lines, except for glyphs that are non-full ascenders or non-full descenders.

By clearance, I mean the (vertical) distance between a base and a mark, measured between the ideal bounding boxes of the base and the mark. For some designed fonts, the clearance is consistent for each vertical reference line, so that all glyphs that live at that reference line are separated from all marks with the same clearance; e.g. dots should not dance below the baseline for different letters that sit on the baseline, and acutes and graves sit at about the same height for different letters with similar top vertical reference line.

In roman styles of Libertinus, the clearance is encoded in the base anchor. The Y coordinate of an above-anchor is the top of the base's ideal bounding box plus the clearance of the nearest vertical reference line, and the Y coordinate of a below-anchor is the bottom of the ideal bounding box minus that clearance. For most glyphs that edge of the ideal bounding box is the reference line itself. For a non-full ascender or non-full descender it is not, but the clearance is the same; see the section on vertical reference lines below. The X coordinate of the anchor is at some horizontal optical center of the base glyph, but the meaning of the optical center depends on the glyph letterform, or whatever the designer thinks looks best. For a mostly symmetric letterform in the roman styles, a candidate for this X coordinate is the midpoint of the bounding box of the glyph.

The anchor of a mark is at the horizontal center of the mark, if the mark is symmetric. An asymmetric mark, such as the acute or grave, has its anchor a few units to one side of the center of its outline. Vertically, the anchor of an above-mark is about 140 units above the bottom of the mark outline. That is near the top of a tall mark, such as the circumflex, and above the top of a short mark, such as the macron or dot. Because the above-anchor of a base at the x-height is 216 units above the x-height, the bottom of a tall mark is about 75 units above the base, and a short mark sits a little higher.

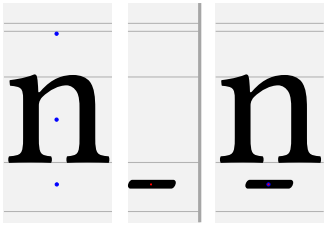


Figure 1: Libertinus n and macron with anchors.

Other fonts use a different geometric model for anchors. In Noto, the base's below-anchor is at the baseline, the overlay-anchor is inside the glyph, and the above-anchor is at the x-height. See the blue dots in the left panel of Figure 2; from the bottom to the top, the mark's below-anchor is the blue dot on the baseline, the overlay-anchor is blue dot inside the base glyph outline, and the above-anchor is the blue dot at the x-height. In Noto, the mark's below-anchor is at the origin $(0, 0)$, and the mark glyph outline is drawn below the origin. In the middle panel of Figure 2, the mark's below-anchor is the red square at the origin. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors. So in Noto the clearance is encoded in the mark anchor, and in Libertinus it is encoded in the base anchor.

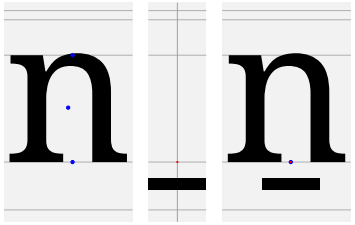


Figure 2: Noto n and macron with anchors.

Italic anchor model

In italic, the Libertinus anchor model is similar to the roman styles, but it is applied along the italic axis. The italic lane of a base glyph is its advance-width box, slanted by the italic angle θ (12°). The midline of the italic lane is the line $X = W/2 + Y \tan\theta$, where W is the advance width of the glyph. See the gray band and the dashed blue line in the left panel of Figure 3.

The Y coordinate of an above- or below-anchor is set exactly as in the roman styles: the top or bottom of the glyph's ideal bounding box plus or minus the clearance of the nearest vertical reference line, measured vertically. So marks sit at the same height in roman and italic. The X coordinate is the point of the midline at that height: $ax = W/2 + ay \tan\theta$ and $bx = W/2 + by \tan\theta$.

The height in this formula is the height of the anchor itself, that is, the edge of the ideal bounding box plus or minus the clearance, not the height of the reference line or of the ideal bounding box. So the Y coordinate of the anchor must be determined before the X coordinate can be calculated. In italic, the above-anchor is at $W/2 + 137$ for a letter at the x-height ($ay = 645$), at $W/2 + 181$ for a capital ($ay = 850$), and at $W/2 + 189$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 23$ for a letter on the baseline ($by = -110$) and at $W/2 - 68$ for a descender ($by = -319$). An error of 10 units in the Y coordinate moves the X coordinate by about 2 units.

The anchor of a mark lies on the mark's own slanted axis, through the center of its outline if the mark is symmetric. See the dashed red lines in the middle panel of Figure 3. That is, the mark anchor is at $X = C + Y \tan\theta$, where Y is the height of the anchor, as in the roman styles, and C is where the axis of the mark meets the baseline: the horizontal center of the mark outline after the slant is removed. An asymmetric mark keeps the same offset from its axis as in the roman styles. Superimposing the anchors lays the axis of the mark on the midline of the base, as in the right panel. The mark is displaced from the base along the italic axis, not vertically above or below it.

The midline is a geometric center, not an optical center, but it is a good candidate for the anchor X coordinate of a mostly symmetric letterform. Asymmetric letterforms, and letters that overhang their advance width such as f and j, need curated anchors.

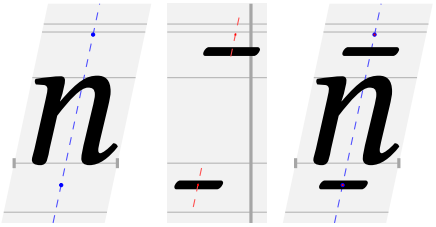


Figure 3: Libertinus italic n, macron and macron below, with anchors. The gray band is the italic lane and the dashed blue line is its midline.

Semibold italic anchor model

Semibold italic has its own anchor model, which is not the italic anchor model, and the patch keeps it. The outlines are slanted (the italic angle θ is $11.5°$), but the anchors are, almost all, the anchors of regular, copied unchanged: 42 of the 44 above- and below-mark anchors, and 319 of the 384 above- and below-anchors of bases, have exactly the coordinates of regular.

So the anchors do not follow the slant. A slanted outline moves further right the higher it goes, but an anchor copied from regular stays where it was. An above-anchor therefore sits left of where the italic anchor model would put it, by roughly the amount the slant has moved the outline at that height. This is true of the base and of the mark alike: the base's above-anchor is left of the midline of its italic lane, and the mark's anchor is left of the mark's own slanted axis, by about the same amount. When the two anchors are superimposed, the two errors cancel, and the axis of the mark lands near the midline of the base, as it would in italic. See Figure 4, where the filled symbols are the designer's anchors and the hollow symbols are the positions given by the italic anchor model: the above-anchor of n and the anchor of the macron are both far left of the hollow symbols, but by nearly the same distance.

The semibold italic anchor model is therefore the italic anchor model with both anchors shifted left by the same constant K: the base anchor is at $X = W/2 + Y \tan\theta - K$ and the mark anchor is at $X = C + Y \tan\theta - K$, with $\tan\theta = 0.2035$ and the same Y coordinates as in the other styles. Above-anchors are high, where the slant has moved the outline far to the right, so their shift is large: $K_a = 150$ (the median of the above-marks; most are between 132 and 168). Below-anchors are just under the baseline, where the slant has moved the outline a little to the left, so their shift is small and negative: $K_b = -12$ (most below-marks are between -23 and -4). The italic anchor model is the case $K = 0$.

The candidates for a base are $ax = W/2 + ay \tan\theta - 150$ and $bx = W/2 + by \tan\theta + 12$. So the above-anchor is at $W/2 - 19$ for a letter at the x-height ($ay = 645$), at $W/2 + 23$ for a capital ($ay = 850$), and at $W/2 + 31$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 10$ for a letter on the baseline ($by = -110$) and at $W/2 - 53$ for a descender ($by = -319$). For semibold italic n, $W = 606$, so $ax = 284$ and $bx = 293$; the font has $ax = 264$ and $bx = 265$, the values of regular n, and the macron lands 29 units left of the midline.

The cancellation is only approximate, because the designer's anchors do not follow the slant at all, whereas the model's anchors do, apart from the constant K. A mark's shift is fixed by the height of its own anchor: for a mark, $K = Y \tan\theta$ plus a constant, where Y is the height of the mark anchor. But the designer's base anchors are near $W/2 - 9$ at every height, so they do not move right for a taller base, such as a capital or an ascender. The error of a mark on a base therefore grows with the difference in height of the two anchors, by 2 units for every 10. The median distance of a designer's base anchor from the candidate is about 20 units for a lowercase letter and 30 units for a capital.

The patch does not move this style to the italic anchor model. It keeps the designer's anchors where they conform to the semibold italic anchor model, and corrects only the anchors that do not, so that they conform to it: a base anchor is moved to the candidate above, and a mark anchor is moved to $X = C + Y \tan\theta - 150$ for an above-mark or $X = C + Y \tan\theta + 12$ for a below-mark. An anchor set by the italic anchor model would be wrong here: a base anchor without the shift would put every above-mark about 150 units too far right, because the marks carry the shift.

The following do not conform to the semibold italic anchor model and probably need to be patched, some of them manually.

- Capitals. The designer's above-anchors are about 25 units left of the candidate, since they are 205 units higher than the anchors of lowercase letters but no further right. The worst are B, D, I, N, O, R, W (40 to 60 units left) and L (159 units left; its anchor is over the stem, as in regular).

- The capital forms of the grave, acute, circumflex and breve (`grave.cap`, `acute.cap`, `circumflex.cap`, `breve.cap`), which the font substitutes over capitals. Their mark anchors were set with the slant (K between -13 and 29), but the above-anchors of capitals were not, so these four marks land 160 to 190 units left of the middle of a capital. The other above-marks have no capital form in use and land 20 to 50 units left. The capital forms need mark anchors at $X = C + Y \tan\theta - 150$; `caron.cap` and `dieresis.cap` are not substituted.
- Above-anchors that were set with the slant, as in italic, so that an above-mark lands 130 to 190 units too far right: a, s, h and J.
- Other base anchors more than 40 units from the candidate. Above: c, g, Æ, Ɖ, ƕ, Ƣ, Ƥ, ɳ, ɗ, ɦ, ʄ, Ħ, ɵ, ɧ. Below: A, C, D, J, a, d, f, j, q, r, t, Æ, ĸ, ƕ, Ƣ, Ƥ, ɳ, ɾ, ɑ, ɖ, ɗ, ɤ, ɥ, ɯ, ɻ, Ħ, ɵ. Several of these are copies of the anchors of another glyph. The anchors of ɥ and ɥ are transposed, as in the other styles.
- Marks whose shift is far from the median, so that no base anchor can be right for them and for the other marks: dieresis U+0308 ($K = 111$, so it lands about 40 units left of the other above-marks), comma above U+0313 (125), inverted breve U+0311 (132), hook above U+0309 and double grave U+030F (168, about 18 units right), koronis U+0343 (371), right tack below U+0319 (-30) and tilde below U+0330 ($+10$). The double acute U+030B (156) conforms.
- Tilde U+0303, one of the two mark anchors that is not a copy of regular; its shift is 139.
- The Hebrew ligatures U+05F0, U+05F1, U+05F2, which have above- and below-anchors in this lookup table in semibold italic only.
- Letters that overhang their advance width, such as f and j, which need curated anchors in every italic style.

The anchors of precomposed glyphs were not compared with these candidates.

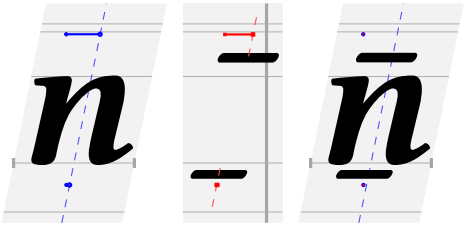


Figure 4: Libertinus semibold italic n, macron and macron below, with the designer's anchors (filled) and the positions given by the italic anchor model (hollow). The above-anchor of n is 170 units left of the midline and the anchor of the macron is 141 units left of its axis.

Mark anchor model

The anchor of a mark follows the same model in every style. Horizontally, it lies on the mark's own axis, $X = C + Y \tan\theta - K$ ($\theta = 0$ and $K = 0$ in the roman styles, $K = 0$ in italic, $K = 150$ above and -12 below in semibold italic). An asymmetric mark, such as the grave, acute, double acute or double grave, keeps the offset from its axis that it has in regular. Vertically, the anchor is at a fixed distance from the mark's outline: the anchor of an above-mark is a certain height above the bottom of the outline, and the anchor of a below-mark is a certain depth below the top of the outline. These distances differ from mark to mark (a short mark such as the macron sits a little higher than a tall one), but since the base anchors have the same Y coordinates in every style, a mark sits at the same height in every style only if its distance is the same as in regular.

So each mark can be checked in two ways: its X offset from its axis, compared with regular, and the distance between its anchor and its outline, compared with regular. A third, weaker check is the designer's precomposed glyphs, measured on bases whose own anchors conform to the model (o, n, u). The precomposed glyphs are not consistent between styles (the macron of ō is 18 units lower than the combining macron in regular, but 32 units higher in italic), so they can confirm a finding but not settle one. In what follows, "lower" and "higher" and "left" and "right" describe where the mark lands on a base whose anchor conforms. A mark lands lower when its anchor is further above its outline, and left when its anchor is further right.

Status. The to-do item "Audit mark anchors to fit roman and italic anchor models" has not been done. There is no audit of mark anchors, and no stage of the patch moves the designer's mark anchors to the model. The patch changes mark anchors in four places, and one change of base anchors (in stage 1) serves the marks.

- Stage 1, semibold only: the designer's second above-anchor class `top_punkt` (class 1) is merged into the above class. Its 12 marks (dot above, diaeresis, hook above, ring above, U+0312, U+0313, U+0314, U+031A, U+0351, U+0357, U+0363 and `hookabovecomb.cap`) move to class 0, and their anchors are raised by 96 units, the difference between the two classes' base anchors. This does not move these marks on a lowercase letter, but it fixes their coverage: before, they attached only to the 24 bases with a `top_punkt` anchor; now they attach to every base with an above-anchor. It does not change how high these marks sit compared with regular (see below).
- Stage 3, semibold and semibold italic: the combining marks the style lacks are added, and semibold's U+0358–U+0363, which were copies of the capital accents, are replaced. Their anchors are regular's. In semibold italic, regular's anchor is used with italic's outline moved right by 12 units. There, the added above-marks land 7 to 15 units left of the model ($K = 150$), which is well inside the spread of the designer's own above-marks (K from 111 to 168).
- Stage 4, all styles: the capital forms (`grave.cap`, `acute.cap`, …) get anchors computed from the designer's precomposed capitals, on the axis for a symmetric mark, at the precomposed offset for an asymmetric one, and with $K = 150$ in semibold italic. This fixes semibold italic, whose capital forms had anchors without the shift K and landed 120 to 160 units left, and italic, where only `grave.cap` had an anchor (in the wrong class). In italic the curated `.cap` anchors overrule stage 4; they fit the designer's capital anchors, which are off the midline, so even the symmetric `caron.cap` and `breveinvertedcmb.cap` are 21 units off their axis.
- Stage 7, curated mark anchors in `data/fontanchors_human/`:
  - Regular, left and right tack below (U+0318, U+0319): the designer's anchors were 25 units right and 36 units left of the centre of the outline; the curated anchors are at the centre. The other styles keep the designer's anchors, so the tacks now differ between regular and the other styles.
  - Semibold, grave and acute: 30 units lower (713 to 683), which makes them sit as in regular. The grave and acute tone marks U+0340 and U+0341, which stage 3 adds, still have 713, so they sit 30 units lower than the grave and acute.
  - Italic and semibold italic, tilde below: moved to the centre of the outline. In semibold italic this corrects an offset 22 units different from regular's; in italic it moves the anchor by 8 units, harmlessly.
  - Italic, double acute: the designer's anchor $(-74, 714)$ had regular's offset; the curated $(-57, 714)$ lands the mark 17 units left.
  - Semibold italic, grave $(0, 683)$ and double acute $(-28, 719)$: these were wrong, and have been removed. The designer's anchors, $(-142, 683)$ and $(-219, 714)$, conform to the model. The curated ones lacked the shift K, so on every conforming base the grave landed 142 units left and the double acute 190 units left (ò: 126 units left of the precomposed grave; ő: 150 units left). They had been adjusted on a, whose above-anchor carried the slant and landed every above-mark about 150 units too far right (see the list above Figure 4). The fix: stage 1 now deletes the above-anchors of a, s, h and J (as `"delete"` entries in `data/fontanchors_audit/overrides.py`), stage 6 replaces them with the candidates, and the designer's grave and double acute are back. The designer's a, s, h and J were 137 to 179 units right of the candidate, and h's anchor was at the x-height instead of the ascender. On a, the above-marks now land 7 to 33 units from where the precomposed glyphs put them (before: 113 to 161 units right); on s within 11; the circumflex of ĥ, which landed 219 units too low, now sits 26 units higher than in the precomposed glyph, and that of ĵ 34 units right and 34 higher.

Remaining departures from the model. Compared with regular, after the patch:

- Regular. The marks conform. The dot above U+0307 sits 45 units lower than the dot of the precomposed ȯ. The dot below U+0323 and diaeresis below U+0324 are drawn 43 units lower, relative to their anchors, than in the other three styles. Regular's precomposed ạ agrees with the combining dot below, but in ụ the diaeresis is 35 units higher. Whether regular or the other styles should change is a design decision.
- Italic. The marks conform. The tilde is 11 units off its axis. The dot above sits 10 units higher than in regular, which is still 11 units lower than in the precomposed ȯ.
- Semibold. Horizontally, the marks are within about 12 units of regular's offsets. Vertically, about a quarter of the above-marks sit lower than in regular: U+0357 32, U+0351 20, double acute 19, diaeresis 14, dot above 13 (ȯ confirms, 21), hook above 13, macron 13 (ō confirms, 12), circumflex 10. The tilde sits 12 units higher (õ confirms, 14), and so does the breve below. Semibold's x-height is 5 units higher than regular's, so differences under 10 do not matter.

- Semibold italic. Horizontally, the marks whose K is far from 150 are listed above Figure 4 (dieresis lands 38 units left; comma above, inverted breve, hook above, double grave, koronis, right tack below); none is patched. Vertically, the designer drew some outlines at a different height from regular's but kept regular's anchors: the macron sits 32 units lower than in regular (ō confirms, 32), the macron below 52 units higher (n̲ confirms, 26), the dot above 21 units higher (which agrees with the precomposed ȯ), the up tack below U+031D 12 units lower and the down tack below U+031E 12 units higher.

Changes needed.

- Semibold italic: move the anchors of the marks whose K is far from 150 to the model; move the macron's anchor 32 units down and the macron below's anchor 52 units up.
- Semibold: lower the anchors of the above-marks that sit low by the amounts listed, so that they sit as in regular; give U+0340 and U+0341 the curated height of the grave and acute (683).
- Regular: lower the anchor of the dot above by about 45 units (and by about 11 in italic and 21 in semibold) to match the precomposed glyphs; decide between regular and the other styles for the dot below and diaeresis below, and for the tacks below.
- Italic: remove the curated double acute, restoring the designer's anchor.
- Implement these as an audit of mark anchors (the counterpart of the anchor audit of base anchors) and a patch stage, rather than as more curated anchors, so that the check can be repeated.

Æ: a letter the midpoint cannot centre

Over Æ, accents do not go at the midpoint of the glyph. The left leg of the A stops short of the top, so the top edge runs from the apex of the A to the end of the arm of the E, and its centre lies well to the right. Other fonts agree: Charis SIL, Gentium Plus, EB Garamond, Junicode, DejaVu Serif and Andika put the above-anchor at the macron of Ǣ, within 5 units, and below they use the common stroke or the same X. In Libertinus the designer's above-anchors missed the accents of Ǣ and Ǽ by 55–88 and 16–68 units, so the curated above-anchors lie between the two (555, 674, 555, 545 for regular, italic, semibold, semibold italic). In the italic styles the macron and the acute cannot both be met, and each lands about 30 units from its precomposed position. Below, semibold's anchor is curated under the common stroke, at 510. The designer's 549 of the other three styles stands: nothing collides there.

Mark outline zones

Figures 5, 6, 7 and 8 overlay the outlines of all the above and below marks of the patched fonts on a solid o, at three times the scale of the anchor model figures. Each mark is placed so that its anchor is on the anchor of the o for its class (the blue dots), which is where the shaper puts it. The common marks are black outlines. The other marks are gray with a light fill, so the shading builds up where many marks overlap. The grid is in font units of the o, with lines every 100 units. In semibold italic the above marks lie well to the right of the blue dot, because both the base and the mark anchor carry the shift K.

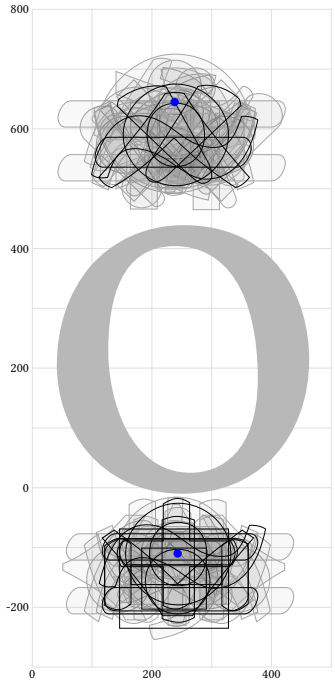


Figure 5: Mark zones, regular.

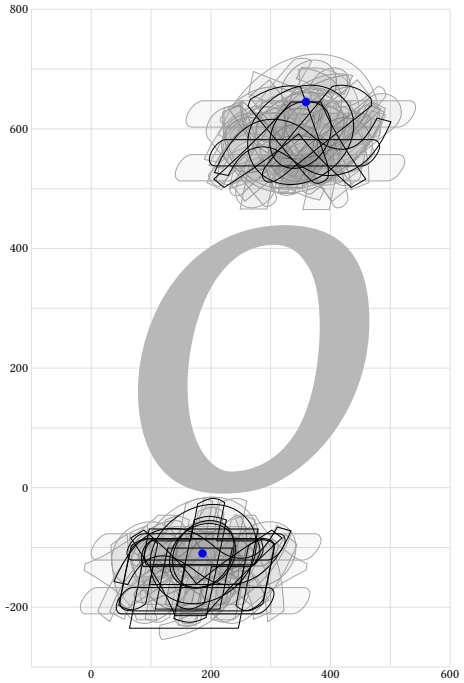


Figure 6: Mark zones, italic.

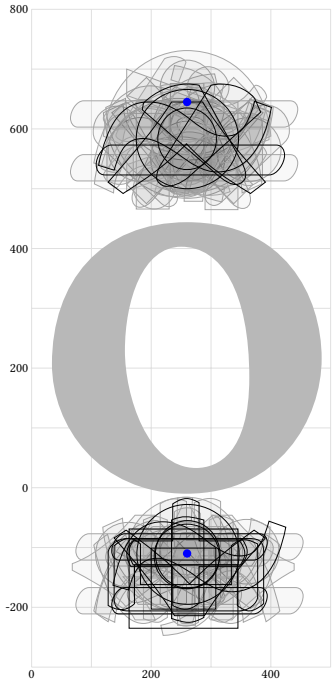


Figure 7: Mark zones, semibold.

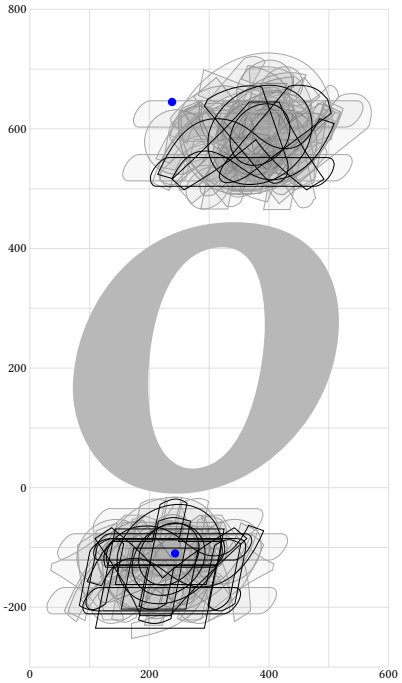


Figure 8: Mark zones, semibold italic.

Font metrics per glyph

See above- and below-anchor coordinates for A–Z, a–z, and IPA letters for Libertinus serif fonts (regular, italic, semibold, semibold italic) in the following tables (Tables 1–11), and the mark anchors of above- and below-marks (Tables 12–15). Recall that Language Science Press uses semibold Libertinus fonts, e.g. `LibertinusSerif-Semibold.otf`, for boldface styles.

Table row headings

hex: Unicode character codepoint in 4-hex.

cat: Unicode character category. L = letter. Lu = uppercase (capital) letter, Ll = lowercase (small) letter, Lo = other letter. Used to determine expected ideal bounding box. M = mark. Mn = nonspacing mark.

reg, it, sb, si: regular, italic, semibold, semibold italic. In that row, the character is printed alone, with combining dot above, and with combining macron below, to show anchor positioning (via combining grapheme joiner) when the base has an anchor; otherwise HarfBuzz default positioning applies.

patch: Use the patched Libertinus font to print the character alone, with combining dot above, and with combining macron below to show anchor positioning of the patched font (via combining grapheme joiner). Where the original font has no above- or below-anchor, the patched font has the candidate (a?, b?), unless a human-curated anchor overrules it; existing anchors are not moved.

ideal: top and bottom ideal bounding box reference lines, with offset from Y_{max} or Y_{min} for non-full descender or non-full ascender. c = capline, x = x-height, a = ascender, b = baseline, d = descender.

a: X, Y coordinates of the above-anchor.

a?: X, Y coordinates of the above-anchor candidate.

aδ: X, Y coordinate differences between above-anchor and above-anchor candidate

b: X, Y coordinates of the below-anchor.

b?: X, Y coordinates of the below-anchor candidate.

bδ: X, Y coordinate differences between below-anchor and below-anchor candidate

In the tables of marks, each column is a mark and the rows are as follows.

reg, it, sb, si, patch: the mark on n and on H (via combining grapheme joiner), in the original and patched fonts.

m: X, Y coordinates of the mark anchor. A class in parentheses is an anchor class other than above (0) or below (2): semibold puts the dot above, diaeresis, ring, hook above and comma above in class 1, whose base anchors are lower (n has 550 in class 1 and 646 in class 0). This is the designer's anchor class `top_punkt`; it places the marks at the same height as class 0, but only 24 bases have it. The patched semibold merges it into class 0.

m?: X, Y coordinates of the mark-anchor candidate. Y is the bottom of the mark outline plus the style's median distance of an above-mark anchor above it (regular 143, italic 142, semibold 157 in class 0 and 60 in class 1, semibold italic 147), or the top of the outline minus the median distance of a below-mark anchor below it (regular 46, italic 48, semibold 54, semibold italic 54). X is on the mark's own axis at that height, $C + Y \tan\theta - K$.

mδ: X, Y coordinate differences between mark anchor and mark-anchor candidate. An asymmetric mark, such as the acute or grave, has an X difference by design.

Table 1: Font metrics for A–Z 1/2

hex	0041	0042	0043	0044	0045	0046	0047	0048	0049	004A	004B	004C	004D
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu
reg	AȦẠ	BḂḄ	CĊÇ	DḊḌ	EĖẸ	FḞF	GĠG	HḢḤ	IİỊ	JJ̇J	KḴḲ	LĿḶ	MṀṂ
patch	AȦẠ	BḂḄ	CĊÇ	DḊḌ	EĖẸ	FḞF	GĠG	HḢḤ	IİỊ	JJ̇J	KḴḲ	LĿḶ	MṀṂ
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+60	c, b	c, b	c, b
a	354, 850	267, 850	340, 850	306, 850	281, 850	284, 850	365, 850	361, 850	150, 850	182, 850	336, 850	151, 850	431, 850
a?	346, 850	280, 850	324, 850	336, 850	271, 850	244, 850	352, 850	362, 850	148, 850	120, 850	323, 850	260, 850	418, 850
aδ	8, 0	-14, 0	16, 0	-30, 0	10, 0	40, 0	14, 0	-1, 0	2, 0	62, 0	13, 0	-110, 0	13, 0
b	308, -110	268, -110	363, -110	309, -110	287, -110	209, -110	367, -110	362, -110	147, -110	109, -319	342, -110	262, -110	393, -110
b?	346, -110	280, -110	324, -110	336, -110	271, -110	244, -110	352, -110	362, -110	148, -110	120, -257	323, -110	260, -110	418, -110
bδ	-38, 0	-12, 0	39, 0	-27, 0	16, 0	-36, 0	16, 0	0, 0	-1, 0	-11, -62	19, 0	2, 0	-25, 0
it	*AȦẠ*	*BḂḄ*	*CĊÇ*	*DḊḌ*	*EĖẸ*	*FḞF*	*GĠG*	*HḢḤ*	*IİỊ*	*JJ̇J*	*KḴḲ*	*LĿḶ*	*MṀṂ*
patch	*AȦẠ*	*BḂḄ*	*CĊÇ*	*DḊḌ*	*EĖẸ*	*FḞF*	*GĠG*	*HḢḤ*	*IİỊ*	*JJ̇J*	*KḴḲ*	*LĿḶ*	*MṀṂ*
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c+32, d+32	c, b	c, b	c, b
a	505, 850	447, 850	500, 850	396, 850	413, 850	406, 850	513, 850	481, 850	316, 850	392, 850	466, 850	389, 850	557, 850
a?	514, 850	459, 850	489, 850	514, 850	444, 850	409, 850	513, 850	517, 850	321, 850	388, 850	499, 850	440, 850	583, 850
aδ	-9, 0	-12, 0	11, 0	-118, 0	-31, 0	-3, 0	0, 0	-36, 0	-5, 0	4, 0	-33, 0	-51, 0	-26, 0
b	308, -108	238, -108	333, -110	279, -107	269, -107	131, -111	345, -110	320, -107	127, -109	109, -319	292, -110	252, -110	363, -110
b?	310, -110	255, -110	285, -110	310, -110	240, -110	205, -110	309, -110	313, -110	117, -110	146, -285	295, -110	236, -110	379, -110
bδ	-2, 2	-17, 2	48, 0	-31, 3	29, 3	-74, -1	36, 0	7, 3	10, 1	-37, -34	-3, 0	16, 0	-16, 0
sb	**AȦẠ**	**BḂḄ**	**CĊÇ**	**DḊḌ**	**EĖẸ**	**FḞF**	**GĠG**	**HḢḤ**	**IİỊ**	**JJ̇J**	**KḴḲ**	**LĿḶ**	**MṀṂ**
patch	**AȦẠ**	**BḂḄ**	**CĊÇ**	**DḊḌ**	**EĖẸ**	**FḞF**	**GĠG**	**HḢḤ**	**IİỊ**	**JJ̇J**	**KḴḲ**	**LĿḶ**	**MṀṂ**
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+50	c, b	c, b	c, b
a										203, 806			
a?	350, 850	300, 850	332, 850	352, 850	283, 850	268, 850	368, 850	362, 850	160, 850	131, 850	340, 850	267, 850	428, 850
aδ										72, -44			
b								362, -110		109, -319			
b?	350, -110	300, -110	332, -110	352, -110	283, -110	268, -110	368, -110	362, -110	160, -110	131, -267	340, -110	267, -110	428, -110
bδ								0, 0		-22, -52			
si	***AȦẠ***	***BḂḄ***	***CĊÇ***	***DḊḌ***	***EĖẸ***	***FḞF***	***GĠG***	***HḢḤ***	***IİỊ***	***JJ̇J***	***KḴḲ***	***LĿḶ***	***MṀṂ***
patch	***AȦẠ***	***BḂḄ***	***CĊÇ***	***DḊḌ***	***EĖẸ***	***FḞF***	***GĠG***	***HḢḤ***	***IİỊ***	***JJ̇J***	***KḴḲ***	***LĿḶ***	***MṀṂ***
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c+34, d+68	c, b	c, b	c, b
a	354, 850	267, 850	340, 850	306, 850	273, 850	276, 850	365, 850	361, 850	150, 850	432, 850	336, 850	149, 850	431, 850
a?	382, 850	319, 850	341, 850	367, 850	292, 850	269, 850	371, 850	396, 850	194, 850	249, 850	369, 850	310, 850	441, 850
aδ	-28, 0	-52, 0	-1, 0	-61, 0	-19, 0	7, 0	-6, 0	-35, 0	-44, 0	183, 0	-33, 0	-161, 0	-10, 0
b	308, -110	268, -110	363, -110	279, -110	279, -110	201, -110	367, -110	365, -110	147, -110	109, -319	342, -110	260, -110	393, -110
b?	349, -110	286, -110	308, -110	334, -110	259, -110	236, -110	338, -110	363, -110	161, -110	187, -249	336, -110	277, -110	408, -110
bδ	-41, 0	-18, 0	55, 0	-55, 0	20, 0	-35, 0	29, 0	2, 0	-14, 0	-78, -70	6, 0	-17, 0	-15, 0

Table 2: Font metrics for A–Z 2/2

hex	004E	004F	0050	0051	0052	0053	0054	0055	0056	0057	0058	0059	005A
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu
reg	N̈N̤	O̊O̤	P̈P̤	Q̊Q̤	R̈R̤	S̈S̤	T̈T̤	ŮṲ	V̊V̤	W̊W̤	ẌX̤	Ÿ Y̤	Z̈Z̤
patch	N̈N̤	O̊O̤	P̈P̤	Q̊Q̤	R̈R̤	S̈S̤	T̈T̤	ŮṲ	V̊V̤	W̊W̤	ẌX̤	Ÿ Y̤	Z̈Z̤
ideal	c, b	c, b	c, b	c, d+23	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b
a	346, 850	335, 850	277, 850	348, 850	269, 850	232, 850	307, 850	346, 850	336, 850	450, 850	332, 850	295, 850	320, 850
a?	353, 850	352, 850	260, 850	359, 850	301, 850	236, 850	297, 850	331, 850	326, 850	474, 850	323, 850	288, 850	310, 850
aδ	-7, 0	-16, 0	18, 0	-11, 0	-32, 0	-4, 0	10, 0	15, 0	10, 0	-24, 0	9, 0	8, 0	10, 0
b	348, -110	350, -110	240, -110		258, -110	234, -110	300, -110	328, -110	318, -110	479, -110	307, -110	295, -110	287, -110
b?	353, -110	352, -110	260, -110	359, -294	301, -110	236, -110	297, -110	331, -110	326, -110	474, -110	323, -110	288, -110	310, -110
bδ	-5, 0	-2, 0	-20, 0		-43, 0	-2, 0	3, 0	-3, 0	-8, 0	5, 0	-16, 0	8, 0	-22, 0
it	*N̈N̤*	*O̊O̤*	*P̈P̤*	*Q̊Q̤*	*R̈R̤*	*S̈S̤*	*T̈T̤*	*ŮṲ*	*V̊V̤*	*W̊W̤*	*ẌX̤*	*Ÿ Y̤*	*Z̈Z̤*
patch	*N̈N̤*	*O̊O̤*	*P̈P̤*	*Q̊Q̤*	*R̈R̤*	*S̈S̤*	*T̈T̤*	*ŮṲ*	*V̊V̤*	*W̊W̤*	*ẌX̤*	*Ÿ Y̤*	*Z̈Z̤*
ideal	c, b	c, b	c, b	c, d+23	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b
a	476, 850	515, 850	426, 850	498, 850	419, 850	402, 850	457, 850	486, 850	481, 850	610, 850	482, 850	435, 850	450, 850
a?	514, 850	515, 850	430, 850	515, 850	458, 850	408, 850	453, 850	498, 850	479, 850	635, 850	495, 850	457, 850	470, 850
aδ	-38, 0	0, 0	-4, 0	-17, 0	-39, 0	-6, 0	4, 0	-12, 0	2, 0	-25, 0	-13, 0	-22, 0	-20, 0
b	328, -109	310, -113	140, -110		248, -111	204, -111	277, -109	308, -109	263, -110	439, -107	287, -107	270, -109	267, -107
b?	310, -110	311, -110	226, -110	272, -294	254, -110	204, -110	249, -110	294, -110	275, -110	431, -110	291, -110	253, -110	266, -110
bδ	18, 1	-1, -3	-86, 0		-6, -1	0, -1	28, 1	14, 1	-12, 0	8, 3	-4, 3	17, 1	1, 3
sb	**NNN̤**	**OOO̤**	**PPP̤**	**Q̊Q̤**	**R̈R̤**	**S̈S̤**	**T̈T̤**	**UUṲ**	**V̊V̤**	**W̊W̤**	**ẌX̤**	**Ÿ Y̤**	**Z̈Z̤**
patch	**N̈N̤**	**O̊O̤**	**P̈P̤**	**Q̊Q̤**	**R̈R̤**	**S̈S̤**	**T̈T̤**	**ŮṲ**	**V̊V̤**	**W̊W̤**	**ẌX̤**	**Ÿ Y̤**	**Z̈Z̤**
ideal	c, b	c, b	c, b	c, d	c, b	c, b	c+28, b	c, b	c, b	c, b	c, b	c, b	c, b
a		363, 802		363, 802	304, 808	252, 802	298, 802		331, 805	493, 805	364, 804	316, 805	321, 803
a?	354, 850	366, 850	278, 850	366, 850	320, 850	238, 850	299, 850	339, 850	318, 850	489, 850	354, 850	308, 850	309, 850
aδ		-2, -48		-2, -48	-16, -42	14, -48	-1, -48		12, -45	4, -45	10, -46	8, -45	12, -47
b					292, -111	234, -111	301, -107		303, -108	470, -107	318, -107	316, -109	308, -107
b?	354, -110	366, -110	278, -110	366, -319	320, -110	238, -110	299, -110	339, -110	318, -110	489, -110	354, -110	308, -110	309, -110
bδ					-28, -1	-4, -1	2, 3		-16, 2	-19, 3	-36, 3	8, 1	-1, 3
si	***N̈N̤***	***O̊O̤***	***P̈P̤***	***Q̊Q̤***	***R̈R̤***	***S̈S̤***	***T̈T̤***	***ŮṲ***	***V̊V̤***	***W̊W̤***	***ẌX̤***	***Ÿ Y̤***	***Z̈Z̤***
patch	***N̈N̤***	***O̊O̤***	***P̈P̤***	***Q̊Q̤***	***R̈R̤***	***S̈S̤***	***T̈T̤***	***ŮṲ***	***V̊V̤***	***W̊W̤***	***ẌX̤***	***Ÿ Y̤***	***Z̈Z̤***
ideal	c, b	c, b	c, b	c, d	c, b	c, b	c+28, b	c, b	c, b	c, b	c, b	c, b	c, b
a	346, 850	335, 850	316, 850	348, 850	269, 850	232, 850	307, 850	346, 850	336, 850	450, 850	332, 850	295, 850	320, 850
a?	394, 850	390, 850	284, 850	390, 850	315, 850	250, 850	309, 850	364, 850	328, 850	501, 850	362, 850	319, 850	322, 850
aδ	-48, 0	-55, 0	32, 0	-42, 0	-46, 0	-18, 0	-2, 0	-18, 0	8, 0	-51, 0	-30, 0	-24, 0	-2, 0
b	348, -110	350, -110	240, -110		258, -110	234, -110	300, -110	328, -110	318, -110	479, -110	307, -110	295, -110	287, -110
b?	361, -110	357, -110	251, -110	314, -319	282, -110	217, -110	276, -110	331, -110	295, -110	468, -110	329, -110	286, -110	289, -110
bδ	-13, 0	-7, 0	-11, 0		-24, 0	17, 0	24, 0	-3, 0	23, 0	11, 0	-22, 0	9, 0	-2, 0

Table 3: Font metrics for a–z 1/2

hex	0061	0062	0063	0064	0065	0066	0067	0068	0069	006A	006B	006C	006D
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	aȧạ	bḃḅ	cċc̣	dḋḍ	eėẹ	fḟf̣	gġg̣	hḣḥ	iiị	jjj̣	kk̇ḳ	l̇l̇ḷ	ṁṁṃ
patch	aȧạ	bḃḅ	cċc̣	dḋḍ	eėẹ	fḟf̣	gġg̣	hḣḥ	iiị	jjj̣	kk̇ḳ	l̇l̇ḷ	ṁṁṃ
ideal	x, b	a, b	x, b	a, b	x, b	a, b	x+33, d	a, b	a-51, b	a-51, d+9	a, b	a, b	x, b
a	209, 645		242, 645	218, 645	243, 645		231, 645	299, 645	137, 795		306, 784	135, 885	392, 645
a?	246, 645	232, 885	218, 645	270, 885	222, 645	208, 885	256, 676	270, 885	142, 834	76, 834	261, 885	134, 885	402, 645
aδ	-36, 0		24, 0	-52, -240	21, 0		-26, -31	28, -240	-5, -39		45, -101	1, 0	-10, 0
b	196, -110	234, -110	236, -110	221, -110	234, -110	132, -110	229, -319	272, -110	136, -110	109, -319	290, -214	135, -110	408, -110
b?	246, -110	232, -110	218, -110	270, -110	222, -110	208, -110	256, -319	270, -110	142, -110	76, -308	261, -110	134, -110	402, -110
bδ	-50, 0	2, 0	18, 0	-50, 0	12, 0	-76, 0	-28, 0	2, 0	-6, 0	34, -11	29, -104	1, 0	6, 0
it	*aȧạ*	*bḃḅ*	*cċc̣*	*dḋḍ*	*eėẹ*	*fḟf̣*	*gġg̣*	*hḣḥ*	*iiị*	*jjj̣*	*kk̇ḳ*	*l̇l̇ḷ*	*ṁṁṃ*
patch	*aȧạ*	*bḃḅ*	*cċc̣*	*dḋḍ*	*eėẹ*	*fḟf̣*	*gġg̣*	*hḣḥ*	*iiị*	*jjj̣*	*kk̇ḳ*	*l̇l̇ḷ*	*ṁṁṃ*
ideal	x, b	a, b	x, b	a, b	x, b	a, d	x, d	a, b	a-80, b	a-80, d+36	a, b	a, b	x, b
a	341, 647		343, 645	218, 645	332, 645		360, 645	409, 645			405, 646	292, 890	522, 645
a?	380, 645	428, 890	332, 645	434, 890	338, 645	346, 890	376, 645	449, 890	310, 810	302, 810	432, 890	322, 890	529, 645
aδ	-39, 2		11, 0	-216, -245	-6, 0		-16, 0	-40, -245			-27, -244	-30, 0	-7, 0
b	174, -108	214, -108	163, -113	221, -107	158, -108	222, -111	197, -319	229, -111	136, -107	0, -319	223, -107	115, -107	382, -105
b?	220, -110	216, -110	171, -110	221, -110	177, -110	89, -319	171, -319	236, -110	115, -110	70, -281	220, -110	110, -110	368, -110
bδ	-46, 2	-2, 2	-8, -3	0, 3	-19, 2	133, 208	26, 0	-7, -1	21, 3	-70, -38	3, 3	5, 3	14, 5
sb	**aȧạ**	**bḃḅ**	**cċc̣**	**dḋḍ**	**eėẹ**	**fḟf̣**	**gġg̣**	**hḣḥ**	**iiị**	**jjj̣**	**kk̇ḳ**	**l̇l̇ḷ**	**ṁṁṃ**
patch	**aȧạ**	**bḃḅ**	**cċc̣**	**dḋḍ**	**eėẹ**	**fḟf̣**	**gġg̣**	**hḣḥ**	**iiị**	**jjj̣**	**kk̇ḳ**	**l̇l̇ḷ**	**ṁṁṃ**
ideal	x, b	a, b-13	x, b	a, b	x, b	a, b	x+33, d	a, b	a-48, b	a-48, d	a, b	a, b	x, b
a							231, 645		137, 795			142, 885	396, 645
a?	255, 645	236, 885	224, 645	274, 885	232, 645	226, 885	264, 676	275, 885	156, 837	78, 837	262, 885	148, 885	408, 645
aδ							-32, -31		-18, -42			-6, 0	-12, 0
b							229, -319	277, -110	136, -107	109, -319		135, -110	396, -105
b?	255, -110	236, -121	224, -110	274, -110	232, -110	226, -110	264, -319	275, -110	156, -110	78, -319	262, -110	148, -110	408, -110
bδ							-34, 0	2, 0	-20, 3	31, 0		-14, 0	-12, 5
si	***aȧạ***	***bḃḅ***	***cċc̣***	***dḋḍ***	***eėẹ***	***fḟf̣***	***gġg̣***	***hḣḥ***	***iiị***	***jjj̣***	***kk̇ḳ***	***l̇l̇ḷ***	***ṁṁṃ***
patch	***aȧạ***	***bḃḅ***	***cċc̣***	***dḋḍ***	***eėẹ***	***fḟf̣***	***gġg̣***	***hḣḥ***	***iiị***	***jjj̣***	***kk̇ḳ***	***l̇l̇ḷ***	***ṁṁṃ***
ideal	x, b	a, b	x, b	a, b	x, b	a, d	x, d	a, b	a-67, b	a-67, d+25	a, b	a, b	x, b
a	379, 645		242, 645	218, 645	243, 645		291, 645	469, 645	137, 795		306, 784	174, 890	392, 645
a?	251, 645	290, 890	203, 645	309, 890	218, 645	216, 890	220, 645	331, 890	180, 823	192, 823	316, 890	194, 890	407, 645
aδ	128, 0		39, 0	-91, -245	25, 0		71, 0	138, -245	-43, -28		-10, -106	-20, 0	-15, 0
b	196, -110	234, -110	236, -110	221, -110	234, -110	132, -110	199, -319	272, -110	136, -110	67, -319	290, -214	135, -110	400, -110
b?	260, -110	248, -110	212, -110	268, -110	227, -110	132, -319	186, -319	290, -110	153, -110	127, -292	275, -110	153, -110	415, -110
bδ	-64, 0	-14, 0	24, 0	-47, 0	7, 0	0, 209	13, 0	-18, 0	-17, 0	-60, -27	15, -104	-18, 0	-15, 0

Table 4: Font metrics for a–z 2/2

hex	006E	006F	0070	0071	0072	0073	0074	0075	0076	0077	0078	0079	007A
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	nn̰	oo̰	pp̰	qq̰	rr̰	ss̰	tt̰	uṵ	vv̰	ww̰	xx̰	yy̰	zz̰
patch	nn̰	oo̰	pp̰	qq̰	rr̰	ss̰	tt̰	uṵ	vv̰	ww̰	xx̰	yy̰	zz̰
ideal	x, b	x, b	x, d	x, d	x, b	x, b	a-116, b	x, b	x, b	x, b	x, b	x, d	x, b
a	264, 646	238, 645	271, 646	238, 645	191, 646	206, 645		255, 645	256, 646	345, 646	256, 646	262, 645	220, 646
a?	276, 645	252, 645	242, 645	268, 645	190, 645	199, 645	163, 769	271, 645	249, 645	373, 645	245, 645	260, 645	220, 645
aδ	-12, 1	-14, 0	28, 1	-30, 0	1, 1	7, 0		-16, 0	7, 1	-28, 1	11, 1	2, 0	0, 1
b	265, -110	243, -110	179, -319	348, -319	135, -110	196, -110	163, -110	239, -110	241, -110	352, -110	224, -110	242, -319	207, -110
b?	276, -110	252, -110	242, -319	268, -319	190, -110	199, -110	163, -110	271, -110	249, -110	373, -110	245, -110	260, -319	220, -110
bδ	-10, 0	-9, 0	-64, 0	80, 0	-55, 0	-3, 0	0, 0	-32, 0	-8, 0	-21, 0	-21, 0	-18, 0	-14, 0
it	*nn̰*	*oo̰*	*pp̰*	*qq̰*	*rr̰*	*ss̰*	*tt̰*	*uṵ*	*vv̰*	*ww̰*	*xx̰*	*yy̰*	*zz̰*
patch	*nn̰*	*oo̰*	*pp̰*	*qq̰*	*rr̰*	*ss̰*	*tt̰*	*uṵ*	*vv̰*	*ww̰*	*xx̰*	*yy̰*	*zz̰*
ideal	x, b	x, b	x+62, d	x+17, d	x, b	x, b	x+132, b	x, b	x, b	x, b	x, b	x, d	x+15, b
a	394, 646	359, 645	411, 646	369, 645	312, 646	327, 645		386, 646		348, 646	366, 646	367, 644	360, 646
a?	396, 645	361, 645	394, 705	386, 660	316, 645	314, 645	318, 775	398, 645	373, 645	481, 645	375, 645	389, 645	358, 658
aδ	-2, 1	-2, 0	17, -59	-17, -15	-4, 1	13, 0		-12, 1		-133, 1	-9, 1	-22, -1	2, -12
b	235, -111	186, -107	248, -110	169, -110	113, -110	169, -111	133, -107	226, -111		355, -110	224, -110	368, -107	187, -111
b?	236, -110	200, -110	177, -319	178, -319	155, -110	153, -110	130, -110	237, -110	213, -110	321, -110	214, -110	184, -319	195, -110
bδ	-1, -1	-14, 3	71, 209	-9, 209	-42, 0	16, -1	3, 3	-11, -1		34, 0	10, 0	184, 212	-8, -1
sb	**nn̰**	**oo̰**	**pp̰**	**qq̰**	**rr̰**	**ss̰**	**tt̰**	**uṵ**	**vv̰**	**ww̰**	**xx̰**	**yy̰**	**zz̰**
patch	**nn̰**	**oo̰**	**pp̰**	**qq̰**	**rr̰**	**ss̰**	**tt̰**	**uṵ**	**vv̰**	**ww̰**	**xx̰**	**yy̰**	**zz̰**
ideal	x, b	x, b	x, d	x, d	x, b	x, b	a-103, b	x, b	x, b	x, b	x, b	x, d	x, b
a	264, 646							264, 646					
a?	290, 645	260, 645	254, 645	273, 645	212, 645	201, 645	168, 782	276, 645	262, 645	384, 645	260, 645	264, 645	222, 645
aδ	-26, 1							-12, 1					
b	265, -111						163, -107	248, -111					
b?	290, -110	260, -110	254, -319	273, -319	212, -110	201, -110	168, -110	276, -110	262, -110	384, -110	260, -110	264, -319	222, -110
bδ	-24, -1						-5, 3	-28, -1					
si	***nn̰***	***oo̰***	***pp̰***	***qq̰***	***rr̰***	***ss̰***	***tt̰***	***uṵ***	***vv̰***	***ww̰***	***xx̰***	***yy̰***	***zz̰***
patch	***nn̰***	***oo̰***	***pp̰***	***qq̰***	***rr̰***	***ss̰***	***tt̰***	***uṵ***	***vv̰***	***ww̰***	***xx̰***	***yy̰***	***zz̰***
ideal	x, b	x, b	x+36, d	x+24, d	x, b	x, b	a-125, b	x, b	x, b	x, b	x, b	x, d	x+23, b
a	264, 646	238, 645	271, 646	238, 645	191, 646	326, 645		255, 645	256, 646	345, 646		262, 645	220, 646
a?	284, 645	235, 645	257, 679	272, 667	203, 645	190, 645	189, 765	262, 645	220, 645	332, 645	240, 645	236, 645	214, 666
aδ	-20, 1	3, 0	14, -33	-34, -22	-12, 1	136, 0		-7, 0	36, 1	13, 1		26, 0	6, -20
b	265, -110	243, -110	179, -319	348, -319	135, -110	196, -110	133, -107	239, -110	241, -110	352, -110		242, -319	207, -110
b?	293, -110	243, -110	216, -319	233, -319	212, -110	198, -110	173, -110	270, -110	228, -110	341, -110	248, -110	202, -319	218, -110
bδ	-28, 0	0, 0	-37, 0	115, 0	-77, 0	-2, 0	-40, 3	-31, 0	13, 0	11, 0		40, 0	-11, 0

Table 5: Font metrics for Latin and IPA 1/7

hex	00C6	00D0	00D8	00DE	00DF	00E6	00F0	00F8	00FE	0110	0111	0126	0127	0131	0141	0142	014B	0152
cat	Lu	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Lu	Ll	Lu	Ll	Ll	Lu	Ll	Ll	Lu
reg	Æ Ȧ Ӕ	Ð Ḋ Ḏ	Ø Ȯ Ǿ	Þ Ṗ Ꝑ	ß ß̊ ß	æ ȧ ӕ	ð ð̊ ð	ø ȯ ø	þ ṗ þ	Đ Ḋ Ḏ	đ ḋ ḍ	Ħ Ḣ Ḥ	ħ ḣ ḥ	ı i̊ ɨ	Ł Ł̇ Ł	ł l̊ ɨ	ŋ ŋ̇ ŋ	Œ Œ̇ Œ
patch	Æ Ȧ Ӕ	Ð Ḋ Ḏ	Ø Ȯ Ǿ	Þ Ṗ Ꝑ	ß ß̊ ß	æ ȧ ӕ	ð ð̊ ð	ø ȯ ø	þ ṗ þ	Đ Ḋ Ḏ	đ ḋ ḍ	Ħ Ḣ Ḥ	ħ ḣ ḥ	ı i̊ ɨ	Ł Ł̇ Ł	ł l̊ ɨ	ŋ ŋ̇ ŋ	Œ Œ̇ Œ
ideal	c, b	c, b	c, b-13	c, b	a, b	x, b	a, b	x+25, b-28	a, d	c, b	a, b	c, b	a, b	x, b	c, b	a, b	x, d+9	c, b
a	480, 850	341, 850	351, 850	281, 850		350, 645			278, 646					132, 645			264, 646	460, 850
a?	426, 850	333, 850	352, 850	250, 850	259, 885	344, 645	240, 885	250, 668	236, 885	336, 850	270, 885	362, 850	270, 885	142, 645	256, 850	133, 885	240, 645	436, 850
aδ	54, 0	8, 0	0, 0	32, 0		6, 0			42, -239					-10, 0			24, 1	24, 0
b	549, -107		350, -110			327, -109		243, -110						139, -108			265, -111	
b?	426, -110	333, -110	352, -121	250, -110	259, -110	344, -110	240, -110	250, -136	236, -319	336, -110	270, -110	362, -110	270, -110	142, -110	256, -110	133, -110	240, -308	436, -110
bδ	122, 3		-2, 11			-16, 1		-8, 26						-3, 2			25, 197	
it	*Æ Ȧ Ӕ*	*Ð Ḋ Ḏ*	*Ø Ȯ Ǿ*	*Þ Ṗ Ꝑ*	*ß ß̊ ß*	*æ ȧ ӕ*	*ð ð̊ ð*	*ø ȯ ø*	*þ ṗ þ*	*Đ Ḋ Ḏ*	*đ ḋ ḍ*	*Ħ Ḣ Ḥ*	*ħ ḣ ḥ*	*ı i̊ ɨ*	*Ł Ł̇ Ł*	*ł l̊ ɨ*	*ŋ ŋ̇ ŋ*	*Œ Œ̇ Œ*
patch	*Æ Ȧ Ӕ*	*Ð Ḋ Ḏ*	*Ø Ȯ Ǿ*	*Þ Ṗ Ꝑ*	*ß ß̊ ß*	*æ ȧ ӕ*	*ð ð̊ ð*	*ø ȯ ø*	*þ ṗ þ*	*Đ Ḋ Ḏ*	*đ ḋ ḍ*	*Ħ Ḣ Ḥ*	*ħ ḣ ḥ*	*ı i̊ ɨ*	*Ł Ł̇ Ł*	*ł l̊ ɨ*	*ŋ ŋ̇ ŋ*	*Œ Œ̇ Œ*
ideal	c, b	c, b	c+25, b-43	c, b	a, d	x, b	a+6, b	x+48, b-48	a, d-7	c, b	a, b	c, b	a, b	x, b	c, b	a, b	x, d	c, b
a	615, 850					480, 645			278, 646					267, 645			264, 646	575, 850
a?	595, 850	514, 850	515, 850	431, 850	461, 890	481, 645	429, 896	370, 691	439, 890	514, 850	434, 890	517, 850	449, 890	275, 645	440, 850	322, 890	382, 645	587, 850
aδ	20, 0					-1, 0			-161, -244					-8, 0			-118, 1	-12, 0
b	549, -107		310, -113			327, -109		186, -107						139, -108			265, -111	
b?	391, -110	310, -110	302, -151	227, -110	204, -319	320, -110	216, -110	190, -156	181, -324	310, -110	221, -110	313, -110	236, -110	115, -110	236, -110	110, -110	177, -319	383, -110
bδ	158, 3		8, 38			7, 1		-4, 49						24, 2			88, 208	
sb	**Æ Ȧ Ӕ**	**Ð Ḋ Ḏ**	**Ø Ȯ Ǿ**	**Þ Ṗ Ꝑ**	**ß ß̊ ß**	**æ ȧ ӕ**	**ð ð̊ ð**	**ø ȯ ø**	**þ ṗ þ**	**Đ Ḋ Ḏ**	**đ ḋ ḍ**	**Ħ Ḣ Ḥ**	**ħ ḣ ḥ**	**ı i̊ ɨ**	**Ł Ł̇ Ł**	**ł l̊ ɨ**	**ŋ ŋ̇ ŋ**	**Œ Œ̇ Œ**
patch	**Æ Ȧ Ӕ**	**Ð Ḋ Ḏ**	**Ø Ȯ Ǿ**	**Þ Ṗ Ꝑ**	**ß ß̊ ß**	**æ ȧ ӕ**	**ð ð̊ ð**	**ø ȯ ø**	**þ ṗ þ**	**Đ Ḋ Ḏ**	**đ ḋ ḍ**	**Ħ Ḣ Ḥ**	**ħ ḣ ḥ**	**ı i̊ ɨ**	**Ł Ł̇ Ł**	**ł l̊ ɨ**	**ŋ ŋ̇ ŋ**	**Œ Œ̇ Œ**
ideal	c, b	c, b	c+25, b-45	c, b	a, b	x, b	a+19, b	x+41, b-34	a, d	c, b	a, b	c, b	a, b	x, b	c, b	a, b	x, d+36	c, b
a	510, 850					355, 645								151, 645				450, 850
a?	446, 850	349, 850	366, 850	287, 850	284, 885	362, 645	256, 904	260, 684	260, 885	349, 850	274, 885	362, 850	275, 885	156, 645	258, 850	142, 885	248, 645	432, 850
aδ	64, 0					-8, 0								-4, 0				18, 0
b														135, -108				
b?	446, -110	349, -110	366, -153	287, -110	284, -110	362, -110	256, -110	260, -142	260, -319	349, -110	274, -110	362, -110	275, -110	156, -110	258, -110	142, -110	248, -281	432, -110
bδ														-20, 2				
si	***Æ Ȧ Ӕ***	***Ð Ḋ Ḏ***	***Ø Ȯ Ǿ***	***Þ Ṗ Ꝑ***	***ß ß̊ ß***	***æ ȧ ӕ***	***ð ð̊ ð***	***ø ȯ ø***	***þ ṗ þ***	***Đ Ḋ Ḏ***	***đ ḋ ḍ***	***Ħ Ḣ Ḥ***	***ħ ḣ ḥ***	***ı i̊ ɨ***	***Ł Ł̇ Ł***	***ł l̊ ɨ***	***ŋ ŋ̇ ŋ***	***Œ Œ̇ Œ***
patch	***Æ Ȧ Ӕ***	***Ð Ḋ Ḏ***	***Ø Ȯ Ǿ***	***Þ Ṗ Ꝑ***	***ß ß̊ ß***	***æ ȧ ӕ***	***ð ð̊ ð***	***ø ȯ ø***	***þ ṗ þ***	***Đ Ḋ Ḏ***	***đ ḋ ḍ***	***Ħ Ḣ Ḥ***	***ħ ḣ ḥ***	***ı i̊ ɨ***	***Ł Ł̇ Ł***	***ł l̊ ɨ***	***ŋ ŋ̇ ŋ***	***Œ Œ̇ Œ***
ideal	c, b	c, b	c+25, b-40	c, b	a, d	x, b	a+19, b	x+22, b-30	a, d	c, b	a, b	c, b	a, b	x, b	c, b	a, b	x, d+16	c, b
a	500, 850	341, 850	351, 850	281, 850		350, 645			278, 646					132, 645			264, 646	480, 850
a?	459, 850	367, 850	390, 850	303, 850	325, 890	339, 645	291, 909	239, 665	300, 890	367, 850	309, 890	396, 850	331, 890	144, 645	300, 850	190, 890	253, 645	450, 850
aδ	41, 0	-26, 0	-39, 0	-22, 0		11, 0			-22, -244					-12, 0			11, 1	30, 0
b	549, -107		350, -110			327, -109		243, -110						139, -108			265, -111	
b?	426, -110	334, -110	349, -148	270, -110	241, -319	347, -110	246, -110	237, -138	216, -319	334, -110	268, -110	363, -110	290, -110	153, -110	267, -110	148, -110	222, -301	417, -110
bδ	123, 3		1, 38			-20, 1		6, 28						-14, 2			43, 190	

Table 6: Font metrics for Latin and IPA 2/7

hex	0153	017F	0181	0182	0186	0187	0189	018A	018B	018E	018F	0190	0191	0192	0193	0194	0196	0197
cat	Ll	Ll	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Ll	Lu	Lu	Lu	Lu
reg	œɶœ̧	ſſ̧	Ɓ̧Ƃ̧	Ƃ̧Ƃ̧	Ɔ̧Ɔ̧	Ƈ̧Ƈ̧	Ɖ̧Ɗ̧	Ɗ̧Ƌ̧	ƋƋ̧	Ɛ̧Ɛ̧	Ə̧Ə̧	Ɛ̧Ɛ̧	Ƒ̧Ƒ̧	ƒƒ̧	Ɠ̧Ɠ̧	Ɣ̧Ƴ̧	Ɩ̧Ɩ̧	Ɨ̧Ɨ̧
patch	œɶœ̧	ſẛ̧	Ɓ̧Ƃ̧	Ƃ̧Ƃ̧	Ɔ̧Ɔ̧	Ƈ̧Ƈ̧	Ɖ̧Ɗ̧	Ɗ̧Ƌ̧	ƋƋ̧	Ɛ̧Ɛ̧	Ə̧Ə̧	Ɛ̧Ɛ̧	Ƒ̧Ƒ̧	ƒƒ̧	Ɠ̧Ɠ̧	Ɣ̧Ƴ̧	Ɩ̧Ɩ̧	Ɨ̧Ɨ̧
ideal	x, b	a, b	c, b	c, b	c, b	c+98, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+60	a, d	c+98, b	c, d	a, b	c, b
a	405, 645		327, 805					306, 805								331, 805		
a?	388, 645	190, 885	303, 850	274, 850	322, 850	369, 850	333, 850	368, 850	292, 850	280, 850	326, 850	256, 850	183, 850	201, 885	372, 850	308, 850	152, 885	148, 850
aδ	18, 0		24, -45					-62, -45								22, -45		
b			328, -108					369, -107										
b?	388, -110	190, -110	303, -110	274, -110	322, -110	369, -110	333, -110	368, -110	292, -110	280, -110	326, -110	256, -110	183, -257	201, -319	372, -110	308, -319	152, -110	148, -110
bδ			25, 2					0, 3										
it	*œɶœ̧*	*ſſ̧*	*Ɓ̧Ƃ̧*	*Ƃ̧Ƃ̧*	*Ɔ̧Ɔ̧*	*Ƈ̧Ƈ̧*	*Ɖ̧Ɗ̧*	*Ɗ̧Ƌ̧*	*ƋƋ̧*	*Ɛ̧Ɛ̧*	*Ə̧Ə̧*	*Ɛ̧Ɛ̧*	*Ƒ̧Ƒ̧*	*ƒƒ̧*	*Ɠ̧Ɠ̧*	*Ɣ̧Ƴ̧*	*Ɩ̧Ɩ̧*	*Ɨ̧Ɨ̧*
patch	*œɶœ̧*	*ſẛ̧*	*Ɓ̧Ƃ̧*	*Ƃ̧Ƃ̧*	*Ɔ̧Ɔ̧*	*Ƈ̧Ƈ̧*	*Ɖ̧Ɗ̧*	*Ɗ̧Ƌ̧*	*ƋƋ̧*	*Ɛ̧Ɛ̧*	*Ə̧Ə̧*	*Ɛ̧Ɛ̧*	*Ƒ̧Ƒ̧*	*ƒƒ̧*	*Ɠ̧Ɠ̧*	*Ɣ̧Ƴ̧*	*Ɩ̧Ɩ̧*	*Ɨ̧Ɨ̧*
ideal	x, b	a, d	c, b	c, b	c+14, b	c+98, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+60	a, d+60	c+98, b	c, d+20	a, b	c, b
a	480, 645		327, 805					306, 805								331, 805		
a?	490, 645	339, 890	488, 850	465, 850	503, 850	503, 850	534, 850	553, 850	450, 850	465, 850	515, 850	439, 850	419, 850	414, 890	541, 850	462, 850	322, 890	321, 850
aδ	-10, 0		-161, -45					-247, -45								-131, -45		
b			328, -108					369, -107										
b?	329, -110	82, -319	284, -110	261, -110	299, -110	299, -110	330, -110	349, -110	246, -110	261, -110	311, -110	235, -110	184, -257	170, -257	337, -110	218, -297	110, -110	117, -110
bδ			44, 2					20, 3										
sb	**œɶœ̧**	**ſſ̧**	**Ɓ̧Ƃ̧**	**Ƃ̧Ƃ̧**	**Ɔ̧Ɔ̧**	**Ƈ̧Ƈ̧**	**Ɖ̧Ɗ̧**	**Ɗ̧Ƌ̧**	**ƋƋ̧**	**Ɛ̧Ɛ̧**	**Ə̧Ə̧**	**Ɛ̧Ɛ̧**	**Ƒ̧Ƒ̧**	**ƒƒ̧**	**Ɠ̧Ɠ̧**	**Ɣ̧Ƴ̧**	**Ɩ̧Ɩ̧**	**Ɨ̧Ɨ̧**
patch	**œɶœ̧**	**ſẛ̧**	**Ɓ̧Ƃ̧**	**Ƃ̧Ƃ̧**	**Ɔ̧Ɔ̧**	**Ƈ̧Ƈ̧**	**Ɖ̧Ɗ̧**	**Ɗ̧Ƌ̧**	**ƋƋ̧**	**Ɛ̧Ɛ̧**	**Ə̧Ə̧**	**Ɛ̧Ɛ̧**	**Ƒ̧Ƒ̧**	**ƒƒ̧**	**Ɠ̧Ɠ̧**	**Ɣ̧Ƴ̧**	**Ɩ̧Ɩ̧**	**Ɨ̧Ɨ̧**
ideal	x, b	a, b	c, b	c, b	c, b	c+98, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+36	a, d+62	c+98, b	c, d+22	a, b	c, b
a	430, 645				275, 833													
a?	402, 645	217, 885	316, 850	268, 850	314, 850	368, 850	349, 850	381, 850	292, 850	302, 850	353, 850	269, 850	236, 850	221, 885	374, 850	294, 850	170, 885	158, 850
aδ	28, 0				-38, -17													
b																		
b?	402, -110	217, -110	316, -110	268, -110	314, -110	368, -110	349, -110	381, -110	292, -110	302, -110	353, -110	269, -110	236, -281	221, -255	374, -110	294, -295	170, -110	158, -110
bδ																		
si	***œɶœ̧***	***ſſ̧***	***Ɓ̧Ƃ̧***	***Ƃ̧Ƃ̧***	***Ɔ̧Ɔ̧***	***Ƈ̧Ƈ̧***	***Ɖ̧Ɗ̧***	***Ɗ̧Ƌ̧***	***ƋƋ̧***	***Ɛ̧Ɛ̧***	***Ə̧Ə̧***	***Ɛ̧Ɛ̧***	***Ƒ̧Ƒ̧***	***ƒƒ̧***	***Ɠ̧Ɠ̧***	***Ɣ̧Ƴ̧***	***Ɩ̧Ɩ̧***	***Ɨ̧Ɨ̧***
patch	***œɶœ̧***	***ſẛ̧***	***Ɓ̧Ƃ̧***	***Ƃ̧Ƃ̧***	***Ɔ̧Ɔ̧***	***Ƈ̧Ƈ̧***	***Ɖ̧Ɗ̧***	***Ɗ̧Ƌ̧***	***ƋƋ̧***	***Ɛ̧Ɛ̧***	***Ə̧Ə̧***	***Ɛ̧Ɛ̧***	***Ƒ̧Ƒ̧***	***ƒƒ̧***	***Ɠ̧Ɠ̧***	***Ɣ̧Ƴ̧***	***Ɩ̧Ɩ̧***	***Ɨ̧Ɨ̧***
ideal	x, b	a+13, d-9	c, b	c, b	c, b	c+98, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+36	a, d+62	c+98, b	c, d+22	a, b	c, b
a	415, 645		327, 805					306, 805								331, 805		
a?	384, 645	209, 903	359, 850	303, 850	357, 850	359, 850	388, 850	424, 850	304, 850	320, 850	369, 850	293, 850	288, 850	269, 890	395, 850	316, 850	194, 890	181, 850
aδ	31, 0		-32, -45					-118, -45								15, -45		
b			328, -108					369, -107										
b?	393, -110	121, -326	326, -110	270, -110	324, -110	326, -110	355, -110	391, -110	271, -110	287, -110	336, -110	260, -110	220, -281	198, -255	362, -110	245, -295	153, -110	148, -110
bδ			2, 2					-22, 3										

Table 7: Font metrics for Latin and IPA 3/7

hex	0198	019C	019D	019F	01B7	0237	0250	0251	0252	0253	0254	0255	0256	0257	0258	0259	025A	025B
cat	Lu	Lu	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	KKK̦	ШШШ̦	ƝƝƝ̦	ƟƟƟ̦	ȝȝȝ	jjj̦	ɐɐɐ̦	ɑɑɑ̦	ɒɒɒ̦	ɓɓɓ̦	ɔɔɔ̦	ɕɕɕ̦	ɖɖɖ̦	ɗɗɗ̦	ɘɘɘ̦	əəə̦	ɚɚɚ̦	ɛɛɛ̦
patch	KḰK̦	ШШ́Ш̦	ƝƝ́Ɲ̦	ƟƟ́Ɵ̦	ȝȝ́ȝ̦	jjj̦	ɐɐɐ̦	ɑɑɑ̦	ɒɒɒ̦	ɓɓɓ̦	ɔɔɔ̦	ɕɕɕ̦	ɖɖɖ̦	ɗɗɗ̦	ɘɘɘ̦	əəə̦	ɚɚɚ̦	ɛɛɛ̦
ideal	c, b	c, b	c, d+45	c, b	c+22, d+21	x, d+9	x, b	x, b	x, b	a, b	x, b	x, b-106	a, d+44	a, b	x, b	x, b	x+45, b	x, b
a		431, 801				140, 645	228, 645	228, 645	264, 645	480, 630	216, 645	209, 645	218, 645	218, 645	204, 645	195, 645	229, 645	209, 645
a?	339, 850	434, 850	324, 850	352, 850	303, 850	74, 645	216, 645	272, 645	247, 645	265, 885	216, 645	222, 645	314, 885	322, 885	220, 645	232, 645	306, 688	202, 645
aδ		-2, -49				66, 0	12, 0	-44, 0	17, 0	215, -255	0, 0	-13, 0	-96, -240	-104, -240	-16, 0	-37, 0	-78, -43	8, 0
b		450, -108		350, -110	272, -319		229, -114	212, -114	259, -114	212, -114	177, -114	212, -114	212, -114	212, -114	212, -114	186, -114	212, -114	212, -114
b?	339, -110	434, -110	324, -272	352, -110	303, -296	74, -308	216, -110	272, -110	247, -110	265, -110	216, -110	222, -214	314, -273	322, -110	220, -110	232, -110	306, -110	202, -110
bδ		16, 2		-2, 0	-31, -23		12, -4	-60, -4	12, -4	-53, -4	-38, -4	-10, 100	-102, 159	-110, -4	-8, -4	-46, -4	-94, -4	10, -4
it	*KKK̦*	*ШШШ̦*	*ƝƝƝ̦*	*ƟƟƟ̦*	*ȝȝȝ*	*jjj̦*	*ɐɐɐ̦*	*ɑɑɑ̦*	*ɒɒɒ̦*	*ɓɓɓ̦*	*ɔɔɔ̦*	*ɕɕɕ̦*	*ɖɖɖ̦*	*ɗɗɗ̦*	*ɘɘɘ̦*	*əəə̦*	*ɚɚɚ̦*	*ɛɛɛ̦*
patch	*KḰK̦*	*ШШ́Ш̦*	*ƝƝ́Ɲ̦*	*ƟƟ́Ɵ̦*	*ȝȝ́ȝ*	*jjj̦*	*ɐɐɐ̦*	*ɑɑɑ̦*	*ɒɒɒ̦*	*ɓɓɓ̦*	*ɔɔɔ̦*	*ɕɕɕ̦*	*ɖɖɖ̦*	*ɗɗɗ̦*	*ɘɘɘ̦*	*əəə̦*	*ɚɚɚ̦*	*ɛɛɛ̦*
ideal	c, b	c, b	c, d+45	c, b	c+22, d+21	x, d+36	x+16, b	x, b	x, b	a+6, b	x, b	x, b-100	a+9, d+44	a, b	x, b	x, b	x+44, b	x, b
a		431, 801				140, 645					322, 654		218, 645	218, 645	204, 645			322, 654
a?	499, 850	609, 850	528, 850	515, 850	485, 850	267, 645	368, 659	380, 645	391, 645	431, 896	332, 645	352, 645	465, 899	464, 890	356, 645	356, 645	418, 687	338, 645
aδ		-178, -49				-127, 0					-10, 9		-247, -254	-246, -245	-152, 0			-16, 9
b		450, -108					229, -114	212, -114	243, -114	212, -114	172, -114	212, -114	212, -114	212, -114	212, -114	212, -114	212, -114	212, -114
b?	295, -110	405, -110	290, -272	311, -110	241, -296	70, -281	205, -110	220, -110	231, -110	218, -110	171, -110	171, -208	215, -273	251, -110	195, -110	195, -110	248, -110	178, -110
bδ		45, 2					24, -4	-8, -4	12, -4	-6, -4	1, -4	41, 94	-3, 159	-39, -4	17, -4	17, -4	-36, -4	34, -4
sb	**KKK̦**	**ШШШ̦**	**ƝƝƝ̦**	**ƟƟƟ̦**	**ȝȝȝ**	**jjj̦**	**ɐɐɐ̦**	**ɑɑɑ̦**	**ɒɒɒ̦**	**ɓɓɓ̦**	**ɔɔɔ̦**	**ɕɕɕ̦**	**ɖɖɖ̦**	**ɗɗɗ̦**	**ɘɘɘ̦**	**əəə̦**	**ɚɚɚ̦**	**ɛɛɛ̦**
patch	**KḰK̦**	**ШШ́Ш̦**	**ƝƝ́Ɲ̦**	**ƟƟ́Ɵ̦**	**ȝȝ́ȝ**	**jjj̦**	**ɐɐɐ̦**	**ɑɑɑ̦**	**ɒɒɒ̦**	**ɓɓɓ̦**	**ɔɔɔ̦**	**ɕɕɕ̦**	**ɖɖɖ̦**	**ɗɗɗ̦**	**ɘɘɘ̦**	**əəə̦**	**ɚɚɚ̦**	**ɛɛɛ̦**
ideal	c, b	c, b	c, d+47	c, b	c+29, d+24	x, d	x, b-14	x, b	x, b	a, b	x, b-14	x, b-101	a, d+46	a, b	x, b	x, b	x+50, b	x, b
a						163, 645					199, 645							229, 645
a?	338, 850	444, 850	336, 850	366, 850	316, 850	78, 645	207, 645	276, 645	251, 645	268, 885	199, 645	232, 645	322, 885	324, 885	230, 645	228, 645	319, 693	212, 645
aδ						85, 0					0, 0							16, 0
b					272, -319		228, -114				184, -114					212, -114		
b?	338, -110	444, -110	336, -270	366, -110	316, -293	78, -319	207, -122	276, -110	251, -110	268, -110	199, -122	232, -209	322, -271	324, -110	230, -110	228, -110	319, -110	212, -110
bδ					-44, -26		21, 8				-15, 8					-16, -4		
si	***KKK̦***	***ШШШ̦***	***ƝƝƝ̦***	***ƟƟƟ̦***	***ȝȝȝ***	***jjj̦***	***ɐɐɐ̦***	***ɑɑɑ̦***	***ɒɒɒ̦***	***ɓɓɓ̦***	***ɔɔɔ̦***	***ɕɕɕ̦***	***ɖɖɖ̦***	***ɗɗɗ̦***	***ɘɘɘ̦***	***əəə̦***	***ɚɚɚ̦***	***ɛɛɛ̦***
patch	***KḰK̦***	***ШШ́Ш̦***	***ƝƝ́Ɲ̦***	***ƟƟ́Ɵ̦***	***ȝȝ́ȝ***	***jjj̦***	***ɐɐɐ̦***	***ɑɑɑ̦***	***ɒɒɒ̦***	***ɓɓɓ̦***	***ɔɔɔ̦***	***ɕɕɕ̦***	***ɖɖɖ̦***	***ɗɗɗ̦***	***ɘɘɘ̦***	***əəə̦***	***ɚɚɚ̦***	***ɛɛɛ̦***
ideal	c, b	c, b	c, d+47	c, b	c+28, d+24	x, d+25	x, b	x, b	x, b	a, b	x, b	x, b-101	a, d+46	a, b	x, b	x, b	x+50, b	x, b
a		431, 801				140, 645		228, 645	264, 645		160, 645		218, 645	218, 645	204, 645			209, 645
a?	353, 850	463, 850	383, 850	387, 850	329, 850	121, 645	221, 645	247, 645	247, 645	284, 890	199, 645	204, 645	302, 890	318, 890	207, 645	218, 645	299, 693	194, 645
aδ		-32, -49				19, 0		-19, 0	17, 0		-39, 0		-84, -245	-100, -245	-3, 0			15, 0
b		450, -108			236, -319		229, -114	212, -114	259, -114	212, -114	177, -114	212, -114	212, -114	212, -114	212, -114		212, -114	212, -114
b?	320, -110	430, -110	317, -270	354, -110	258, -293	93, -292	230, -110	256, -110	256, -110	243, -110	207, -110	192, -209	227, -271	276, -110	216, -110	227, -110	298, -110	203, -110
bδ		20, 2			-22, -26		-1, -4	-44, -4	3, -4	-31, -4	-30, -4	20, 95	-15, 157	-64, -4	-4, -4		-86, -4	9, -4

Table 8: Font metrics for Latin and IPA 4/7

hex	025C	025D	025E	025F	0260	0261	0262	0263	0264	0265	0266	0267	0268	0269	026A	026B	026C	026D
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ɜ ɜ̣	ɝ ɝ̣	ɞ ɞ̣	ɟ ɟ̣	ɠ ɠ̣	ɡ ɡ̣	ɢ ɢ̣	ɣ ɣ̣	ɤ ɤ̣	ɥ ɥ̣	ɦ ɦ̣	ɧ ɧ̣	ɨ ɨ̣	ɩ ɩ̣	ɪ ɪ̣	ɫ ɫ̣	ɬ ɬ̣	ɭ ɭ̣
patch	ɜ ɜ̣	ɝ ɝ̣	ɞ ɞ̣	ɟ ɟ̣	ɠ ɠ̣	ɡ ɡ̣	ɢ ɢ̣	ɣ ɣ̣	ɤ ɤ̣	ɥ ɥ̣	ɦ ɦ̣	ɧ ɧ̣	ɨ ɨ̣	ɩ ɩ̣	ɪ ɪ̣	ɫ ɫ̣	ɬ ɬ̣	ɭ ɭ̣
ideal	x, b	a-90, b	x, b	x, d+26	a, d	x, d	x, b	x, d+30	x, b	x, d	a, b	a, d+11	a-51, b	x, b	x, b	a, b	a, b	a, d+44
a	194, 645	209, 645	239, 645	169, 645	209, 645	239, 645	265, 645	259, 645	248, 645	268, 645	299, 565	299, 565	155, 795	123, 645	133, 645	209, 645	209, 645	209, 645
a?	196, 645	234, 795	242, 645	124, 645	320, 885	228, 645	259, 645	258, 645	221, 645	272, 645	262, 885	228, 885	150, 834	170, 645	136, 645	144, 885	182, 885	182, 885
aδ	-2, 0	-26, -150	-2, 0	44, 0	-112, -240	11, 0	6, 0	1, 0	27, 0	-4, 0	36, -320	72, -320	6, -39	-47, 0	-3, 0	65, -240	27, -240	27, -240
b	184, -114	212, -114	256, -114	196, -110	236, -110	226, -320	248, -110	251, -277	216, -114	219, -129	240, -114	240, -114	134, -114	153, -114	139, -114	146, -114	218, -120	196, -110
b?	196, -110	234, -110	242, -110	124, -291	320, -319	228, -319	259, -110	258, -287	221, -110	272, -319	262, -110	228, -306	150, -110	170, -110	136, -110	144, -110	182, -110	182, -273
bδ	-12, -4	-22, -4	14, -4	72, 181	-84, 209	-2, -1	-11, 0	-7, 10	-5, -4	-52, 190	-22, -4	12, 192	-16, -4	-17, -4	3, -4	2, -4	36, -10	14, 163
it	*ɜ ɜ̣*	*ɝ ɝ̣*	*ɞ ɞ̣*	*ɟ ɟ̣*	*ɠ ɠ̣*	*ɡ ɡ̣*	*ɢ ɢ̣*	*ɣ ɣ̣*	*ɤ ɤ̣*	*ɥ ɥ̣*	*ɦ ɦ̣*	*ɧ ɧ̣*	*ɨ ɨ̣*	*ɩ ɩ̣*	*ɪ ɪ̣*	*ɫ ɫ̣*	*ɬ ɬ̣*	*ɭ ɭ̣*
patch	*ɜ ɜ̣*	*ɝ ɝ̣*	*ɞ ɞ̣*	*ɟ ɟ̣*	*ɠ ɠ̣*	*ɡ ɡ̣*	*ɢ ɢ̣*	*ɣ ɣ̣*	*ɤ ɤ̣*	*ɥ ɥ̣*	*ɦ ɦ̣*	*ɧ ɧ̣*	*ɨ ɨ̣*	*ɩ ɩ̣*	*ɪ ɪ̣*	*ɫ ɫ̣*	*ɬ ɬ̣*	*ɭ ɭ̣*
ideal	x, b	x+132, b	x+17, b-25	x, d	a+6, d+31	x+17, d+31	x, b	x, d+45	x, b+2	x, d	a, b	a, d	a-80, b	x, b	x, b	a, b	a, b	a, d+30
a							265, 645			258, 645	299, 565			123, 645	128, 645			
a?	331, 645	413, 775	378, 660	305, 645	431, 896	381, 660	388, 645	387, 645	338, 645	396, 645	449, 890	449, 890	316, 810	266, 645	281, 645	333, 890	368, 890	341, 890
aδ							-123, 0			-138, 0	-150, -325			-143, 0	-153, 0			
b	184, -114	212, -114	256, -114				248, -114		177, -114	219, -129	240, -114		134, -114	153, -114	139, -114	146, -114	218, -120	
b?	170, -110	225, -110	209, -133	100, -319	180, -286	180, -286	227, -110	192, -272	178, -106	191, -319	236, -110	192, -319	121, -110	106, -110	120, -110	120, -110	156, -110	91, -287
bδ	14, -4	-13, -4	47, 19				21, -4		-1, -8	28, 190	4, -4		13, -4	47, -4	19, -4	26, -4	62, -10	
sb	**ɜ ɜ̣**	**ɝ ɝ̣**	**ɞ ɞ̣**	**ɟ ɟ̣**	**ɠ ɠ̣**	**ɡ ɡ̣**	**ɢ ɢ̣**	**ɣ ɣ̣**	**ɤ ɤ̣**	**ɥ ɥ̣**	**ɦ ɦ̣**	**ɧ ɧ̣**	**ɨ ɨ̣**	**ɩ ɩ̣**	**ɪ ɪ̣**	**ɫ ɫ̣**	**ɬ ɬ̣**	**ɭ ɭ̣**
patch	**ɜ ɜ̣**	**ɝ ɝ̣**	**ɞ ɞ̣**	**ɟ ɟ̣**	**ɠ ɠ̣**	**ɡ ɡ̣**	**ɢ ɢ̣**	**ɣ ɣ̣**	**ɤ ɤ̣**	**ɥ ɥ̣**	**ɦ ɦ̣**	**ɧ ɧ̣**	**ɨ ɨ̣**	**ɩ ɩ̣**	**ɪ ɪ̣**	**ɫ ɫ̣**	**ɬ ɬ̣**	**ɭ ɭ̣**
ideal	x, b	a-128, b	x, b	x, d	a+6, d+33	x, d	x+23, b	x, d+32	x, b	x, d	a, b	a, d	a-67, b	x, b	x, b	a, b	a, b	a, d
a															146, 645			
a?	216, 645	276, 757	249, 645	144, 645	336, 891	235, 645	266, 666	272, 645	202, 645	278, 645	274, 885	243, 885	152, 818	164, 645	152, 645	156, 885	190, 885	194, 885
aδ															-6, 0			
b													134, -114		156, -114	146, -114	218, -120	
b?	216, -110	276, -110	249, -110	144, -319	336, -284	235, -319	266, -110	272, -285	202, -110	278, -319	274, -110	243, -319	152, -110	164, -110	152, -110	156, -110	190, -110	194, -319
bδ													-18, -4		4, -4	-10, -4	28, -10	
si	***ɜ ɜ̣***	***ɝ ɝ̣***	***ɞ ɞ̣***	***ɟ ɟ̣***	***ɠ ɠ̣***	***ɡ ɡ̣***	***ɢ ɢ̣***	***ɣ ɣ̣***	***ɤ ɤ̣***	***ɥ ɥ̣***	***ɦ ɦ̣***	***ɧ ɧ̣***	***ɨ ɨ̣***	***ɩ ɩ̣***	***ɪ ɪ̣***	***ɫ ɫ̣***	***ɬ ɬ̣***	***ɭ ɭ̣***
patch	***ɜ ɜ̣***	***ɝ ɝ̣***	***ɞ ɞ̣***	***ɟ ɟ̣***	***ɠ ɠ̣***	***ɡ ɡ̣***	***ɢ ɢ̣***	***ɣ ɣ̣***	***ɤ ɤ̣***	***ɥ ɥ̣***	***ɦ ɦ̣***	***ɧ ɧ̣***	***ɨ ɨ̣***	***ɩ ɩ̣***	***ɪ ɪ̣***	***ɫ ɫ̣***	***ɬ ɬ̣***	***ɭ ɭ̣***
ideal	x, b	a-128, b	x, b	x, d	a, d+33	x, d	x+23, b	x, d+32	x, b	x, d	a, b	a, d	a-67, b	x, b	x, b	a, b	a, b	a, d
a	194, 645						265, 645			258, 645	299, 565	299, 565	137, 795	123, 645	128, 645			
a?	194, 645	261, 762	232, 645	141, 645	309, 890	238, 645	248, 666	255, 645	185, 645	258, 645	302, 890	301, 890	180, 823	138, 645	137, 645	187, 890	223, 890	187, 890
aδ	0, 0						17, -21			0, 0	-3, -325	-2, -325	-43, -28	-15, 0	-9, 0			
b	184, -114	212, -114	256, -114				248, -110		177, -114	219, -129	240, -114	240, -114	136, -110	153, -114	139, -114	146, -114	218, -120	
b?	203, -110	246, -110	241, -110	107, -319	232, -284	204, -319	252, -110	228, -285	194, -110	224, -319	260, -110	217, -319	153, -110	147, -110	145, -110	145, -110	181, -110	103, -319
bδ	-19, -4	-34, -4	15, -4				-4, 0		-17, -4	-5, 190	-20, -4	23, 205	-17, 0	6, -4	-6, -4	1, -4	37, -10	

Table 9: Font metrics for Latin and IPA 5/7

hex	026E	026F	0270	0271	0272	0273	0274	0275	0276	0277	0278	0279	027A	027B	027C	027D	027E	027F
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ɮɮ	ɯɯ	ɰɰ	ɱɱ	ɲɲ	ɳɳ	ɴɴ	ɵɵ	ɶɶ	ɷɷ	ɸɸ	ɹɹ	ɺɺ	ɻɻ	ɼɼ	ɽɽ	ɾɾ	ɿɿ
patch	ɮɮ	ɯɯ	ɰɰ	ɱɱ	ɲɲ	ɳɳ	ɴɴ	ɵɵ	ɶɶ	ɷɷ	ɸɸ	ɹɹ	ɺɺ	ɻɻ	ɼɼ	ɽɽ	ɾɾ	ɿɿ
ideal	a, d	x, b-13	x, d	x, d+11	x, d+11	x, d+11	x, b	x, b	x, b	x, b	a, d	x, b-13	a-51, b-13	x, d+58	x, d	x, d	x, b	x, b
a	209, 645	407, 645	209, 645	383, 645	264, 646	263, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645		222, 703	209, 645	209, 645	209, 645
a?	250, 885	390, 645	379, 645	366, 645	234, 645	296, 645	278, 645	252, 645	340, 645	357, 645	286, 885	178, 645	180, 834	226, 645	186, 645	186, 645	174, 645	170, 645
aδ	-42, -240	16, 0	-170, 0	18, 0	30, 1	-32, 0	-70, 0	-43, 0	-132, 0	-148, 0	-76, -240	32, 0	30, -189		36, 58	22, 0	34, 0	38, 0
b	196, -110	410, -114	294, -114	383, -105	265, -111	262, -112	247, -114	196, -110	196, -110	341, -114	196, -110	190, -114	203, -114	196, -110	127, -305	196, -110	196, -110	196, -110
b?	250, -319	390, -121	379, -319	366, -306	234, -306	296, -306	278, -110	252, -110	340, -110	357, -110	286, -319	178, -121	180, -121	226, -259	186, -319	186, -319	174, -110	170, -110
bδ	-54, 209	20, 7	-85, 205	18, 201	30, 195	-34, 194	-32, -4	-56, 0	-144, 0	-16, -4	-90, 209	12, 7	24, 7	-30, 149	-60, 14	10, 209	22, 0	26, 0
it	*ɮɮ*	*ɯɯ*	*ɰɰ*	*ɱɱ*	*ɲɲ*	*ɳɳ*	*ɴɴ*	*ɵɵ*	*ɶɶ*	*ɷɷ*	*ɸɸ*	*ɹɹ*	*ɺɺ*	*ɻɻ*	*ɼɼ*	*ɽɽ*	*ɾɾ*	*ɿɿ*
patch	*ɮɮ*	*ɯɯ*	*ɰɰ*	*ɱɱ*	*ɲɲ*	*ɳɳ*	*ɴɴ*	*ɵɵ*	*ɶɶ*	*ɷɷ*	*ɸɸ*	*ɹɹ*	*ɺɺ*	*ɻɻ*	*ɼɼ*	*ɽɽ*	*ɾɾ*	*ɿɿ*
ideal	a, d+8	x, b-15	x, d	x+21, d	x+19, d	x, d	x, b	x+14, b	x, b	x, b-18	a, d+32	x, b	a-71, b-19	x, d+58	x+19, d+34	x+17, d+38	x, b	x, b
a		407, 645		383, 645	264, 646	263, 645									222, 703			
a?	447, 890	517, 645	515, 645	519, 664	400, 662	367, 645	418, 645	381, 657	457, 645	494, 645	481, 890	315, 645	362, 819	352, 645	329, 662	323, 660	311, 645	307, 645
aδ		-110, 0		-136, -19	-136, -16	-104, 0									-107, 41			
b		410, -114	294, -114	383, -105	265, -111	262, -112	247, -114			341, -114		190, -114	203, -114		127, -305			
b?	192, -309	354, -123	310, -319	310, -319	191, -319	162, -319	258, -110	218, -110	296, -110	330, -126	231, -285	155, -110	161, -127	160, -259	128, -283	124, -279	151, -110	147, -110
bδ		56, 9	-16, 205	73, 214	74, 208	100, 207	-11, -4			11, 12		35, -4	42, 13		-1, -22			
sb	**ɮɮ**	**ɯɯ**	**ɰɰ**	**ɱɱ**	**ɲɲ**	**ɳɳ**	**ɴɴ**	**ɵɵ**	**ɶɶ**	**ɷɷ**	**ɸɸ**	**ɹɹ**	**ɺɺ**	**ɻɻ**	**ɼɼ**	**ɽɽ**	**ɾɾ**	**ɿɿ**
patch	**ɮɮ**	**ɯɯ**	**ɰɰ**	**ɱɱ**	**ɲɲ**	**ɳɳ**	**ɴɴ**	**ɵɵ**	**ɶɶ**	**ɷɷ**	**ɸɸ**	**ɹɹ**	**ɺɺ**	**ɻɻ**	**ɼɼ**	**ɽɽ**	**ɾɾ**	**ɿɿ**
ideal	a, d	x-2, b-17	x, d	x, d	x, d	x, d	x, b	x, b	x, b	x, b	a, d+34	x-2, b-17	a-69, b	x, d+60	x, d+36	x, d	x, b	x, b
a		377, 645																
a?	274, 885	354, 641	388, 645	363, 645	246, 645	306, 645	290, 645	260, 645	334, 645	369, 645	297, 885	144, 641	205, 816	238, 645	206, 645	198, 645	186, 645	182, 645
aδ		23, 4																
b		380, -114						229, -114				190, -114						
b?	274, -319	354, -125	388, -319	363, -319	246, -319	306, -319	290, -110	260, -110	334, -110	369, -110	297, -283	144, -125	205, -110	238, -257	206, -281	198, -319	186, -110	182, -110
bδ		26, 11						-30, -4				46, 11						
si	***ɮɮ***	***ɯɯ***	***ɰɰ***	***ɱɱ***	***ɲɲ***	***ɳɳ***	***ɴɴ***	***ɵɵ***	***ɶɶ***	***ɷɷ***	***ɸɸ***	***ɹɹ***	***ɺɺ***	***ɻɻ***	***ɼɼ***	***ɽɽ***	***ɾɾ***	***ɿɿ***
patch	***ɮɮ***	***ɯɯ***	***ɰɰ***	***ɱɱ***	***ɲɲ***	***ɳɳ***	***ɴɴ***	***ɵɵ***	***ɶɶ***	***ɷɷ***	***ɸɸ***	***ɹɹ***	***ɺɺ***	***ɻɻ***	***ɼɼ***	***ɽɽ***	***ɾɾ***	***ɿɿ***
ideal	a, d	x-3, b	x, d	x, d	x, d	x, d	x, b	x, b	x-1, b	x, b	a, d+34	x-3, b	a-69, b	x, d+60	x, d+36	x, d	x, b	x, b
a		407, 645		383, 645	264, 646	263, 645									222, 703			
a?	301, 890	372, 640	373, 645	373, 645	254, 645	254, 645	275, 645	229, 645	313, 642	350, 645	328, 890	171, 640	217, 821	194, 645	181, 645	177, 645	167, 645	167, 645
aδ		35, 5		10, 0	10, 1	9, 0									41, 58			
b		410, -114	294, -114	383, -105	265, -111	262, -112	247, -114			341, -114		190, -114	203, -114		127, -305			
b?	217, -319	382, -110	339, -319	339, -319	220, -319	220, -319	283, -110	238, -110	322, -110	359, -110	251, -283	180, -110	190, -110	172, -257	155, -281	143, -319	176, -110	176, -110
bδ		28, -4	-45, 205	44, 214	45, 208	42, 207	-36, -4			-18, -4		10, -4	13, -4		-28, -24			

Table 10: Font metrics for Latin and IPA 6/7

hex	0280	0281	0282	0283	0284	0285	0286	0287	0288	0289	028A	028B	028C	028D	028E	028F	0290	0291
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ʀ̀ʀ̄	ʁ̀ʁ̄	ʂ̀ʂ̄	ʃ̀ʃ̄	ʄ̀ʄ̄	ɱ̀	ʃ̀ʃ̄	ɨ̀ɨ̄	ʈ̀ʈ̄	ʉ̀ʉ̄	ʊ̀ʊ̄	ʋ̀ʋ̄	ʌ̀ʌ̄	ʍ̀ʍ̄	ʎ̀ʎ̄	ʏ̀ʏ̄	ʐ̀ʐ̄	ʑ̀ʑ̄
patch	ʀ̀ʀ̄	ʁ̀ʁ̄	ʂ̀ʂ̄	ʃ̀ʃ̄	ʄ̀ʄ̄	ɱ̀	ʃ̀ʃ̄	ɨ̀ɨ̄	ʈ̀ʈ̄	ʉ̀ʉ̄	ʊ̀ʊ̄	ʋ̀ʋ̄	ʌ̀ʌ̄	ʍ̀ʍ̄	ʎ̀ʎ̄	ʏ̀ʏ̄	ʐ̀ʐ̄	ʑ̀ʑ̄
ideal	x, b	x, b	x, d+56	a, d+18	a, d+18	x+121, d+18	a, d+7	x, d+79	a-114, d+11	x, b	x, b	x, b	x, b	x, b	a-37, b	x, b	x, d+47	x+19, d+110
a	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	259, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645
a?	238, 645	238, 645	199, 645	140, 885	140, 885	170, 764	150, 885	150, 645	158, 771	271, 645	254, 645	235, 645	244, 645	367, 645	248, 848	236, 645	262, 645	218, 662
aδ	-30, 0	-30, 0	10, 0	68, -240	68, -240	38, -119	60, -240	59, 0	52, -126	-62, 0	-44, 0	24, 0	-35, 0	-158, 0	-39, -203	-27, 0	-54, 0	-8, -17
b	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	258, -114	251, -114	196, -110	196, -110	196, -110	239, -107	196, -110	196, -110
b?	238, -110	238, -110	199, -261	140, -299	140, -299	170, -299	150, -310	150, -238	158, -306	271, -110	254, -110	235, -110	244, -110	367, -110	248, -110	236, -110	262, -270	218, -207
bδ	-42, 0	-42, 0	-3, 151	56, 189	56, 189	26, 189	46, 200	46, 128	38, 196	-75, 0	4, -4	16, -4	-48, 0	-171, 0	-52, 0	3, 3	-66, 160	-22, 97
it	*ʀ̀ʀ̄*	*ʁ̀ʁ̄*	*ʂ̀ʂ̄*	*ʃ̀ʃ̄*	*ʄ̀ʄ̄*	*ɱ̀*	*ʃ̀ʃ̄*	*ɨ̀ɨ̄*	*ʈ̀ʈ̄*	*ʉ̀ʉ̄*	*ʊ̀ʊ̄*	*ʋ̀ʋ̄*	*ʌ̀ʌ̄*	*ʍ̀ʍ̄*	*ʎ̀ʎ̄*	*ʏ̀ʏ̄*	*ʐ̀ʐ̄*	*ʑ̀ʑ̄*
patch	*ʀ̀ʀ̄*	*ʁ̀ʁ̄*	*ʂ̀ʂ̄*	*ʃ̀ʃ̄*	*ʄ̀ʄ̄*	*ɱ̀*	*ʃ̀ʃ̄*	*ɨ̀ɨ̄*	*ʈ̀ʈ̄*	*ʉ̀ʉ̄*	*ʊ̀ʊ̄*	*ʋ̀ʋ̄*	*ʌ̀ʌ̄*	*ʍ̀ʍ̄*	*ʎ̀ʎ̄*	*ʏ̀ʏ̄*	*ʐ̀ʐ̄*	*ʑ̀ʑ̄*
ideal	x, b	x, b	x+15, d+52	a+10, d+18	a+10, d+18	a+10, d+18	a+10, d+34	x, d+78	a-114, d+6	x, b	x, b-13	x, b-16	x, b	x, b	a-52, b	x, b	x+19, d+41	x+19, d+110
a												259, 645						
a?	358, 645	349, 645	317, 658	343, 900	343, 900	343, 900	343, 900	288, 645	314, 776	398, 645	392, 645	382, 645	383, 645	506, 645	433, 838	372, 645	346, 662	355, 662
aδ												-123, 0						
b											258, -114	251, -114				239, -107		
b?	197, -110	189, -110	121, -265	88, -299	88, -299	88, -299	92, -283	100, -239	83, -311	238, -110	229, -121	218, -124	222, -110	346, -110	231, -110	212, -110	147, -276	171, -207
bδ											29, 7	33, 10				27, 3		
sb	**ʀ̀ʀ̄**	**ʁ̀ʁ̄**	**ʂ̀ʂ̄**	**ʃ̀ʃ̄**	**ʄ̀ʄ̄**	**ɱ̀**	**ʃ̀ʃ̄**	**ɨ̀ɨ̄**	**ʈ̀ʈ̄**	**ʉ̀ʉ̄**	**ʊ̀ʊ̄**	**ʋ̀ʋ̄**	**ʌ̀ʌ̄**	**ʍ̀ʍ̄**	**ʎ̀ʎ̄**	**ʏ̀ʏ̄**	**ʐ̀ʐ̄**	**ʑ̀ʑ̄**
patch	**ʀ̀ʀ̄**	**ʁ̀ʁ̄**	**ʂ̀ʂ̄**	**ʃ̀ʃ̄**	**ʄ̀ʄ̄**	**ɱ̀**	**ʃ̀ʃ̄**	**ɨ̀ɨ̄**	**ʈ̀ʈ̄**	**ʉ̀ʉ̄**	**ʊ̀ʊ̄**	**ʋ̀ʋ̄**	**ʌ̀ʌ̄**	**ʍ̀ʍ̄**	**ʎ̀ʎ̄**	**ʏ̀ʏ̄**	**ʐ̀ʐ̄**	**ʑ̀ʑ̄**
ideal	x, b	x, b	x, d+53	a, d+21	a, d+21	a, d+21	a, d	x, d+67	a-103, d	x, b	x, b	x, b	x, b	x, b	a-36, b	x, b	x, d+41	x+23, d+107
a											266, 645							
a?	236, 645	220, 645	201, 645	152, 885	152, 885	152, 885	162, 885	157, 645	182, 782	274, 645	267, 645	256, 645	231, 645	356, 645	243, 849	244, 645	254, 645	230, 666
aδ											-1, 0							
b																		
b?	236, -110	220, -110	201, -264	152, -296	152, -296	152, -296	162, -319	157, -250	182, -319	274, -110	267, -110	256, -110	231, -110	356, -110	243, -110	244, -110	254, -276	230, -210
bδ																		
si	***ʀ̀ʀ̄***	***ʁ̀ʁ̄***	***ʂ̀ʂ̄***	***ʃ̀ʃ̄***	***ʄ̀ʄ̄***	***ɱ̀***	***ʃ̀ʃ̄***	***ɨ̀ɨ̄***	***ʈ̀ʈ̄***	***ʉ̀ʉ̄***	***ʊ̀ʊ̄***	***ʋ̀ʋ̄***	***ʌ̀ʌ̄***	***ʍ̀ʍ̄***	***ʎ̀ʎ̄***	***ʏ̀ʏ̄***	***ʐ̀ʐ̄***	***ʑ̀ʑ̄***
patch	***ʀ̀ʀ̄***	***ʁ̀ʁ̄***	***ʂ̀ʂ̄***	***ʃ̀ʃ̄***	***ʄ̀ʄ̄***	***ɱ̀***	***ʃ̀ʃ̄***	***ɨ̀ɨ̄***	***ʈ̀ʈ̄***	***ʉ̀ʉ̄***	***ʊ̀ʊ̄***	***ʋ̀ʋ̄***	***ʌ̀ʌ̄***	***ʍ̀ʍ̄***	***ʎ̀ʎ̄***	***ʏ̀ʏ̄***	***ʐ̀ʐ̄***	***ʑ̀ʑ̄***
ideal	x-1, b	x, b	x, d+52	a, d+21	a, d+21	a, d+21	a, d	x, d+67	a-103, d	x, b	x-1, b	x, b	x, b	x, b	a-56, b	x, b	x+23, d+38	x+23, d+107
a												259, 645						
a?	214, 642	214, 645	178, 645	195, 890	195, 890	195, 890	189, 890	149, 645	172, 787	255, 645	248, 642	241, 645	239, 645	363, 645	286, 834	228, 645	206, 666	212, 666
aδ												18, 0						
b											258, -114	251, -114				239, -107		
b?	223, -110	223, -110	155, -265	116, -296	116, -296	116, -296	105, -319	129, -250	109, -319	264, -110	257, -110	249, -110	247, -110	371, -110	256, -110	237, -110	176, -279	196, -210
bδ											1, -4	2, -4				2, 3		

Table 11: Font metrics for Latin and IPA 7/7

hex	0292	0293	0294	0295	0296	0297	0299	029B	029C	029D	029F	03B2
cat	Ll	Ll	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ʒ̇ʒ̲	ʒ̇ʒ̲	ʔ̇ʔ̲	ʕ̇ʕ̲	ʝ̇ʝ̲	ɕ̇ɕ̲	ʙ̇ʙ̲	ɠ̇ɠ̲	ʜ̇ʜ̲	ʝ̇ʝ̲	ʟ̇ʟ̲	β̇β̲
patch	ʒ̇ʒ̲	ʒ̇ʒ̲	ʔ̇ʔ̲	ʕ̇ʕ̲	ʝ̇ʝ̲	ɕ̇ɕ̲	ʙ̇ʙ̲	ɠ̇ɠ̲	ʜ̇ʜ̲	ʝ̇ʝ̲	ʟ̇ʟ̲	β̇β̲
ideal	x, d	x, d	c, b	c, b	c, b	x, d+11	x, b	a-121, b	x, b	a-51, d+55	x, b	a, d
a	218, 646	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	
a?	216, 645	220, 645	211, 850	221, 850	208, 850	221, 645	232, 645	294, 764	293, 645	126, 834	207, 645	262, 885
aδ	2, 1	-11, 0	-2, -205	-12, -205	1, -205	-12, 0	-24, 0	-85, -119	-84, 0	83, -189	2, 0	
b	185, -319	196, -110	202, -107	230, -107	196, -110	196, -110	196, -110	196, -110	305, -110	196, -110	196, -110	275, -114
b?	216, -319	220, -319	211, -110	221, -110	208, -110	221, -306	232, -110	294, -110	293, -110	126, -262	207, -110	262, -319
bδ	-32, 0	-24, 209	-9, 3	9, 3	-12, 0	-25, 196	-36, 0	-98, 0	12, 0	70, 152	-11, 0	14, 205
it	*ʒ̇ʒ̲*	*ʒ̇ʒ̲*	*ʔ̇ʔ̲*	*ʕ̇ʕ̲*	*ʝ̇ʝ̲*	*ɕ̇ɕ̲*	*ʙ̇ʙ̲*	*ɠ̇ɠ̲*	*ʜ̇ʜ̲*	*ʝ̇ʝ̲*	*ʟ̇ʟ̲*	*β̇β̲*
patch	*ʒ̇ʒ̲*	*ʒ̇ʒ̲*	*ʔ̇ʔ̲*	*ʕ̇ʕ̲*	*ʝ̇ʝ̲*	*ɕ̇ɕ̲*	*ʙ̇ʙ̲*	*ɠ̇ɠ̲*	*ʜ̇ʜ̲*	*ʝ̇ʝ̲*	*ʟ̇ʟ̲*	*β̇β̲*
ideal	x+15, d	x, d-80	c+27, b	c+38, b	c, b-23	x, d	x, b	a-94, b	x, b	a-80, d+63	x, b	a, d
a	328, 646											
a?	353, 658	355, 645	373, 850	373, 850	373, 850	411, 645	352, 645	410, 796	416, 645	303, 810	355, 645	425, 890
aδ	-25, -12											
b	130, -319		202, -107	230, -107					273, -110			275, -114
b?	145, -319	133, -397	169, -110	169, -110	164, -131	206, -319	192, -110	217, -110	256, -110	77, -254	194, -110	168, -319
bδ	-15, 0		33, 3	61, 3					17, 0			107, 205
sb	**ʒ̇ʒ̲**	**ʒ̇ʒ̲**	**ʔ̇ʔ̲**	**ʕ̇ʕ̲**	**ʝ̇ʝ̲**	**ɕ̇ɕ̲**	**ʙ̇ʙ̲**	**ɠ̇ɠ̲**	**ʜ̇ʜ̲**	**ʝ̇ʝ̲**	**ʟ̇ʟ̲**	**β̇β̲**
patch	**ʒ̇ʒ̲**	**ʒ̇ʒ̲**	**ʔ̇ʔ̲**	**ʕ̇ʕ̲**	**ʝ̇ʝ̲**	**ɕ̇ɕ̲**	**ʙ̇ʙ̲**	**ɠ̇ɠ̲**	**ʜ̇ʜ̲**	**ʝ̇ʝ̲**	**ʟ̇ʟ̲**	**β̇β̲**
ideal	x+20, d	x+15, d-96	c, b	c, b	c, b	x, d	x, b	a-121, b	x, b	a-64, d+65	x, b	a, d
a	234, 645											
a?	227, 663	236, 658	206, 850	206, 850	206, 850	273, 645	246, 645	290, 764	316, 645	132, 821	219, 645	290, 885
aδ	7, -18											
b	193, -319								351, -110			
b?	227, -319	236, -413	206, -110	206, -110	206, -110	273, -319	246, -110	290, -110	316, -110	132, -252	219, -110	290, -319
bδ	-34, 0								35, 0			
si	***ʒ̇ʒ̲***	***ʒ̇ʒ̲***	***ʔ̇ʔ̲***	***ʕ̇ʕ̲***	***ʝ̇ʝ̲***	***ɕ̇ɕ̲***	***ʙ̇ʙ̲***	***ɠ̇ɠ̲***	***ʜ̇ʜ̲***	***ʝ̇ʝ̲***	***ʟ̇ʟ̲***	***β̇β̲***
patch	***ʒ̇ʒ̲***	***ʒ̇ʒ̲***	***ʔ̇ʔ̲***	***ʕ̇ʕ̲***	***ʝ̇ʝ̲***	***ɕ̇ɕ̲***	***ʙ̇ʙ̲***	***ɠ̇ɠ̲***	***ʜ̇ʜ̲***	***ʝ̇ʝ̲***	***ʟ̇ʟ̲***	***β̇β̲***
ideal	x+24, d	x+15, d-96	c, b	c, b	c, b	x, d+15	x, b	a-94, b	x+28, b	a-78, d+65	x, b	a, d
a	218, 646											
a?	211, 667	213, 658	227, 850	227, 850	227, 850	268, 645	213, 645	278, 796	342, 671	158, 812	211, 645	302, 890
aδ	7, -21											
b	141, -319		202, -107	230, -107					327, -110			275, -114
b?	172, -319	157, -413	194, -110	194, -110	194, -110	237, -302	221, -110	256, -110	345, -110	103, -252	219, -110	218, -319
bδ	-31, 0		8, 3	36, 3					-18, 0			57, 205

Table 12: Font metrics for above marks 1/2

hex	0300	0301	0302	0303	0304	0306	0307	0308	0309	030A	030B	030C	030D	030F	0310
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	ǹH̀	ńH́	n̂Ĥ	ñH̃	n̄H̄	n̆H̆	ṅḢ	n̈Ḧ	n̊Ḣ	n̊Ḧ	ő́Ḧ	n̆Ḧ	ṅḢ	n̋H̋	n̊̃Ḧ
patch	ǹH̀	ńH́	n̂Ĥ	ñH̃	n̄H̄	n̆H̆	ṅḢ	n̈Ḧ	n̊Ḣ	n̊Ḧ	ő́Ḧ	n̆Ḧ	ṅḢ	n̋H̋	n̊̃Ḧ
m	-142, 683	-191, 682	-139, 697	-149, 697	-209, 644	-161, 681	-180, 667	-210, 645	-160, 765	-184, 708	-219, 714	-205, 676	-155, 688	-140, 758	-251, 675
m?	-159, 690	-160, 691	-140, 697	-149, 711	-212, 678	-161, 672	-180, 667	-210, 669	-161, 755	-182, 711	-170, 700	-204, 676	-154, 691	-168, 724	-251, 672
mδ	17, -7	-31, -9	1, 0	0, -14	3, -34	0, 9	0, 0	0, -24	1, 10	-2, -3	-49, 14	-1, 0	-1, -3	28, 34	0, 3
it	*ǹH*	*ńH*	*n̂H*	*ñH̃*	*n̄H̄*	*n̆H*	*ṅḢ*	*n̈Ḧ*	*n̊Ḣ*	*n̊Ḧ*	*ő́Ḧ*	*n̆Ḧ*	*ṅḢ*	*n̋H̋*	*n̊̃Ḧ*
patch	*ǹH̀*	*ńH́*	*n̂Ĥ*	*ñH̃*	*n̄H̄*	*n̆H̆*	*ṅḢ*	*n̈Ḧ*	*n̊Ḣ*	*n̊Ḧ*	*ő́Ḧ*	*n̆Ḧ*	*ṅḢ*	*n̋H̋*	*n̊̃Ḧ*
m	-3, 683	-52, 682	3, 697	-17, 697	-78, 645	-22, 681	-40, 689	-79, 645	-4, 765	-40, 708	-74, 714	-67, 676	-15, 688	14, 758	-114, 675
m?	-18, 689	-18, 690	3, 696	-3, 716	-71, 680	-24, 671	-35, 698	-74, 669	-8, 754	-37, 712	-26, 699	-66, 675	-13, 690	-19, 723	-114, 671
mδ	15, -6	-34, -8	0, 1	-14, -19	-7, -35	2, 10	-5, -9	-5, -24	4, 11	-3, -4	-48, 15	-1, 1	-2, -2	33, 35	0, 4
sb	**ǹH**	**ńH**	**n̂H**	**ñH̃**	**n̄H̄**	**n̆H**	**ṅḢ**	**n̈Ḧ**	**n̊Ḣ**	**n̊Ḧ**	**ő́Ḧ**	**n̆Ḧ**	**ṅḢ**	**n̋H̋**	**n̊̃Ḧ**
patch	**ǹH̀**	**ńH́**	**n̂Ĥ**	**ñH̃**	**n̄H̄**	**n̆H̆**	**ṅḢ**	**n̈Ḧ**	**n̊Ḣ**	**n̊Ḧ**	**ő́Ḧ**	**n̆Ḧ**	**ṅḢ**	**n̋H̋**	**n̊̃Ḧ**
m	-127, 713	-180, 713	-127, 700	-133, 679	-205, 625	-149, 684	-169, 603 (1)	-198, 557 (1)	-162, 682 (1)	-170, 646 (1)	-197, 733	-193, 676	-137, 719	-127, 758	-239, 675
m?	-144, 702	-144, 699	-128, 704	-136, 719	-200, 660	-149, 680	-168, 603	-186, 580	-148, 672	-170, 630	-158, 714	-192, 686	-137, 696	-152, 737	-235, 680
mδ	17, 11	-36, 14	1, -4	3, -40	-5, -35	0, 4	-1, 0	-12, -23	-14, 10	0, 16	-39, 19	-1, -10	0, 23	25, 21	-4, -5
si	***ǹH̀***	***ńH́***	***n̂Ĥ***	***ñH̃***	***n̄H̄***	***n̆H̆***	***ṅḢ***	***n̈Ḧ***	***n̊Ḣ***	***n̊Ḧ***	***ő́Ḧ***	***n̆Ḧ***	***ṅḢ***	***n̋H̋***	***n̊̃Ḧ***
patch	***ǹH̀***	***ńH́***	***n̂Ĥ***	***ñH̃***	***n̄H̄***	***n̆H̆***	***ṅḢ***	***n̈Ḧ***	***n̊Ḣ***	***n̊Ḧ***	***ő́Ḧ***	***n̆Ḧ***	***ṅḢ***	***n̋H̋***	***n̊̃Ḧ***
m	-142, 683	-191, 682	-139, 697	-134, 700	-209, 645	-161, 681	-180, 667	-210, 645	-160, 765	-184, 708	-219, 714	-205, 676	-155, 688	-140, 758	-251, 675
m?	-154, 694	-152, 694	-136, 697	-141, 717	-217, 651	-162, 673	-177, 692	-243, 671	-145, 759	-174, 717	-166, 704	-205, 676	-147, 686	-156, 728	-248, 673
mδ	12, -11	-39, -12	-3, 0	7, -17	8, -6	1, 8	-3, -25	33, -26	-15, 6	-10, -9	-53, 10	0, 0	-8, 2	16, 30	-3, 2

Table 13: Font metrics for above marks 2/2

hex	0311	0312	033D	033E	033F	030E	0346	034A	034B	034C	0350	0351	0352	0357	035B
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	n̑H̑	n̑H̑	n̑H̑	n̑H̑	n̄H̄	n̋H̋	n̑H̑	n̑H̑	n̊H̊	n̄H̄	n̆H̆	n̓H̓	n̂Ĥ	n̓H̓	n̆H̆
patch	n̑H̑	n̑H̑	n̑H̑	n̑H̑	n̄H̄	n̋H̋	n̑H̑	n̑H̑	n̊H̊	n̄H̄	n̆H̆	n̓H̓	n̂Ĥ	n̓H̓	n̆H̆
m	-205, 645	-110, 605	-265, 605	-265, 605	-235, 611	-245, 719	-251, 621	-215, 645	-205, 645	-235, 645	-201, 645	-171, 645	-251, 645	-171, 645	-164, 645
m?	-208, 641	-102, 611	-260, 582	-247, 584	-240, 622	-244, 682	-255, 624	-217, 637	-217, 611	-238, 647	-183, 643	-156, 652	-251, 652	-141, 662	-174, 632
mδ	3, 4	-8, -6	-5, 23	-18, 21	5, -11	-1, 37	4, -3	2, 8	12, 34	3, -2	-18, 2	-15, -7	0, -7	-30, -17	10, 13
it	*n̑H̑*	*n̑H̑*	*n̑H̑*	*n̑H̑*	*n̄H̄*	*n̋H̋*	*n̑H̑*	*n̑H̑*	*n̊H̊*	*n̄H̄*	*n̆H̆*	*n̓H̓*	*n̂Ĥ*	*n̓H̓*	*n̆H̆*
patch	*n̑H̑*	*n̑H̑*	*n̑H̑*	*n̑H̑*	*n̄H̄*	*n̋H̋*	*n̑H̑*	*n̑H̑*	*n̊H̊*	*n̄H̄*	*n̆H̆*	*n̓H̓*	*n̂Ĥ*	*n̓H̓*	*n̆H̆*
m	-74, 645	13, 605	-142, 605	-142, 605	-111, 611	-99, 719	-125, 621	-84, 645	-74, 645	-104, 645	-70, 645	-40, 645	-120, 645	-40, 645	-33, 645
m?	-76, 640	24, 610	-142, 581	-128, 583	-113, 621	-105, 681	-128, 623	-87, 636	-92, 610	-107, 646	-53, 642	-23, 651	-118, 651	-6, 661	-46, 631
mδ	2, 5	-11, -5	0, 24	-14, 22	2, -10	6, 38	3, -2	3, 9	18, 35	3, -1	-17, 3	-17, -6	-2, -6	-34, -16	13, 14
sb	**n̑H̑**	**n̑H**	**n□H□**	**n□H□**	**n□H□**	**n̋H̋**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n̓H‘**	**n□H□**	**n̓H’**	**n̆H**
patch	**n̑H̑**	**n̑H̑**	**n̑H̑**	**n̑H̑**	**n̄H̄**	**n̋H̋**	**n̑H̑**	**n̑H̑**	**n̊H̊**	**n̄H̄**	**n̆H̆**	**n̓H̓**	**n̂Ĥ**	**n̓H̓**	**n̆H̆**
m	-205, 645	-97, 504 (1)				-232, 719						100, 547 (1)		100, 549 (1)	-180, 639
m?	-220, 650	-102, 528				-232, 696						106, 547		112, 547	-184, 681
mδ	15, -5	5, -24				0, 23						-6, 0		-12, 2	4, -42
si	***n̑H̑***	***n̑H̑***	***n□H□***	***n□H□***	***n□H□***	***n̋H̋***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***
patch	***n̑H̑***	***n̑H̑***	***n̑H̑***	***n̑H̑***	***n̄H̄***	***n̋H̋***	***n̑H̑***	***n̑H̑***	***n̊H̊***	***n̄H̄***	***n̆H̆***	***n̓H̓***	***n̂Ĥ***	***n̓H̓***	***n̆H̆***
m	-205, 645	-110, 605				-245, 719									
m?	-226, 644	-116, 614				-242, 686									
mδ	21, 1	6, -9				-3, 33									

Table 14: Font metrics for below marks 1/2

hex	0316	0317	0318	0319	031C	0320	0323	0324	0325	0326	0329	032A	032B	032C	032D	032E
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	n̖H	n̗H	n̘H	n̙H	n̚H	n̛H	n̜H	n̝H	n̞H	n̟H	n̠H	n̡H	n̢H	n̤H	n̥H	n̦H
patch	n̖H	n̗H	n̘H	n̙H	n̚H	n̛H	n̜H	n̝H	n̞H	n̟H	n̠H	n̡H	n̢H	n̤H	n̥H	n̦H
m	-219, -133	-210, -132	-224, -148	-285, -147	-216, -146	-253, -148	-216, -113	-218, -116	-218, -116	-255, -82	-257, -122	-253, -113	-251, -108	-204, -116	-195, -133	-251, -110
m?	-248, -124	-224, -114	-249, -102	-249, -101	-262, -98	-252, -170	-216, -152	-217, -152	-217, -78	-245, -87	-256, -106	-255, -134	-252, -119	-204, -123	-195, -128	-252, -109
mδ	29, -9	14, -18	25, -46	-36, -46	46, -48	-1, 22	0, 39	-1, 36	-1, -38	-10, 5	-1, -16	2, 21	1, 11	0, 7	0, -5	1, -1
it	*n̖H*	*n̗H*	*n̘H*	*n̙H*	*n̚H*	*n̛H*	*n̜H*	*n̝H*	*n̞H*	*n̟H*	*n̠H*	*n̡H*	*n̢H*	*n̤H*	*n̥H*	*n̦H*
patch	*n̖H*	*n̗H*	*n̘H*	*n̙H*	*n̚H*	*n̛H*	*n̜H*	*n̝H*	*n̞H*	*n̟H*	*n̠H*	*n̡H*	*n̢H*	*n̤H*	*n̥H*	*n̦H*
m	-246, -133	-237, -132	-254, -148	-315, -147	-246, -146	-283, -148	-239, -113	-242, -116	-242, -116	-272, -82	-282, -122	-276, -113	-273, -108	-228, -116	-222, -133	-275, -110
m?	-274, -126	-248, -116	-270, -104	-270, -103	-281, -100	-287, -172	-238, -111	-239, -111	-233, -82	-263, -89	-278, -108	-283, -136	-277, -121	-229, -125	-221, -130	-269, -111
mδ	28, -7	11, -16	16, -44	-45, -44	35, -46	4, 24	-1, -2	-3, -5	-9, -34	-9, 7	-4, -14	7, 23	4, 13	1, 9	-1, -3	-6, 1
sb	**n̖H**	**n̗H**	**n̘H**	**n̙H**	**n̚H**	**n̛H**	**n̜H**	**n̝H**	**n̞H**	**n̟H**	**n̠H**	**n̡H**	**n̢H**	**n̤H**	**n̥H**	**n̦H**
patch	**n̖H**	**n̗H**	**n̘H**	**n̙H**	**n̚H**	**n̛H**	**n̜H**	**n̝H**	**n̞H**	**n̟H**	**n̠H**	**n̡H**	**n̢H**	**n̤H**	**n̥H**	**n̦H**
m	-219, -133	-210, -132	-224, -148	-285, -147	-216, -146	-253, -149	-216, -113	-218, -116	-218, -116	-255, -73	-257, -124	-253, -113	-251, -108	-208, -116	-195, -133	-239, -110
m?	-248, -132	-224, -122	-249, -110	-249, -110	-262, -106	-251, -179	-216, -118	-218, -113	-216, -88	-244, -86	-256, -116	-255, -142	-252, -127	-204, -131	-195, -136	-240, -105
mδ	29, -1	14, -10	25, -38	-36, -37	46, -40	-2, 30	0, 5	0, -3	-2, -28	-11, 13	-1, -8	2, 29	1, 19	-4, 15	0, 3	1, -5
si	***n̖H***	***n̗H***	***n̘H***	***n̙H***	***n̚H***	***n̛H***	***n̜H***	***n̝H***	***n̞H***	***n̟H***	***n̠H***	***n̡H***	***n̢H***	***n̤H***	***n̥H***	***n̦H***
patch	***n̖H***	***n̗H***	***n̘H***	***n̙H***	***n̚H***	***n̛H***	***n̜H***	***n̝H***	***n̞H***	***n̟H***	***n̠H***	***n̡H***	***n̢H***	***n̤H***	***n̥H***	***n̦H***
m	-219, -133	-210, -132	-224, -148	-285, -147	-216, -146	-253, -148	-216, -113	-218, -116	-218, -116	-255, -82	-257, -122	-253, -113	-251, -108	-204, -116	-195, -133	-251, -110
m?	-251, -132	-225, -122	-246, -110	-259, -110	-260, -106	-263, -179	-214, -117	-216, -113	-211, -88	-237, -86	-256, -116	-260, -142	-253, -127	-206, -127	-199, -136	-244, -113
mδ	32, -1	15, -10	22, -38	-26, -37	44, -40	10, 31	-2, 4	-2, -3	-7, -28	-18, 4	-1, -6	7, 29	2, 19	2, 11	4, 3	-7, 3

Table 15: Font metrics for below marks 2/2

hex	032F	031D	031E	031F	0330	0331	0333	0339	033A	033B	033C	0347	0348	0349	034D	034E
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯
patch	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯
m	-250, -116	-249, -143	-247, -106	-249, -145	-213, -70	-244, -110	-244, -110	-216, -146	-253, -113	-246, -113	-251, -108	-244, -110	-197, -110	-192, -110	-222, -110	-227, -110
m?	-251, -125	-249, -112	-249, -112	-252, -99	-205, -68	-240, -133	-240, -123	-178, -98	-255, -134	-252, -118	-252, -119	-240, -123	-195, -94	-189, -111	-228, -92	-228, -74
mδ	1, 9	0, -31	2, 6	3, -46	-8, -2	-4, 23	-4, 13	-38, -48	2, 21	6, 5	1, 11	-4, 13	-2, -16	-3, 1	6, -18	1, -36
it	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*
patch	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*
m	-274, -116	-278, -143	-269, -106	-278, -145	-227, -70	-266, -110	-266, -110	-246, -146	-276, -113	-269, -113	-273, -108	-266, -110	-219, -110	-214, -110	-244, -110	-249, -110
m?	-276, -127	-271, -114	-272, -114	-271, -101	-220, -74	-268, -135	-266, -125	-198, -100	-283, -136	-277, -120	-275, -121	-266, -125	-214, -96	-213, -113	-247, -94	-243, -76
mδ	2, 11	-7, -29	3, 8	-7, -44	-7, 4	2, 25	0, 15	-48, -46	7, 23	8, 7	2, 13	0, 15	-5, -14	-1, 3	3, -16	-6, -34
sb	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**
patch	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**
m	-250, -116	-249, -143	-247, -106	-249, -146	-213, -70	-244, -58										
m?	-263, -127	-249, -121	-249, -120	-251, -108	-192, -70	-240, -89										
mδ	13, 11	0, -22	2, 14	2, -38	-21, 0	-4, 31										
si	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***
patch	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***
m	-250, -116	-249, -143	-247, -106	-249, -145	-213, -70	-244, -110										
m?	-265, -131	-252, -132	-247, -108	-249, -108	-184, -77	-234, -89										
mδ	15, 15	3, -11	0, 2	0, -37	-29, 7	-10, -21										

Vertical references, anchor Y coordinates and reference, and clearance

A font has at least five vertical reference lines where glyph outlines live inside: descender height, baseline, x-height, ascender height, capital height. The values or ranges of these Y coordinates are in the first five rows of Table 16. These vertical aspects are the gray horizontal lines of Figure 1. These vertical aspects are displayed in a FontForge glyph outline window as black horizontal guidelines with red or blue shaded bands for ranges. I denote these as yra, yrc, yrx, $yr0$, yrd because they are Y values, and r indicates a reference value. a, c, x, 0, d indicate the five vertical aspects. These values (and the ranges that set the bands) are in the SFD, in a line that starts with `BlueValues`, e.g. in Libertinus serif regular,

```
BlueValues 47 [-12 0 429 442 460 472 600 610 645 658 688 698]
OtherBlues 11 [-238 -227]
```

means $yra = 698, [688, 698]$; $yrc = 645, [645, 658]$; $yrx = 429, [429, 442]$; $yr0 = 0, [-12, 0]$; and $yrd = -232, [-238, -227]$.

In the anchor model for Libertinus, the Y coordinate of an above-anchor (ay) or a below-anchor (by) is at a stable value depending on where the glyph outline lives relative to a vertical aspect. See the rows a and b in Tables 1–4 and find the common values according to the glyph's vertical aspects. There is no set name in font design for these above- or below-anchor Y coordinate reference values by ascender aspect, so I will denote them $ayra$, $ayrc$, $ayrx$, $byr0$, $byrd$, These are the middle five rows of Table 16. For example, all glyphs that sit on the baseline have below-anchor Y coordinate $by = -110$, and thus $byr0 = -110$. All capitals A–Z have $ay = 850$, and thus $ayrc = 850$. Any glyph that descends to the descender value $yrd = -232$ or descender range $[-238, -227]$ should have $by = byrd = -319$.

The clearance is defined as the difference between the anchor reference value and the vertical aspect Y value. I will denote these clearances as $ay\delta a$, $ay\delta c$, $ay\delta x$, $by\delta 0$, $by\delta d$. So, $ay\delta a = ayra - yra$ etc. For example, capital letters A–Z have a capital height $yrc = 645$, and all capital letters A–Z have an above-anchor Y coordinate at $ayrc = 850$. So the above-anchor clearance for capitals is $ay\delta c = ayrc - yec = 850 - 645 = 205$. Because of the weird anchor model for Libertinus, this clearance value is not the exact vertical distance between a base and a mark, but this clearance value is still useful.

A glyph may not be a full ascender or a full descender, and its bounding box may not live closely within two vertical aspects. In such a case, the anchor Y coordinate is set by looking for the nearest, or most suitable, vertical aspect, and considering the clearance of that vertical aspect relative to the bounding box of the glyph.

Here is the heuristic algorithm for calculating the Y coordinate of the above- or below-anchor (ay or by) of a glyph that is not a full ascender or descender.

Consider a glyph that is not a full ascender. Find the nearest, or most suitable, vertical aspect – ascender a, capital c, or x-height x. Then $ay = Y_{\text{max}} + ay\delta[acx] - \epsilon$, where Y_{max} is the maximum Y coordinate of bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

Capitals are the exception: their top reference line is always the capline, and their above-anchor always has the capital reference Y ($ay = 850$), even where the outline rises above the capline. In humanist letterforms capitals do not have ascenders: an ascender is a stroke of the pen, not merely a high Y_{max}. What rises above the capline in Libertinus is away from the horizontal center of the glyph, where the mark sits: the top-left stroke of J (italic and semibold italic, 34 units above the capline), the serifs of T (semibold and semibold italic, 28), the slash of Ø (25), the top of Ȝ (up to 29), and the hooks of C̓ and G̓ (98). A mark at the capital reference Y does not collide with these. The one capital with a genuine ascender is the capital iota ɩ (U+0196), whose stem reaches the ascender line; it takes the ascender reference ($ay = 885$; italic styles 890). The glottal stop ʔ (U+0294) and pharyngeal ʕ (U+0295), counted here as capitals, rise above the capline only in italic (672 and 683); there they have curated anchors at $ay = 850 + (Y_{\text{max}} - 645) - 10$, with the bowl overshoot correction $\epsilon = 10$ instead of the default 2, which gives 867 and 878.

Consider a glyph that is not a full descender. Find the nearest, or most suitable, vertical aspect – baseline 0 or desender d. Then $by = Y_{\text{min}} + by\delta[0d] + \epsilon$, where Y_{min} in the minimum Y coordinate of the bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

The capital anchor Y reference and clearance of semibold are the patched values. The designer set semibold's capitals near 805 (clearance 160) and compensated only in the capital alternates of the marks, so a mark without a capital alternate crashed into the capital: in 225 of the 780 combinations of A–Z with 30 above marks, the bottom of the mark was below the top of the letter. The patch moves the capitals to 850 and adds 45 to the mark anchors of the capital alternates, which stay where they were. That leaves 4 such combinations, all on T, whose serif tips rise to 673 away from the marks; under the marks the gap is 21–26.

For the vertical aspect, the first single value is the optical value of the vertical aspect. In practice, this is the y-coordinate of the central control point of a (vertical) stem that meets that aspect. In the table, the range for the vertical aspects are the BlueValues set by the designer in the SFD and seen as guidelines in a FontForge glyph outline window view.

I need to keep track of serif and bowl corrections relative to the (single) vertical aspect value. In Libertinus serif regular, a glyph that meets an aspect with a stem needs a 2 unit correction. For example, the letter x meets the baseline and x-height aspects, with stems, but $y_{\text{min}} = 0 - 2 = -2$ and $y_{\text{max}} = 429 + 2 = 431$. For bowls, the corrections tend to be different for each

Table 16: Vertical aspects, anchor Y references, and clearances of Libertinus

	v. aspect	regular	italic	semibold	semibold italic
yra	ascender	$698, [688, 698]$	$698, [688, 698]$	$698, [690, 698]$	$698, [696, 705]$
yrc	capital	$645, [645, 658]$	$645, [645, 658]$	$645, [645, 662]$	$645, [645, 662]$
yrx	x-height	$429, [429, 442]$	$429, [429, 442]$	$434, [433, 447]$	$434, [434, 447]$
$yr0$	baseline	$0, [-12, 0]$	$0, [-12, 1]$	$0, [-12, 1]$,	$0, [-20, 0]$
yrd	descender	$-232, [-238, -227]$	$-232, [-238, -227]$	$-232, [-238, -212]$	$-232, [-239, -219]$
$ayra$	ascender	885	890	885	890
$ayrc$	capital	850	850	850	850
$ayrx$	x-height	645	645	645	645
$byr0$	baseline	-110	-110	-110	-110
$byrd$	descender	-319	-319	-319	-319
$ay\delta a$	ascender	187	192	187	192
$ay\delta c$	capital	205	205	205	205
$ay\delta x$	x-height	216	216	211	211
$by\delta 0$	baseline	110	110	110	110
$by\delta d$	descender	87	87	87	87

aspect. For Libertinus serif regular, a glyph that is a bowl full-descender (usually) has a 6 unit overshoot, i.e. $y_{\text{min}} = -232 - 6 = -238$. A bowl on the baseline (usually) has a 10 unit overshoot, i.e. $y_{\text{min}} = 0 - 10 = -10$. A bowl at the x-height aspect (usually) has a 10 unit overshoot, i.e. $y_{\text{max}} = 429 + 10 = 439$. A bowl at the capital aspect (usually) has a 13 unit overshoot, i.e. $y_{\text{max}} = 645 + 13 = 658$. For full ascenders, there is no correction in practice, regardless of the shape of the glyph's ascender, because $y_{\text{max}} = 698$ for all such glyphs.

That is, these stem and bowl corrections are 2 and 6, 10, 10, 13.

For the other font styles (italic, semibold, semibold italic), these corrections are remarkably consistent for all aspects, even though e.g. italic small letters do not really have flat terminal stems. But I think that the correction principle is valid for glyphs that are not within the threshold of an expected vertical aspect. For a non-full descender, if y_{min} is closest to the descender aspect, and the glyph looks like a bowl at this lowest point, then the correction is 6; otherwise I assume that the glyph must look like a flat stem and the correction is 2. But for a non-full descender that is closer to the baseline, the bowl correction is 10. Full descender bowls tend to be flatter because the lower threhold of the descender really is a hard lower bound.

Copied anchors

Regular, above

(281, 850) ĖĚÊṖP̌P̂

(336, 850) ḰǨK̂V̇V̌V̂

(346, 850) ṄŇN̂ÙŬÛ

(209, 645) ȧăâạ̈ç̈ç̇ç̆ç̈ḙ̈ḛ̆ḛ̂ȝ̈ȝ̆ȝ̂ģ̈ǧg̈ḫ̈ḫ̇ḫ̆ḫ̂ḽ̈ḽ̇ḽ̆ḽ̂ḳ̈ḳ̇ḳ̆ḳ̂ụ̈ǔq̈ṇ̈ṅ̇ň̆n̂ȯ̈ȯ̇ȯ̆ȯ̂œ̈œ̇œ̆œ̂ȫ̇ȫ̆ȫ̂ṗ̈ṗ̇ṗ̆ṗ̂ɨ̈ɨ̇ɨ̆ɨ̂ḽ̈ḽ̇ḽ̆ḽ̂ṛ̈ṛ̇ṛ̆ṛ̂ṟ̈ṟ̇ṟ̆ṟ̂ṙ̈ṙ̇ṙ̆ṙ̂ṝ̈ṝ̇ṝ̆ṝ̂ṟ̈ṟ̇ṟ̆ṟ̂ṛ̈ṛ̇ṛ̆ṛ

(228, 645) ėěêàǎâ
(239, 645) ȅěȇġǧĝ
(259, 645) ẏẙŷùǔû
(299, 565) ɦȟȟȟȟȟ

Regular, below

(287, -110) ẹ̲ę̲ę̲ẹ̲ẓ̲ẓ̲
(109, -319) j̰j̰j̰j̰j̰
(350, -110) ọ̲ǫ̲ǫ̲ǫ̲ǫ̲ọ̲ǫ̲ǫ̲
(234, -110) ṣ̲ş̲s̲ḅ̲b̲b̲ẹ̲ę̲
(196, -110) ạa̲aṣ̲s̲s̲ɪɪɪllllklklɵɵɵɵœœœɸɸɸɸɾɾɾ

Semibold, above

(363, 802) ȮȎȏȪȫȰȱ
(264, 646) ṅṉṋṷṻṹ

Semibold, below

(109, -319) J̩J̩j̩j̩j̩

Semibolditalic, above

No clusters detected.

Semibolditalic, below

No clusters detected.
ƀ ƅ ƀ ƅ̀ ƀ ƅ ƀ Ɓ Ɓ Ɓ Ƃ̀ Ƃ Ɓ Ƃ Ɗ Ɗ Ɗ Ɗ̀ Ɗ Ɗ Ɗ Ɯ Ɯ̀ Ɯ Ɯ̀ Ɯ̄ Ɯ̀ Ɯ Ɯ̀ j̩ j̩ j̩ j̩̄ j̩ j̩̄ j̩ ɵ̀ ɵ ɵ̀ ɵ̄ ɵ̄ ɵ̄ ɵ̀ ɵ̀ ɵ̀ ɵ̄ ɵ̄ ɵ̀ ɯ̀ ɯ ɯ̀ ɯ̄ ɯ̀ ɯ̄ ɯ̀ ɯ̄

Anchor audit

This section lists the above- and below-anchors of the Latin and IPA letters that still need a decision, in the patched fonts. The letters are those of stage 6, `BASE_LATIN` and `BASE_IPA`. Other scripts, precomposed letters and superscripts are not audited. What the patch has already fixed does not appear.

The tests ask whether marks fit, not where an anchor came from. They are in `anchor_audit.py`; the tests for copied anchors are left out here, since a copy that fits is not a problem. Semibold italic is judged by its own model: its shift K is added back before x is compared with the glyph.

Each anchor is compared with the candidate font (`fonts/*-candidate.otf`): the patch run again with every Latin and IPA above- and below-anchor deleted, so that stages 6–8 refill them. That is what a `"delete"` override gives. Curated anchors come back unchanged, and a derived letter inherits from its base letter. Each anchor falls in one group, by the action it needs:

Replace. The anchor fails: a mark overlaps the glyph, or the anchor is in the wrong place. The candidate moves it and passes. If the right-hand samples look better, add `{cls: "delete"}` to `data/fontanchors_audit/overrides.py`, with class 0 for above and 2 for below.

Curate. The anchor fails, and the candidate is the same or fails too. Add a curated anchor to `data/fontanchors_human/{style}.py`.

Judge. The anchor passes, but a test is suspicious (a mark crowds the glyph, for instance), and the candidate differs. Delete it only if the candidate is clearly better.

Kept. Add `{cls: "keep"}` to the overrides for an anchor that is right as it stands, such as a deliberate design choice. It leaves the groups above and is listed here, so the list shrinks as decisions are made.

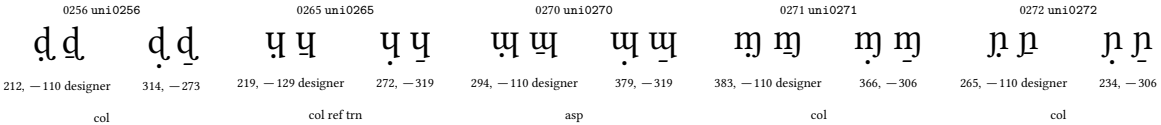
Each cell shows the codepoint and glyph name, as `overrides.py` wants it. Below that are dot, macron and circumflex above (dot and macron below), positioned by anchor through the combining grapheme joiner. A pair of samples shows the patched font on the left and the candidate font on the right, each with its anchor. The patched anchor says where it came from: *designer*, *curated* or *scripted*. A scripted anchor that fails usually follows from its base letter, in which case the override belongs on the base letter. The tags name the tests the patched anchor fails or is flagged by.

	Replace	Curate	Judge	Kept	No anchor
Regular	0	0	5	3	0
Italic	6	0	16	1	0
Semibold	0	0	4	1	0
Semibold italic	14	0	22	1	0

Tags: **col** a mark overlaps or crowds the glyph; **in** anchor inside the outline; **sid** anchor on the wrong side of the glyph; **ext** anchor x outside the glyph; **asp** beside an ascender or descender; **ref** y far from the reference values; **clr** floats far from the glyph; **ctr** far from the centre line; **trn** turned letter not following its source; **pre** disagrees with precomposed glyphs.

Regular

Judge: passes, but flagged, and the candidate differs, below (5)

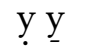


Kept (3)

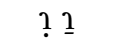
029B uni029B

265, 663 scripted

col

0079 y

242, −319 designer

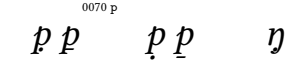
pre

027F uni027F

196, −110 designer

trn

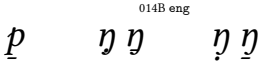
Italic

Replace: fails, and the candidate passes, below (6)

0070 p

248, −110 designer

col asp

177, −319

0014B eng

265, −110 designer

col asp

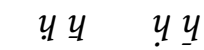
177, −319

0256 uni0256

212, −110 designer

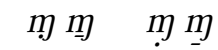
col

215, −273

0265 uni0265

219, −129 designer

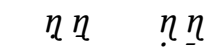
col ref

191, −319

0271 uni0271

383, −110 designer

col asp

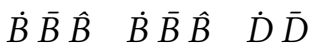
310, −319

0273 uni0273

262, −110 designer

col asp

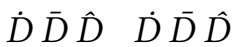
162, −319

Judge: passes, but flagged, and the candidate differs, above (10)

0042 B

447, 850 designer

pre

459, 850

0044 D

396, 850 designer

pre

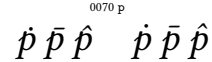
514, 850

0064 d

218, 645 designer

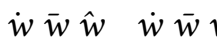
ctr asp pre

434, 890

0070 p

411, 645 designer

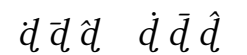
col

394, 705

0077 w

348, 645 designer

pre

481, 645

0256 uni0256

218, 645 designer

ctr

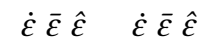
465, 899

0257 uni0257

218, 645 designer

ctr asp

464, 890

025B uni025B

322, 654 designer

ref

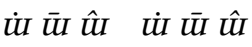
338, 645

0265 uni0265

258, 645 designer

trn

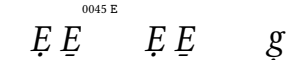
396, 645

026F uni026F

407, 645 designer

trn

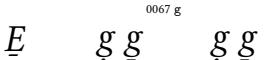
517, 645

Judge: passes, but flagged, and the candidate differs, below (6)

0045 E

269, −110 designer

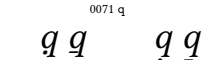
pre

240, −110

0067 g

197, −319 designer

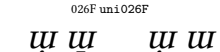
col

171, −319

0071 q

169, −110 designer

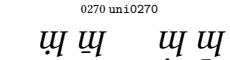
col asp

178, −319

026F uni026F

410, −110 designer

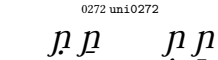
trn

354, −123

0270 uni0270

294, −110 designer

asp

310, −319

0272 uni0272

265, −110 designer

col asp

191, −319

Kept (1)

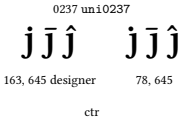
029B uni029B

378, 645 scripted

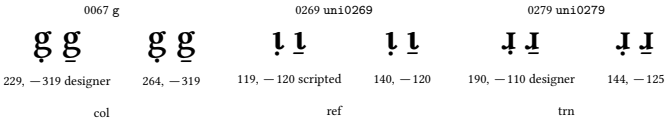
col

Semibold

Judge: passes, but flagged, and the candidate differs, above (1)



Judge: passes, but flagged, and the candidate differs, below (3)

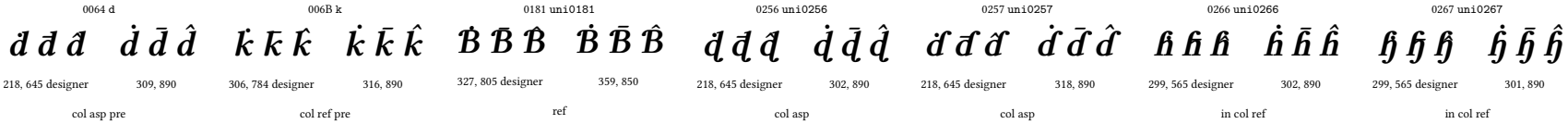


Kept (1)

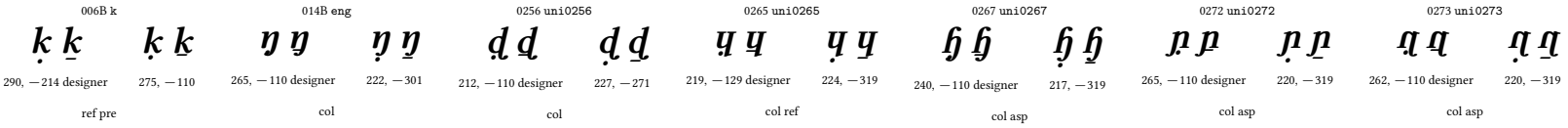


Semibold italic

Replace: fails, and the candidate passes, above (7)

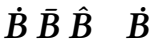


Replace: fails, and the candidate passes, below (7)



Judge: passes, but flagged, and the candidate differs, above (11)

0042 B

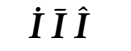


267, 850 designer

319, 850

pre

0049 I



150, 850 designer

194, 850

pre

004B K

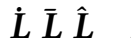


336, 850 designer

369, 850

pre

004C L



149, 850 designer

310, 850

ctr

004E N

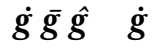


346, 850 designer

394, 850

pre

0067 g

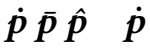


291, 645 designer

220, 645

pre

0070 p

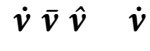


271, 645 designer

257, 679

col

0076 v

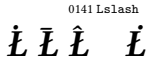


256, 645 designer

220, 645

pre

0141 Lslash

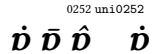


161, 850 scripted

322, 850

ctr

0252 uni0252

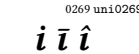


264, 645 designer

247, 645

trn

0269 uni0269



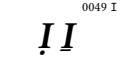
122, 654 scripted

134, 654

ref

Judge: passes, but flagged, and the candidate differs, below (11)

0049 I



147, −110 designer

161, −110

pre

0052 R

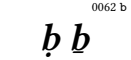


258, −110 designer

282, −110

pre

0062 b

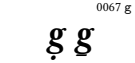


234, −110 designer

248, −110

pre

0067 g

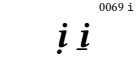


199, −319 designer

186, −319

col

0069 i

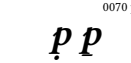


136, −110 designer

153, −110

pre

0070 p

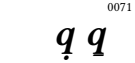


179, −319 designer

216, −319

col

0071 q

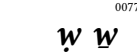


348, −319 designer

233, −319

col

0077 w

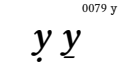


352, −110 designer

341, −110

pre

0079 y



242, −319 designer

202, −319

pre

0270 uni0270

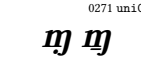


294, −110 designer

339, −319

asp

0271 uni0271



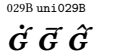
383, −110 designer

339, −319

col asp

Kept (1)

029B uni029B



284, 681 scripted

col ref

Derived letters

Many letters are a base letter plus a bar, stroke, hook, tail, curl or belt: Đ is D with a bar, Ɗ is D with a hook, ɖ is d with a tail. Where the added part stays clear of a mark, the derived letter should attach the mark where its base letter does. The designer often drew the derived letter in a shifted frame: Ɗ is D moved about 80 units right, to make room for the hook. So the derived letter's anchor is the base letter's anchor plus that shift. The heuristic candidate cannot know this: a hook or tail moves the midpoint of the bounding box that the candidate is centred on.

Stage 8 of the patch inherits these anchors. `data/derived_letters.py` lists each derived letter with its base letter and the anchor classes it may inherit: both above and below for a bar or stroke through the letter, or a hook at the top away from the centre; above only for a hook, tail or curl at the bottom; below only for a hook at the top in the way of an above mark (ʒ, ɾ, ʃ). The classes can differ by style: in italic and semibold italic, d has its above-anchor at the x-height, over the bowl beside the ascender, where the bar of đ or the hook of ɗ is in the mark's way, so these inherit only the below-anchor in the italic styles. The same was true of h, whose above-anchor is now at the ascender reference in the italic styles; ħ ɦ still inherit only the below-anchor there. The shift is found by matching outlines: at every height on a 10-unit grid, the positions where the base letter's outline crosses that height are compared with the derived letter's, and the shift that pairs the most crossings within 6 units wins. The share of the base letter's crossings so paired measures how much of the base letter the derived letter really contains. Below 70% the derived letter was redrawn rather than built on the base, and nothing is inherited in that style. Stage 8 runs after the curated anchors, so it inherits the base letter's final anchor, and it never overwrites a curated anchor of the derived letter.

In the tables, each column is a derived letter, with its base letter's codepoint below it. For each style: the derived letter with circumflex above and macron below, in the original and patched fonts (`reg`, `patch`), the base letter likewise in the patched font (`base`), and the outline shift and match (in italics if below 70%). Then, for the above anchor: a, the designer's anchor; a?, the heuristic candidate; a<, the base letter's final anchor plus the shift, in bold where the patched font has it; a!, a curated anchor, which stage 8 keeps. The same for the below anchor (b).

Two letters have an anchor set apart from these rules. The curl of ɕ reaches about -100, so at c's $Y = -110$ most below marks overlap it. Its below-anchor keeps the Y of the anchor model, under the curl (-214 regular, -208 italic, -209 semibold and semibold italic), and takes X from c's final below-anchor plus the shift, following the slant down to that height. ɽ matches r too little to inherit, but its above-anchor takes X from r in the same way, at the x-height reference, 645. Its below-anchor is curated on the axis of its stem, at the descender reference, -319. This is `derived_y` in `data/derived_letters.py`. It does not depend on the outline match, since only X is borrowed.

What the comparison shows:

- In regular and semibold most derived letters are built on the base letter (90% or more of its outline). In italic and semibold italic most were redrawn, and inherit only where the bar or stroke was added to a traced base: Đ Ð đ Ħ ħ Ø ø Ɨ ɨ ʒ ɠ, and some others.
- Several designer anchors are copies: (209, 645) and (196, -110) recur on many IPA letters in regular, including ɖ and ɭ, which have ascenders and now inherit 885 from d and l, and ʈ, which inherits t's anchor (769). ʃ, which descends, had a below-anchor at -110; it now inherits -319 from ɟ.
- The heuristic candidate is pulled sideways by a hook or bar: for Ƒ, 183 against 276 from F; for Ł, 256 against 147 from L (over the stem, as the designer placed it on L).
- Ɓ and Ɗ inherit B's and D's anchors moved about 58 and 82 units right, which the designer's copies lacked.
- The inheritance carries the base letter's own anchor, right or wrong. Where the base letter's anchor is questionable (D in regular is 76 units left of the centre of its body), Đ, Ð, Ð and Ɗ follow it.
- Checked for collisions with a circumflex and a macron: none in what is inherited. ʒ and ɾ inherit only below, because their hook rises into the above mark at the inherited height; italic and semibold italic đ ħ ɗ ɦ inherit only below (see above). ɠ is much better inherited: the designer's anchors put the marks on the hook. Italic ɦ's designer above-anchor (299, 565) put marks inside the hook; it is deleted in stage 1, and stage 6 gives it the ascender candidate.

Table 17: Derived letters and inherited anchors 1/3

hex	00D0	0110	0189	0111	0126	0127	0141	0142	00D8	00F8	019F	0275	0197	0268	0289	026B
base	0044	0044	0044	0064	0048	0068	004C	006C	004F	006F	004F	006F	0049	0069	0075	006C
reg	Đ̣	Đ̣	Đ̣	đ̣	Ħ̣	ħ̣	Ł̣	ł̣	Ø̣	ø̣	Ɵ̣	ɵ̣	ɨ̣	ɨ̣	ʉ̣	ɬ̣
patch	Đ̣	Đ̣	Đ̣	đ̣	Ħ̣	ħ̣	Ł̣	ł̣	Ø̣	ø̣	Ɵ̣	ɵ̣	ɨ̣	ɨ̣	ʉ̣	ɬ̣
base	Ḍ	Ḍ	Ḍ	ḍ	Ḥ	ḥ	Ḷ	ḷ	Ọ	ọ	Ọ	ọ	Ị	i̧	u̧	ḷ
shift	+0, 97%	+0, 97%	+0, 97%	+0, 97%	+0, 95%	+0, 97%	-4, 96%	+0, 96%	+0, 95%	-2, 93%	+0, 97%	+0, 95%	+0, 95%	+14, 93%	+0, 91%	+12, 92%
a	341, 850								351, 850			209, 645		139, 645	209, 645	209, 645
a?	333, 850	336, 850	333, 850	270, 885	362, 850	270, 885	256, 850	133, 885	352, 850	250, 668	352, 850	252, 645	148, 850	150, 834	271, 645	144, 885
a<	**306, 850**	**306, 850**	**306, 850**	**235, 885**	**361, 850**	**282, 885**	**147, 850**	**135, 885**	**335, 850**	**236, 645**	**335, 850**	**238, 645**	**150, 850**	**146, 645**	**255, 645**	147, 885
a!																225, 885
b									350, -110	243, -110	350, -110	196, -110		134, -114	196, -110	146, -114
b?	333, -110	336, -110	333, -110	270, -110	362, -110	270, -110	256, -110	133, -110	352, -121	250, -136	352, -110	252, -110	148, -110	150, -110	271, -110	144, -110
b<	**309, -110**	**309, -110**	**309, -110**	**235, -110**	**362, -110**	**272, -110**	**258, -110**	**135, -110**	**350, -110**	**241, -110**	**350, -110**	**243, -110**	**147, -110**	**150, -110**	**239, -110**	147, -110
b!																225, -110
it	*Đ̣*	*Đ̣*	*Đ̣*	*đ̣*	*Ħ̣*	*ħ̣*	*Ł̣*	*ł̣*	*Ø̣*	*ø̣*	*Ɵ̣*	*ɵ̣*	*ɨ̣*	*ɨ̣*	*ʉ̣*	*ɬ̣*
patch	*Đ̣*	*Đ̣*	*Đ̣*	*đ̣*	*Ħ̣*	*ħ̣*	*Ł̣*	*ł̣*	*Ø̣*	*ø̣*	*Ɵ̣*	*ɵ̣*	*ɨ̣*	*ɨ̣*	*ʉ̣*	*ɬ̣*
base	*Ḍ*	*Ḍ*	*Ḍ*	*ḍ*	*Ḥ*	*ḥ*	*Ḷ*	*ḷ*	*Ọ*	*ọ*	*Ọ*	*ọ*	*Ị*	*i̧*	*u̧*	*ḷ*
shift	+0, 97%	+0, 97%	*+36, 37%*	+0, 97%	+0, 95%	+0, 97%	*+6, 56%*	*+6, 42%*	-4, 96%	+0, 91%	*+2, 42%*	*+6, 38%*	+0, 94%	+14, 93%	*-12, 32%*	+12, 94%
a														267, 645		
a?	514, 850	514, 850	534, 850	434, 890	517, 850	449, 890	440, 850	322, 890	515, 850	370, 691	515, 850	381, 657	321, 850	316, 810	398, 645	333, 890
a<	**396, 850**	**396, 850**	432, 850		**481, 850**		395, 850	298, 890	**511, 850**	**359, 645**	517, 850	365, 645	**316, 850**	**281, 645**	374, 645	**304, 890**
a!																
b									310, -113	186, -107				134, -114		146, -114
b?	310, -110	310, -110	330, -110	221, -110	313, -110	236, -110	236, -110	110, -110	302, -151	190, -156	311, -110	218, -110	117, -110	121, -110	238, -110	120, -110
b<	**279, -110**	**279, -110**	315, -110	**221, -110**	**320, -110**	**229, -110**	258, -110	121, -110	**306, -110**	**178, -148**	312, -110	192, -110	**127, -110**	**150, -110**	214, -110	**127, -110**
b!																
sb	**Đ̣**	**Đ̣**	**Đ̣**	**đ̣**	**Ħ̣**	**ħ̣**	**Ł̣**	**ł̣**	**Ø̣**	**ø̣**	**Ɵ̣**	**ɵ̣**	**ɨ̣**	**ɨ̣**	**ʉ̣**	**ɬ̣**
patch	**Đ̣**	**Đ̣**	**Đ̣**	**đ̣**	**Ħ̣**	**ħ̣**	**Ł̣**	**ł̣**	**Ø̣**	**ø̣**	**Ɵ̣**	**ɵ̣**	**ɨ̣**	**ɨ̣**	**ʉ̣**	**ɬ̣**
base	**Ḍ**	**Ḍ**	**Ḍ**	**ḍ**	**Ḥ**	**ḥ**	**Ḷ**	**ḷ**	**Ọ**	**ọ**	**Ọ**	**ọ**	**Ị**	**i̧**	**u̧**	**ḷ**
shift	+0, 96%	+0, 96%	+0, 96%	+0, 97%	+0, 94%	+0, 97%	+0, 94%	+0, 94%	+0, 94%	+0, 91%	+0, 96%	+0, 96%	+0, 94%	+0, 77%	+0, 91%	+12, 91%
a														151, 645		
a?	349, 850	349, 850	349, 850	274, 885	362, 850	275, 885	258, 850	142, 885	366, 850	260, 684	366, 850	260, 645	158, 850	152, 818	274, 645	156, 885
a<	**352, 850**	**352, 850**	**352, 850**	**274, 885**	**362, 850**	**275, 885**	**267, 850**	**142, 885**	**363, 850**	**260, 645**	**363, 850**	**260, 645**	**160, 850**	**151, 645**	**264, 645**	**154, 885**
a!																
b												229, -114		134, -114		146, -114
b?	349, -110	349, -110	349, -110	274, -110	362, -110	275, -110	258, -110	142, -110	366, -153	260, -142	366, -110	260, -110	158, -110	152, -110	274, -110	156, -110
b<	**352, -110**	**352, -110**	**352, -110**	**274, -110**	**362, -110**	**277, -110**	**267, -110**	**135, -110**	**366, -110**	**260, -110**	**366, -110**	**260, -110**	**160, -110**	**136, -110**	**248, -110**	**147, -110**
b!																
si	***Đ̣***	***Đ̣***	***Đ̣***	***đ̣***	***Ħ̣***	***ħ̣***	***Ł̣***	***ł̣***	***Ø̣***	***ø̣***	***Ɵ̣***	***ɵ̣***	***ɨ̣***	***ɨ̣***	***ʉ̣***	***ɬ̣***
patch	***Đ̣***	***Đ̣***	***Đ̣***	***đ̣***	***Ħ̣***	***ħ̣***	***Ł̣***	***ł̣***	***Ø̣***	***ø̣***	***Ɵ̣***	***ɵ̣***	***ɨ̣***	***ɨ̣***	***ʉ̣***	***ɬ̣***
base	***Ḍ***	***Ḍ***	***Ḍ***	***ḍ***	***Ḥ***	***ḥ***	***Ḷ***	***ḷ***	***Ọ***	***ọ***	***Ọ***	***ọ***	***Ị***	***i̧***	***u̧***	***ḷ***
shift	+0, 96%	+0, 97%	*+50, 45%*	+0, 96%	+12, 94%	+0, 97%	+12, 94%	+16, 94%	-2, 94%	+0, 92%	+0, 96%	*+36, 50%*	-14, 73%	+2, 93%	*+16, 51%*	+28, 74%
a	341, 850								351, 850					276, 645		
a?	367, 850	367, 850	388, 850	309, 890	396, 850	331, 890	300, 850	190, 890	390, 850	239, 665	387, 850	229, 645	181, 850	180, 823	255, 645	187, 890
a<	**306, 850**	**306, 850**	356, 850		**373, 850**		**161, 850**	**190, 890**	**333, 850**	**238, 645**	**335, 850**	274, 645	**136, 850**	**132, 645**	271, 645	**202, 890**
a!																
b									350, -110	243, -110				136, -110		146, -114
b?	334, -110	334, -110	355, -110	268, -110	363, -110	290, -110	267, -110	148, -110	349, -148	237, -138	354, -110	238, -110	148, -110	153, -110	264, -110	145, -110
b<	**279, -110**	**279, -110**	329, -110	**221, -110**	**377, -110**	**272, -110**	**272, -110**	**151, -110**	**348, -110**	**243, -110**	**350, -110**	279, -110	**133, -110**	**138, -110**	255, -110	**163, -110**
b!																

Table 18: Derived letters and inherited anchors 2/3

hex	026C	025F	0269	0181	018A	0187	0193	0198	0253	0257	0266	029B	025D	025A	0284	0191
base	006C	0237	0131	0042	0044	0043	0047	004B	0062	0064	0068	0262	025C	0259	0237	0046
reg																
patch																
base																
shift	+88, 94%	*+12, 65%*	-20, 79%	+58, 94%	+82, 95%	+0, 100%	+0, 100%	-4, 86%	+6, 90%	-2, 85%	*-4, 45%*	+0, 100%	+0, 100%	*-12, 61%*	-20, 76%	-8, 84%
a	209, 645	169, 645	123, 645	327, 805	306, 805				480, 630	218, 645	299, 565	209, 645	209, 645	229, 645	209, 645	
a?	182, 885	124, 645	170, 645	303, 850	368, 850	369, 850	372, 850	339, 850	265, 885	322, 885	262, 885	294, 764	234, 795	306, 688	140, 885	183, 850
a<	**223, 885**	152, 645	**112, 645**	**325, 850**	**388, 850**	**340, 850**	**365, 850**	**332, 850**	285, 885	233, 885	278, 885	**265, 663**				**276, 850**
a!									255, 885	218, 885	263, 885					
b	218, -120	196, -110	153, -114	328, -108	369, -107				212, -114	212, -114	240, -114	196, -110	212, -114	212, -114	196, -110	
b?	182, -110	124, -291	170, -110	303, -110	368, -110	369, -110	372, -110	339, -110	265, -110	322, -110	262, -110	294, -110	234, -110	306, -110	140, -299	183, -257
b<	**223, -110**	121, -319	**119, -119**	**326, -110**	**391, -110**	**363, -110**	**367, -110**	**338, -110**	**240, -110**	**233, -110**	268, -110	**248, -110**	**184, -110**	200, -110	**89, -319**	
b!											262, -110					
it																
patch																
base																
shift	+86, 94%	*+42, 60%*	-18, 92%	+58, 99%	+80, 98%	*+18, 34%*	*+0, 54%*	*+0, 48%*	*-6, 42%*	+6, 78%	+6, 76%	-10, 81%	+0, 99%	*+10, 40%*	*-14, 67%*	*+12, 57%*
a			123, 645	327, 805	306, 805					218, 645	299, 565					
a?	368, 890	305, 645	266, 645	488, 850	553, 850	503, 850	541, 850	499, 850	431, 896	464, 890	449, 890	410, 796	413, 775	418, 687	343, 900	419, 850
a<	**378, 890**	182, 645	**249, 645**	**505, 850**	**476, 850**	518, 850	513, 850	466, 850	423, 896			**378, 645**				418, 850
a!																
b	218, -120		153, -114	328, -108	369, -107				212, -114	212, -114	240, -114		212, -114	212, -114		
b?	156, -110	100, -319	106, -110	284, -110	349, -110	299, -110	337, -110	295, -110	218, -110	251, -110	236, -110	217, -110	225, -110	248, -110	88, -299	184, -257
b<	**201, -110**	104, -320	**121, -110**	**296, -110**	**359, -110**	351, -110	345, -110	292, -110	208, -110	**227, -110**	**235, -110**	**238, -110**	**184, -110**	179, -110	52, -299	
b!																
sb																
patch																
base																
shift	+88, 94%	*+18, 60%*	-16, 83%	+58, 90%	+86, 96%	+0, 99%	+0, 100%	+2, 75%	*+0, 53%*	*-6, 53%*	-2, 71%	+0, 100%	+0, 98%	*+6, 58%*	*-20, 58%*	+16, 70%
a																
a?	190, 885	144, 645	164, 645	316, 850	381, 850	368, 850	374, 850	338, 850	268, 885	324, 885	274, 885	290, 764	276, 757	319, 693	152, 885	236, 850
a<	**230, 885**	181, 645	**135, 645**	**358, 850**	**438, 850**	**332, 850**	**385, 850**	**342, 850**	236, 885	268, 885	**273, 885**	**266, 666**				**284, 850**
a!																
b	218, -120															
b?	190, -110	144, -319	164, -110	316, -110	381, -110	368, -110	374, -110	338, -110	268, -110	324, -110	274, -110	290, -110	276, -110	319, -110	152, -296	236, -281
b<	**223, -110**	96, -319	**119, -120**	**358, -110**	**438, -110**	**332, -110**	**368, -110**	**342, -110**	236, -121	268, -110	**275, -110**	**266, -110**	**216, -110**	218, -110	58, -319	
b!																
si																
patch																
base																
shift	+106, 75%	*-4, 48%*	-12, 77%	*+70, 41%*	*+98, 26%*	*+20, 68%*	*+20, 46%*	*-30, 62%*	-4, 84%	*-2, 50%*	*-4, 55%*	+12, 100%	+0, 98%	*+12, 29%*	*-38, 56%*	*+22, 30%*
a			123, 645	327, 805	306, 805					218, 645	299, 565					
a?	223, 890	141, 645	138, 645	359, 850	424, 850	359, 850	395, 850	353, 850	284, 890	318, 890	302, 890	278, 796	261, 762	299, 693	195, 890	288, 850
a<	**280, 890**	136, 645	**122, 654**	337, 850	404, 850	360, 850	385, 850	306, 850	**286, 890**			**284, 681**				298, 850
a!																
b	218, -120		153, -114	328, -108	369, -107				212, -114	212, -114	240, -114		212, -114	212, -114		
b?	181, -110	107, -319	147, -110	326, -110	391, -110	326, -110	362, -110	320, -110	243, -110	276, -110	260, -110	256, -110	246, -110	298, -110	116, -296	220, -281
b<	**241, -110**	85, -314	**127, -110**	338, -110	377, -110	383, -110	387, -110	312, -110	**230, -110**	219, -110	268, -110	**260, -110**	**184, -110**	198, -110	55, -292	
b!																

Table 19: Derived letters and inherited anchors 3/3

hex	019D	0255	0256	026D	0271	0272	0273	027B	027D	0282	0286	0288	0290	0291	0293
base	004E	0063	0064	006C	006D	006E	006E	0279	0072	0073	0283	0074	007A	007A	0292
reg	N̪N̪	c̪e̪	d̪d̪	l̪l̪	m̪m̪	n̪n̪	n̪n̪	r̪ɹ̪	r̪r̪	s̪s̪	ʃ̪ʃ̪	t̪t̪	z̪z̪	z̪z̪	ʒ̪ʒ̪
patch	N̪N̪	c̪e̪	d̪d̪	l̪l̪	m̪m̪	n̪n̪	n̪n̪	r̪ɹ̪	r̪r̪	s̪s̪	ʃ̪ʃ̪	t̪t̪	z̪z̪	z̪z̪	ʒ̪ʒ̪
base	N̪N̪	c̪c̪	d̪d̪	l̪l̪	m̪m̪	n̪n̪	n̪n̪	r̪ɹ̪	r̪r̪	s̪s̪	ʃ̪ʃ̪	t̪t̪	z̪z̪	z̪z̪	ʒ̪ʒ̪
shift	-2, 90%	-2, 90%	-2, 93%	-2, 89%	*-8, 46%*	*-14, 59%*	*-10, 57%*	*+24, 56%*	*-14, 54%*	-6, 98%	+4, 87%	-6, 71%	+0, 100%	*-12, 61%*	*+2, 34%*
a		209, 645	218, 645	209, 645	383, 645	264, 646	263, 645		209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645
a?	324, 850	222, 645	314, 885	182, 885	366, 645	234, 645	296, 645	226, 645	186, 645	199, 645	150, 885	158, 771	262, 645	218, 662	220, 645
a<	**344, 850**	**240, 645**	**233, 885**	**133, 885**	384, 645	250, 645	254, 645	233, 645	177, 645	**200, 645**	**144, 885**	**133, 769**	**220, 645**	208, 645	220, 645
a!								238, 645							
b		212, -114	212, -114	196, -110	383, -105	265, -111	262, -112	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110
b?	324, -272	222, -214	314, -273	182, -273	366, -306	234, -306	296, -306	226, -259	186, -319	199, -261	150, -310	158, -306	262, -270	218, -207	220, -319
b<															
b!				196, -275			262, -308			196, -263					

it	*N̪N̪*	*c̪e̪*	*d̪d̪*	*l̪l̪*	*m̪m̪*	*n̪n̪*	*n̪n̪*	*r̪ɹ̪*	*r̪r̪*	*s̪s̪*	*ʃ̪ʃ̪*	*t̪t̪*	*z̪z̪*	*z̪z̪*	*ʒ̪ʒ̪*
patch	*N̪N̪*	*c̪e̪*	*d̪d̪*	*l̪l̪*	*m̪m̪*	*n̪n̪*	*n̪n̪*	*r̪ɹ̪*	*r̪r̪*	*s̪s̪*	*ʃ̪ʃ̪*	*t̪t̪*	*z̪z̪*	*z̪z̪*	*ʒ̪ʒ̪*
base	*N̪N̪*	*c̪c̪*	*d̪d̪*	*l̪l̪*	*m̪m̪*	*n̪n̪*	*n̪n̪*	*r̪ɹ̪*	*r̪r̪*	*s̪s̪*	*ʃ̪ʃ̪*	*t̪t̪*	*z̪z̪*	*z̪z̪*	*ʒ̪ʒ̪*
shift	*+16, 46%*	*+0, 69%*	*+24, 41%*	-6, 83%	*-16, 42%*	*+0, 38%*	-6, 87%	*+110, 51%*	*+0, 60%*	*+2, 25%*	-2, 84%	*+4, 68%*	*-14, 37%*	*+6, 37%*	*+16, 35%*
a			218, 645		383, 645	264, 646	263, 645								
a?	528, 850	352, 645	465, 899	341, 890	519, 664	400, 662	367, 645	352, 645	323, 660	317, 658	343, 900	314, 776	346, 662	355, 662	355, 645
a<	492, 850	343, 645	244, 655	**286, 890**	508, 654	396, 654	**388, 645**	425, 645	313, 652	329, 645	**341, 900**	327, 798	346, 645	366, 645	344, 645
a!															
b		212, -114	212, -114		383, -105	265, -111	262, -112								
b?	290, -272	171, -208	215, -273	91, -287	310, -319	191, -319	162, -319	160, -259	124, -279	121, -265	92, -283	83, -311	147, -276	171, -207	133, -397
b<															
b!															

sb	**N̪N̪**	**c̪e̪**	**d̪d̪**	**l̪l̪**	**m̪m̪**	**n̪n̪**	**n̪n̪**	**r̪ɹ̪**	**r̪r̪**	**s̪s̪**	**ʃ̪ʃ̪**	**t̪t̪**	**z̪z̪**	**z̪z̪**	**ʒ̪ʒ̪**
patch	**N̪N̪**	**c̪e̪**	**d̪d̪**	**l̪l̪**	**m̪m̪**	**n̪n̪**	**n̪n̪**	**r̪ɹ̪**	**r̪r̪**	**s̪s̪**	**ʃ̪ʃ̪**	**t̪t̪**	**z̪z̪**	**z̪z̪**	**ʒ̪ʒ̪**
base	**N̪N̪**	**c̪c̪**	**d̪d̪**	**l̪l̪**	**m̪m̪**	**n̪n̪**	**n̪n̪**	**r̪ɹ̪**	**r̪r̪**	**s̪s̪**	**ʃ̪ʃ̪**	**t̪t̪**	**z̪z̪**	**z̪z̪**	**ʒ̪ʒ̪**
shift	*+4, 42%*	*-2, 49%*	-2, 75%	*-20, 46%*	*-10, 56%*	-12, 78%	*-14, 68%*	+54, 71%	*-16, 66%*	+0, 100%	+6, 71%	-6, 73%	+0, 100%	*+0, 63%*	*+6, 32%*
a															
a?	336, 850	232, 645	322, 885	194, 885	363, 645	246, 645	306, 645	238, 645	198, 645	201, 645	162, 885	182, 782	254, 645	230, 666	236, 658
a<	358, 850	222, 645	**272, 891**	122, 885	386, 645	**252, 645**	250, 645	**198, 641**	196, 645	**201, 645**	**158, 885**	**162, 782**	**222, 645**	222, 653	240, 645
a!															
b															
b?	336, -270	232, -209	322, -271	194, -319	363, -319	246, -319	306, -319	238, -257	198, -319	201, -264	162, -319	182, -319	254, -276	230, -210	236, -413
b<															
b!															

si	***N̪N̪***	***c̪e̪***	***d̪d̪***	***l̪l̪***	***m̪m̪***	***n̪n̪***	***n̪n̪***	***r̪ɹ̪***	***r̪r̪***	***s̪s̪***	***ʃ̪ʃ̪***	***t̪t̪***	***z̪z̪***	***z̪z̪***	***ʒ̪ʒ̪***
patch	***N̪N̪***	***c̪e̪***	***d̪d̪***	***l̪l̪***	***m̪m̪***	***n̪n̪***	***n̪n̪***	***r̪ɹ̪***	***r̪r̪***	***s̪s̪***	***ʃ̪ʃ̪***	***t̪t̪***	***z̪z̪***	***z̪z̪***	***ʒ̪ʒ̪***
base	***N̪N̪***	***c̪c̪***	***d̪d̪***	***l̪l̪***	***m̪m̪***	***n̪n̪***	***n̪n̪***	***r̪ɹ̪***	***r̪r̪***	***s̪s̪***	***ʃ̪ʃ̪***	***t̪t̪***	***z̪z̪***	***z̪z̪***	***ʒ̪ʒ̪***
shift	*-24, 65%*	*+2, 50%*	*+0, 52%*	+24, 82%	*-28, 25%*	*-4, 53%*	*-10, 47%*	*+40, 68%*	*-12, 45%*	*+6, 22%*	*+6, 27%*	*-8, 41%*	*-2, 26%*	*-20, 25%*	*+4, 39%*
a			218, 645		383, 645	264, 646	263, 645								
a?	383, 850	204, 645	302, 890	187, 890	373, 645	254, 645	254, 645	194, 645	177, 645	178, 645	189, 890	172, 787	206, 666	212, 666	213, 658
a<	322, 850	244, 645	218, 645	**198, 890**	365, 651	261, 652	254, 645	211, 640	181, 654	196, 645	201, 890	185, 787	218, 645	200, 645	222, 645
a!															
b		212, -114	212, -114		383, -105	265, -111	262, -112								
b?	317, -270	192, -209	227, -271	103, -319	339, -319	220, -319	220, -319	172, -257	143, -319	155, -265	105, -319	109, -319	176, -279	196, -210	157, -413
b<															
b!															

Anchor survey, with patches

I will systematically survey above- and below anchors in the four font font styles, commenting on any features and patching any problems. Refer to Tables 1–4 for anchor values. Missing anchors are readily apparaent as blank entries. Large anchor-midpoint offset values ($ax\delta$, $bx\delta$) indicate possible problems, but these can often be explained by glyph symmetry.

A–Z, regular, above- and below-anchor

The above-anchors of A–Z regular are good.

The above-anchor of D is just right of the major stem, not right enough to be centered over the bowl. For D, $ax = 306$. Consider the precomposed characters Ď (U+010E) and Ḋ (U+1E0A). The designer drew their glyphs with the mark centered at $X = 312$ and 296 respectively. So, this is a deliberate design decision to keep the mark closer to the stem, not centered over the bowl.

Similarly, the above anchor of F is just right of the major stem, not right enough to be centered over cross strokes. For F, $ax = 284$. For Ḟ (U+1E1E), the dot is centered at $X = 254$. So, the designer is deliberately choosing to keep the mark cloder to the stem.

The above-anchor of L is centered over the major stem, which is a valid design choice.

The below-anchors of A–Z regular are good. But J's below anchor is too low, and Q is missing a below-anchor. Even though J and Q arguably do not need below-anchors for linguistic notation, I will set this anchor by the heuristic algorithm.

Patch J regular, below-anchor

J's below-anchor is too low. For J, $by = -319$, but J is not a full descender; its bbox $Y_{\text{min}} = -172$, which is well above $yrd = -232$. Yes, the designer set $by = byrd = -319$ for J in all fout font styles, but this is arguably too low. According to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - by\delta d + \epsilon = -172 - 87 + 5 = -254$, where $\epsilon = 5$ accounts for the overshoot of J's tail.

Patch Q regular, below-anchor

Q has bbox $Y_{\text{min}} = -209$. Again, according to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - by\delta d + \epsilon = -209 - 87 + 5 = -291$, where $\epsilon = 5$ accounts for the overshoot of Q's tail. For bx, observe that O and Q have the same counter outline, literally. So, the below-anchor for Q should be at $bx = 350$, the same as that for O.

a–z, regular, above- and below-anchor

b, f, j, and t are missing above-anchors. Some letters (n, p, r, v, w, x, z) should have ay regularized to 645, not 646. For d, h, and k, the above-anchor is centered above the body, not the ascender; this is good design choice, but care must be taken to avoid collision between an above-mark and an ascender. For d and h, ay is at the x-height anchor reference, and therefore wide marks collide with the ascender. For k, ay is a bit higher, but not at the full ascender anchor references, and therefore the widest marks collide with the ascender.

The below-anchor of k is too low. Otherwise, the below-anchors of a–z are good.

Patch k regular, below-anchor

For k, $by - 214$. But it should be at $byr0 = -110$, like all other letters that sit on the baseline.

Here is this positioning of k in a patched version of the font (with o for comparison only).

Original: k̲ k̲o ok̲

Patched: k̲ k̲o ok̲

Patch b, d, f, h regular, above-anchor

For b, d, f, and h, set $ay = ayra = 885$.

For ax for b, d, f, and h, there are at least two strategies. First, set ax at the bowl apex. So, for f, set $ax = 279$ at the apex of the bowl. But for b, d, and h, you could set ax centered horizontally over the bowl, in which case you also need to decide which two vertical aspects of the bowl should be used for the centering.

Patch j regular, above-anchor

j needs an above anchor for the rare case of an above-mark that is placed over the dot. i does have an above-anchor at $(137, 795)$, and the outlines of the top of

Patch n, p, r, v, w, x, z regular, above-anchor

In my patched version of the font, I regularize ay to $ayrx = 645$ for all letters that sit at the x-height, including n, p, r, v, w, x, z.

Above-anchor, a–z regular

b, f, j, and t are missing the above-anchor. d and h have an above-anchor above the meanline but not above the ascender. l has an above-anchor above the ascender. k has an above-anchor, but it is just below the ascender, so any marks other than the do crash into the ascender. These are all legitimate design choices. There are no above-marks, other than those with precomposed characters, the would ever be combined with b, f, or t.

Patch j dot removal for three above-marks

The dot of i and j should be removed before adding any above-mark. But the dot is not removed for three above-marks: x, turned tilde, double macron. So the GSUB table should be patched to fir the dotless j substitution in this context.

To do.

Capital mark alternates

Libertinus has custom glyphs for some marks (acute, grave, cicrumflex, breve, a.o.) that are designed better for capital letters. These alternates are flatter. GSUB lookups 23 and 24 implement the substitution. This substitution is not fired on some capital consonants (T, V, W, X) because these marks are not expected on these bases.

Libertinus has custom glyphs for some marks that are designed better for superscripts. These alternates are smaller. Regrettably?, no GSUB table implements their substitution.

A cirmumflex Â Ầ $\hat{A}$ $\mathbf{\hat{A}}$ $\boldsymbol{\hat{A}}$

O circumflex Ô Ồ $\hat{O}$ $\mathbf{\hat{O}}$ $\boldsymbol{\hat{O}}$

AE ligature Æ Æ̀ $Æ$ $\mathbf{Æ}$ $\boldsymbol{Æ}$

S̊ T̊ V̊ $\mathring{S}$ $\mathring{T}$ $\mathring{V}$ $\mathbf{\mathring{S}}$ $\mathbf{\mathring{T}}$ $\mathbf{\mathring{V}}$ $\boldsymbol{\mathring{S}}$ $\boldsymbol{\mathring{T}}$ $\boldsymbol{\mathring{V}}$

S̊ T̊ V̊ $\mathring{S}$ $\mathring{T}$ $\mathring{V}$ $\mathbf{\mathring{S}}$ $\mathbf{\mathring{T}}$ $\mathbf{\mathring{V}}$ $\boldsymbol{\mathring{S}}$ $\boldsymbol{\mathring{T}}$ $\boldsymbol{\mathring{V}}$

Regular, above-anchor, capitals, X coordinate

Figure 9 (left panel) shows pairs of bases aligned at their above-anchor. If a capital letter has a bad X coordinate for its above-anchor, then it will look always off, right or left, in its black column relative to the gray letters that it overlays. Note that the above-anchor X coordinate is not in the geometric (horizontal) center of the glyph, by design. The above-anchor X coordinate is always off center by a few UPM units, sometimes left and sometimes right. But it very hard to discern a few UPM units. See the right panel of Figure 9 for an illustration of increasingly bad (spoofed) X coordinates of N's above-anchor.

Note that the above-anchor of L is above the major (vertical) stem of the L. For E, F, P and R, the above-anchor is optically centered above the glyph, not above the major stem.

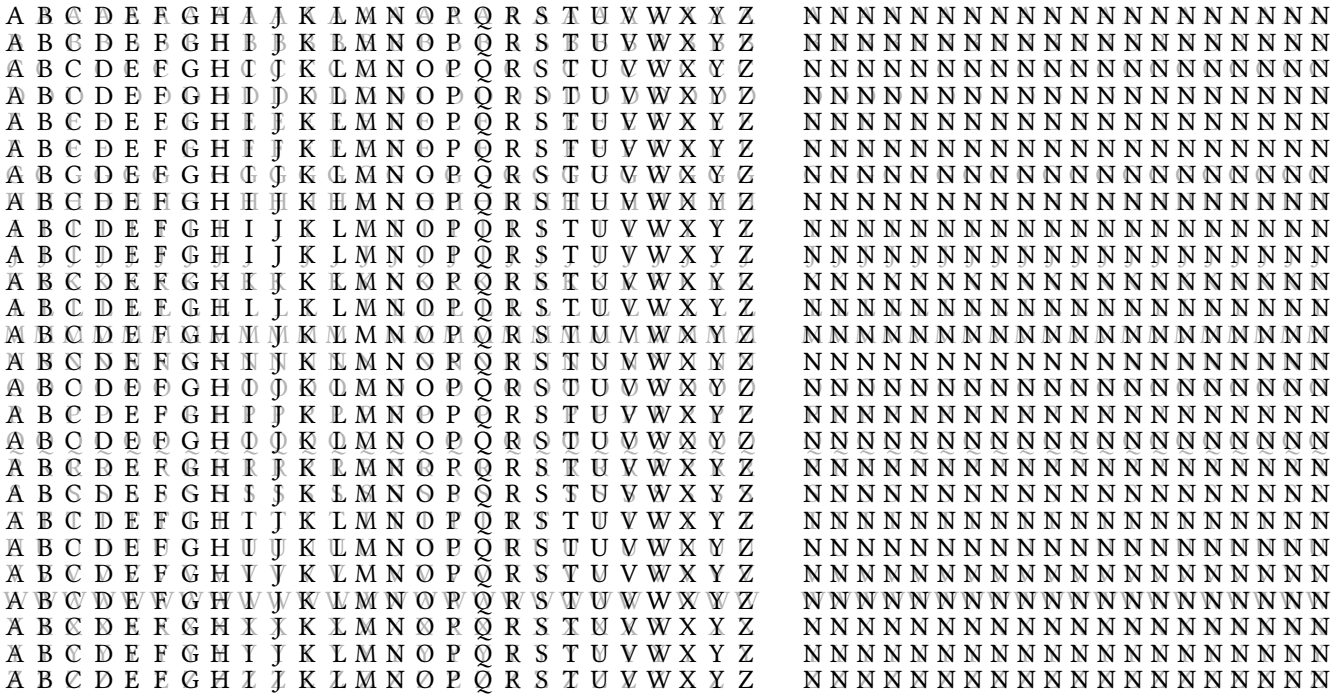


Figure 9: Above-anchor X coordinate alignment. In the left panel, the 26×26 grid shows letters A–Z aligned at the their above-anchor. The right panel shows how hard it is to discern bad X coordinates. The right panel shows N aligned with letters at their above-anchor, starting at the first column with Ns above-anchor as currently set by the font, and then spoofing increasing bad values for the X coordinate of N's above-anchor, by 5 UPM per column. At the rightmost column, it becomes evident that the (spoofed) X coordinate of N's above-anchor is bad, because in that column the black Ns are too far right of the gray letters below them.

Comma-style marks

Libertinus has a some (precomposed) glyphs for Unicode characters (with code points) where the caron or cedilla has a comma-style shape, instead of a wedge or hook as might be naively expected. But all of these are appropriate for Czech/Slovak orthography (caron or *háček*) or Latvian orthography (cedilla, but really the comma-above mark or *komatiņš*, by a known mistake in Unicode).

Czech/Slovak *háček*

In Czech/Slovak orthography, the d, l, L, n and t may take a reduced caron that may look like a comma, not a wedge. In Libertinus, only the d, L, l, and t have the reduced comma-style caron in their precomposed form, and the n has wedge caron precomposed form.

`\char"010F` ď *ď* **ď** ***ď***

`\char"013D` Ľ *Ľ* **Ľ** ***Ľ***

`\char"013E` ľ *ľ* **ľ** ***ľ***

`\char"0148` ň *ň* **ň** ***ň***

`\char"0165` ť *ť* **ť** ***ť***

Libertinus even has a custom glyph for a small capital L with a caron, and it is comma-style.

`\textsc{l\char"030C}` ʟ̬

`\textsc{\char"013E}` ʟ̬

Cedilla and Latvian *komatiņš*

Here are all precomposed characters with a cedilla.

`\char"00C7` Ç *Ç* **Ç** ***Ç***	`\char"00E7` ç *ç* **ç** ***ç***
`\char"1E08` Ḉ *Ḉ* **Ḉ** ***Ḉ***	`\char"1E08` ḉ *ḉ* **ḉ** ***ḉ***
`\char"1E10` Ḑ *Ḑ* **Ḑ** ***Ḑ***	`\char"1E11` ḑ *ḑ* **ḑ** ***ḑ***
`\char"0228` Ȩ *Ȩ* **Ȩ** ***Ȩ***	`\char"0229` ȩ *ȩ* **ȩ** ***ȩ***
`\char"1E1C` Ḝ *Ḝ* **Ḝ** ***Ḝ***	`\char"1E1D` ḝ *ḝ* **ḝ** ***ḝ***
`\char"0122` Ģ *Ģ* **Ģ** ***Ģ***	`\char"0123` ģ *ģ* **ģ** ***ģ***
`\char"1E28` Ḩ *Ḩ* **Ḩ** ***Ḩ***	`\char"1E29` ḩ *ḩ* **ḩ** ***ḩ***
`\char"0136` Ķ *Ķ* **Ķ** ***Ķ***	`\char"0137` ķ *ķ* **ķ** ***ķ***
`\char"013B` Ļ *Ļ* **Ļ** ***Ļ***	`\char"013C` ļ *ļ* **ļ** ***ļ***
`\char"0145` Ņ *Ņ* **Ņ** ***Ņ***	`\char"0146` ņ *ņ* **ņ** ***ņ***
`\char"0156` Ŗ *Ŗ* **Ŗ** ***Ŗ***	`\char"0157` ŗ *ŗ* **ŗ** ***ŗ***
`\char"015E` Ş *Ş* **Ş** ***Ş***	`\char"015F` ş *ş* **ş** ***ş***
`\char"0162` Ţ *Ţ* **Ţ** ***Ţ***	`\char"0163` ţ *ţ* **ţ** ***ţ***

These characters – Ģ ģ Ķ ķ Ļ ļ Ņ ņ Ŗ ŗ – use the Latvian *komatiņš* which is proper for Latvian orthography, but the result of a historical mistake in Unicode naming.

The Unicode characters of D and d with cedilla exist for scholarly and historical reasons, not Latvian orthography. Fonts design these Unicode characters differently. Libertinus designs them inconsistently: comma-style in regular and italic, and hook-style in (semi)bold and (semi)bold italic. I think the cedilla on a D or d should be hook-like, but I do not really know.

The other characters – c, e, h, s, t – with a cedilla should definitely be hook-like, and they are so in Libertinus.

Somewhat mysteriously, if the LaTeX source tries to combine any of these letters – c, d, e, g, h, k, l, n, r, s, t – with the combining cedilla, U+0327, then the precomposed character is produced.

c\char"0327 ç *ç* **ç** ***ç***
d\char"0327 ḑ *ḑ* **ḑ** ***ḑ***
e\char"0327 ȩ *ȩ* **ȩ** ***ȩ***
g\char"0327 ģ *ģ* **ģ** ***ģ***
h\char"0327 ḩ *ḩ* **ḩ** ***ḩ***
k\char"0327 ķ *ķ* **ķ** ***ķ***
l\char"0327 ļ *ļ* **ļ** ***ļ***
n\char"0327 ņ *ņ* **ņ** ***ņ***
r\char"0327 ŗ *ŗ* **ŗ** ***ŗ***
s\char"0327 ş *ş* **ş** ***ş***
t\char"0327 ţ *ţ* **ţ** ***ţ***
This behavior does not make sense for the Latvian *komatiņš* bases – g, k, l, n, r – because any person trying to produce Latvian g with *komatiņš* would not try `g\char"0327` or `\c{g}`.
This behavior is tolerable for the other bases – c, d, e, h, s, t.

IJ

The ij and IJ ligatures are Unicode characters (with code points) and are supported by Libertinus.
IJ `\char"0132` U+0132 capital ligature IJ
ij `\char"0133` U+0133 small ligature ij
Libertinus has custom glyphs for these ligatures with an acute, and the small capital ligature with an acute. The GSUB table that activates these custom glyphs only works when the language is set to Dutch by a font loading command. For example,
```
\newfontfamily\SerifRegularDutch[Language = Dutch]{LibertinusSerif-Regular.otf}
```
ĲĲ́ `{\SerifRegularDutch \char"0132\char"0301}` Dutch capital ligature IJ with acute
ij́ `{\SerifRegularDutch \char"0133\char"0301}` Dutch small ligature ij with acute
ij́ `{\SerifRegularDutch \textsc{\char"0133\char"0301}}` Dutch small capital ligature IJ with acute
In an English LaTeX document, `ij` and `IJ` are kerned by the GPOS table to produce ij and IJ, so they look close together, but they are not substituted by the single glyph for the ligature.

Lowercase i with tilde below and grave above, in italic and semibold italic

First, start with i with tilde below (precomposed).
ḭ *ḭ* **ḭ** ***ḭ*** `\char"1E2D` i with tilde below (precomposed)
You cannot simply put a grave on it because it collides with the dot.
ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"1E2D\char"0300` i with tilde below (precomposed) with grave
Start over with i with grave (precomposed).
ì *ì* **ì** ***ì*** `\char"00EC` i with grave (precomposed)
Try putting a grave below.
ì̗ *ì̗* **ì̗** ***ì̗*** `\char"00EC\char"0330` i with grave (precomposed) with tilde below

This fails, because normalization breaks down the input “i with grave (precomposed) with tilde below” into “i with grave with tilde below”, then into canonical order “i with tilde below with grave”, the recomposes it into “i with tilde below (precomposed) with grave”, which reintroduces i’s dot.

Next, try to block normalization with the combining grapheme joiner (CGJ).

ḭ̀ *ḭ̀* ḭ̀ *ḭ̀* `\char"00EC\cgj\char"0330` i with grave (precomposed) with combining grapheme joiner with tilde below

This looks bad because i with grave (precomposed) does not have a below-anchor and fallback shaping looks bad.

Finally, try starting with dotless i, which has OK above- and below-anchors.

ı̰̀ *ı̰̀* ı̰̀ *ı̰̀* `\char"0131\char"0330\char"0300` dotless i with tilde below with grave

Normalization does not affect the dotless i, so it looks the same with the CGJ.

ı̰̀ *ı̰̀* ı̰̀ *ı̰̀* `\char"0131\cgj\char"0330\char"0300` dotless i with CGJ with tilde below with grave

All together now, for comparison.

ḭìḭ *ḭì* ḭì ḭ *ḭì* `\char"1E2D \char"00EC \char"0131\char"0330\char"0300` i with tilde below (precomposed); i with grave (precomposed); dotless i with tilde below with grave

The same strategy works with other above marks.

ı̰̏ *ı̰̏* ı̰̏ *ı̰̏* `\char"0131\char"0330\char"030F` dotless i with tilde below with double grave

ı̰́ *ı̰́* ı̰́ *ı̰́* `\char"0131\char"0330\char"0301` dotless i with tilde below with acute

ı̰̋ *ı̰̋* ı̰̋ *ı̰̋* `\char"0131\char"0330\char"030B` dotless i with tilde below with double acute

Lowercase a with double grave, followed by floating double grave, in italic

In italic, we want to produce a (precomposed) vowel with a double grave, followed by floating double grave. Ideally, the two double graves should be aligned. We cannot use the combining double grave as the floater, because it has nothing to combine to, and we cannot control its height. So we use a spacing character, specifically the double prime character (‶ `\char"2036`), which looks like a double grave, in italic.

ä‶ `\textit{ä\char"2036}}`

Then we lower it by a few points.

ä‶ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

The distance to move the glyph can be calculated from its SFD splineset, or by trial and error.

Note that only a, e, and i have precomposed glyphs with double acute.

ä‶ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

ë‶ `\textit{ë\raisebox{-0.05em}{\char"2036}}`

ë‶ `\textit{ë\raisebox{-0.045em}{\char"2036}}`

ɛ‶ `\textit{\char"025B\raisebox{-0.05em}{\char"2036}}`

Tilde below, double acute above, double grave above, on vowels, in italic and semibold italic

Consider the tilde below. There is the combining tilde below, U+0330, encoded in LaTeX as the codepoint `\char"0330`. Remember to put the base before the combining mark in LaTeX, e.g. `a\char"0330` a̰. Three vowels (e, i, u) have precomposed characters – lowercase and capital – encoded in LaTeX as these codepoints:

`\char"1E1A \char"1E1B \char"1E2C \char"1E2D \char"1E74 \char"1E75` Ḛ ḛ Ḭ ḭ Ṵ ṵ *Ḛ ḛ Ḭ ḭ Ṵ ṵ* **Ḛ ḛ Ḭ ḭ Ṵ ṵ** ***Ḛ ḛ Ḭ ḭ Ṵ ṵ***

E, e, I, i, U, or u followed by `\char"0330` will produce the precomposed character glyph. A, a, O, and o, as well as any other IPA vowel with a below-anchor, followed by `\char"0330` will be shaped by attaching the combining below tilde to that vowel at their below-anchor. Note the specific case of i. `i\char"0330 = \char"1E2D` ḭ = ị is a dotted i (a precomposed glyph).

Suppose you want a vowel with a tilde below, possibly also with a mark above, especially acute, grave, caron, double acute, or double grave. In the specific case of i, the above mark will crash into the dotted i unless you take care to encode the triplet starting with a dotless i, U+0131. For example, consider the combining acute U+0301.

`i\char"0330\char"0301 = \char"1E2D\char"0301` ḭ ḭ́ = ḭ ḭ́

`\char"0131\char"0330\char"0301` ı̰ ı̰́

Recall that the canonical order for a triplet of a base and two combining marks (one below and one above) is base + combining below mark + combining above mark, not base + above + below.

Consider the wrong order:

`i\char"0301\char"0330` ḭ́ i\char"0301 í i\char"0330 í̱

`\char"00EC\char"0330` ḭ̀ ì

`\char"1E2D\char"0301` ḭ́ ḭ

But note that the position of the combining tilde below is bad in italic and semibold italic. This is because the below-anchor for the combining tilde below mark is bad. For tilde below, the precomposed character glyphs with i, e, and u are sometimes good and sometimes bad in italic and semibold italic.

Precomposed characters are blue and therefore cannot be changed unless their glyph outline is changed. Black combinations of vowels and marks are shaped by anchors and therefore can be changed by setting good anchors.

Use the combining grapheme joiner (CGJ) character U+034F, `\char"034F`, or `\cgj` if loaded by some package, to force the shaper to use combining marks instead of precomposed glyphs.

A E I O U Ɛ Ɔ a e i o u ɛ ɔ *A E I O U Ɛ Ɔ a e i o u ɛ ɔ* **A E I O U Ɛ Ɔ a e i o u ɛ ɔ** ***A E I O U Ɛ Ɔ a e i o u ɛ ɔ***

1 Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ́ *Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ́* **Á É Í Ó Ú Ɛ Ɔ́ á é í ó ú ɛ ɔ́** ***Á É Í Ó Ú Ɛ Ɔ́ á é í ó ú ɛ ɔ́***

2 À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ̀ *À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ̀* **À È Ì Ò Ù Ɛ Ɔ̀ à è ì ò ù ɛ ɔ̀** ***À È Ì Ò Ù Ɛ Ɔ̀ à è ì ò ù ɛ ɔ̀***

3 Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ̆* **Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ ɔ̆** ***Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ ɔ̆***

4 Ő Ű Ɛ Ɔ ő ű ɛ ɔ̋ *Ő Ű Ɛ Ɔ ő ű ɛ ɔ̋* **Ő Ű Ɛ Ɔ̋ ő ű ɛ ɔ̋** ***Ő Ű Ɛ Ɔ̋ ő ű ɛ ɔ̋***

5 Ä Ë Ï Ö Ü Ɛ Ɔ ä ë ï ö ü ɛ ɔ̈ *Ä Ë Ï Ö Ü Ɛ Ɔ ä ë ï ö ü ɛ ɔ̈* **Ä Ë Ï Ö Ü Ɛ Ɔ̈ ä ë ï ö ü ɛ ɔ̈** ***Ä Ë Ï Ö Ü Ɛ Ɔ̈ ä ë ï ö ü ɛ ɔ̈***

6 Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ̨ *Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ̨* **Ą Ę Į Ǫ Ų Ɛ Ɔ̨ ą ę į ǫ ų ɛ ɔ̨** ***Ą Ę Į Ǫ Ų Ɛ Ɔ̨ ą ę į ǫ ų ɛ ɔ̨***

7 Ǻ Ḛ Ḭ Ṍ Ṵ Ɛ Ɔ ǻ ḛ ḭ ṍ ṵ ɛ ɔ̨́ *Ǻ Ḛ Ḭ Ṍ Ṵ Ɛ Ɔ ǻ ḛ ḭ ṍ ṵ ɛ ɔ̨́* **Ǻ Ḛ Ḭ Ṍ Ṵ Ɛ Ɔ̨́ ǻ ḛ ḭ ṍ ṵ ɛ ɔ̨́** ***Ǻ Ḛ Ḭ Ṍ Ṵ Ɛ Ɔ̨́ ǻ ḛ ḭ ṍ ṵ ɛ ɔ̨́***

7c Ǻ Ḛ Ḭ Ṍ Ṵ Ɛ Ɔ ǻ ḛ ḭ ṍ ṵ ɛ ɔ̨́ *Ǻ Ḛ Ḭ Ṍ Ṵ Ɛ Ɔ ǻ ḛ ḭ ṍ ṵ ɛ ɔ̨́* **Ǻ Ḛ Ḭ Ṍ Ṵ Ɛ Ɔ̨́ ǻ ḛ ḭ ṍ ṵ ɛ ɔ̨́** ***Ǻ Ḛ Ḭ Ṍ Ṵ Ɛ Ɔ̨́ ǻ ḛ ḭ ṍ ṵ ɛ ɔ̨́***

8 Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ̨̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ̨̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ̨̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ̨̀***

9 Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ̨̆ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ̨̆* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ̨̆** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ̨̆***

10 Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ̨̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ̨̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ̨̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ̨̋***

11 Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ɛ Ɔ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ɛ ɔ̨̈̀ *Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ɛ Ɔ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ɛ ɔ̨̈̀* **Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ɛ Ɔ̨̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ɛ ɔ̨̈̀** ***Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ɛ Ɔ̨̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ɛ ɔ̨̈̀***

Patched

A E I O U Ɛ Ɔ a e i o u ɛ ɔ *A E I O U Ɛ Ɔ a e i o u ɛ ɔ* **A E I O U Ɛ Ɔ a e i o u ɛ ɔ** ***A E I O U Ɛ Ɔ a e i o u ɛ ɔ***

Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ́ *Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ́* **Á É Í Ó Ú Ɛ Ɔ́ á é í ó ú ɛ ɔ́** ***Á É Í Ó Ú Ɛ Ɔ́ á é í ó ú ɛ ɔ́***

À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ̀ *À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ̀* **À È Ì Ò Ù Ɛ Ɔ̀ à è ì ò ù ɛ ɔ̀** ***À È Ì Ò Ù Ɛ Ɔ̀ à è ì ò ù ɛ ɔ̀***

Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ̆* **Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ ɔ̆** ***Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ ɔ̆***

Ő Ű Ɛ Ɔ ő ű ɛ ɔ̋ *Ő Ű Ɛ Ɔ ő ű ɛ ɔ̋* **Ő Ű Ɛ Ɔ̋ ő ű ɛ ɔ̋** ***Ő Ű Ɛ Ɔ̋ ő ű ɛ ɔ̋***

Ä Ë Ï Ö Ü Ɛ Ɔ ä ë ï ö ü ɛ ɔ̈ *Ä Ë Ï Ö Ü Ɛ Ɔ ä ë ï ö ü ɛ ɔ̈* **Ä Ë Ï Ö Ü Ɛ Ɔ̈ ä ë ï ö ü ɛ ɔ̈** ***Ä Ë Ï Ö Ü Ɛ Ɔ̈ ä ë ï ö ü ɛ ɔ̈***

Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ̨ *Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ̨* **Ą Ę Į Ǫ Ų Ɛ Ɔ̨ ą ę į ǫ ų ɛ ɔ̨** ***Ą Ę Į Ǫ Ų Ɛ Ɔ̨ ą ę į ǫ ų ɛ ɔ̨***

cgj Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę ɪ ǫ ų ɛ ɔ̨ *Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę ɪ ǫ ų ɛ ɔ̨* **Ą Ę Į Ǫ Ų Ɛ Ɔ̨ ą ę ɪ ǫ ų ɛ ɔ̨** ***Ą Ę Į Ǫ Ų Ɛ Ɔ̨ ą ę ɪ ǫ ų ɛ ɔ̨***

Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏***
cgj Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏***
Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̏ Ǫ̏ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̏ ǫ̏***
Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ę̀ Ǫ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ę̀ ǫ̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ę̀ Ǫ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ę̀ ǫ̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ę̀ Ǫ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ę̀ ǫ̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ę̀ Ǫ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ę̀ ǫ̀***
cgj Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ę̀ Ǫ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ę̀ ǫ̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ę̀ Ǫ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ę̀ ǫ̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ę̀ Ǫ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ę̀ ǫ̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ę̀ Ǫ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ę̀ ǫ̀***
Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ę̆ Ǫ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ę̆ ǫ̆ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ę̆ Ǫ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ę̆ ǫ̆* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ę̆ Ǫ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ę̆ ǫ̆** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ę̆ Ǫ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ę̆ ǫ̆***
cgj Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ę̆ Ǫ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ę̆ ǫ̆ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ę̆ Ǫ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ę̆ ǫ̆* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ę̆ Ǫ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ę̆ ǫ̆** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ę̆ Ǫ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ę̆ ǫ̆***
Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̋ Ǫ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̋ ǫ̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̋ Ǫ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̋ ǫ̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̋ Ǫ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̋ ǫ̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̋ Ǫ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̋ ǫ̋***
cgj Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̋ Ǫ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̋ ǫ̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̋ Ǫ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̋ ǫ̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̋ Ǫ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̋ ǫ̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ę̋ Ǫ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ę̋ ǫ̋***
Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ę̈̀ Ǫ̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ę̈̀ ǫ̈̀ *Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ę̈̀ Ǫ̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ę̈̀ ǫ̈̀* **Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ę̈̀ Ǫ̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ę̈̀ ǫ̈̀** ***Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ę̈̀ Ǫ̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ę̈̀ ǫ̈̀***
cgj Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ę̈̀ Ǫ̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ę̈̀ ǫ̈̀ *Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ę̈̀ Ǫ̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ę̈̀ ǫ̈̀* **Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ę̈̀ Ǫ̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ę̈̀ ǫ̈̀** ***Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ę̈̀ Ǫ̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ę̈̀ ǫ̈̀***
Patched not grave
À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* **À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀** ***À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀***
Patched grave
À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* **À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀** ***À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀***

Mark over space

I added a new en-width x-height PUA character, U+E100, with anchors. To float an above mark, say double grave U+030F over a space, write `\char"E100\char"030F` But in practice, the floating mark carries the same tonal specification as the preceding vowel, and therefore must align exactly with the height of that vowel's combining mark. To ensure this mark alignment, insert a combining grapheme joiner, `\cgj`, in the prior vowel + mark combo, to force anchor positioning without substituting a precomposed glyph there. Note that in Libertinus, the two graves in the U+030F glyph are not translates of an identical outline; this is a deliberate design choice by the designer.

Original

`-ke\char"030F\ \char"030F` -kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Patched without CGJ

`-ke\char"030F \char"E100\char"030F` -kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Patched with CGJ

`-ke\cgj\char"030F \char"E100\char"030F` -kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

f with dot below

tiɣə́ ṛfa
tiɣə́ ṛfa
tiɣə́ ṛfa
tiɣə́ ṛfa

ṛ = `\char"1E5B` is the precomposed character U+1E5B Latin small letter r with dot below.

f̣ = `f\char"0323` is f with U+0323 combining dot below.

Here are some letters with a dot below.

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̧ g̦ j̨ p̧ q̧ x̧

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̧ g̦ j̨ p̧ q̧ x̧

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̧ g̦ j̨ p̧ q̧ x̧

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̧ g̦ j̨ p̧ q̧ x̧

The left group of letters – a b d e h i k l m n o r s t u v w y z – have a precomposed character. For the right group of letters – c f g j p q x – the dot is placed according to the below-anchor (if the below-anchor is defined for the letter) or by HarfBuzz fallback shaping (placing the dot below the baseline, regardless of the letter's bounding box).

In all styles – regular, italic, semibold, and semibold italic – the precomposed glyph of y with dot below places the dot beside the descender, even though the y has a below-anchor below the descender.

Do the letters c, f, g, j, p, q, and x have good below-anchors? In regular, the below-anchor of all these combining bases is below the glyph outline (good). In italic, the below-anchor of f, p, and q is beside the descender (bad), the below-anchor of g and j is below the descender (good), and the below-anchor of c and x is below the baseline (good). In semibold, the below-anchor of g and j is below the descender (good), but the other bases – c, f, p, q – have no below-anchor, and HarfBuzz fallback positioning places the combining dot below the baseline, not below the bounding box; for c, f and x this is good, but for p and q this is bad, because of the descenders. In semibold italic, the below-anchor of c, g, j, p and q is below the glyph outline (good), the below-anchor of f is inside the descender (very bad), and x has no below-anchor but HarfBuzz fallback positioning places the combining dot below the baseline (good).

Which dots are used in precomposed characters and how big are the dots? In all styles (except semibold italic), the precomposed characters reference the dot of `uni0307`, combining dot above, U+0307, and take care to translate this above-mark to a position below the letter outline, except for m, which references `dotbelowcomb`, combining dot below, U+0323. In semibold italic, `dotaccent` is used instead of `uni0307`. These two dots are four-point Bézier circles (in regular and semibold, but ovals in italic and semibold italic), but of different diameter. In regular, `uni0307` has diameter 100, and `dotbelowcomb` has diameter 108. That is, the precomposed characters (except m) use a smaller dot, and the non-precomposed letters (and m) take a bigger dot. In semibold, the size difference is reversed: `uni0307` has larger diameter 124, and `dotbelowcomb` has small diameter 108. Actually, semibold `dotbelowcomb` is not a Bézier circle – the horizontal span is 109, and the vertical span is 108. In semibold italic, `dotaccent` is a skewed four-point Bézier oval – the horizontal and vertical spans are 120, offset by $(\pm 6, \pm 7.5)$ from the center — and `dotbelowcomb` is slightly larger, slightly flatter skewed six-point Bézier oval – horizontal span is 122 and the vertical span is 120. So, in semibold italic the two dots in precomposed and combining form are closest in size. Recall that f has a bad below-anchor, inside the descender, but g, j, p and q have a good below-anchor, below the glyph outline.

So, in order to solve the problem of the bad dot below f in italic, I need to decide the scope of the badness to correct. The simplest patch is move the below-anchor in italic and semibold italic below the descender. But this does not solve inconsistent below dot size across styles and letters. In italic and semibold italic, g and j have a (good) below-anchor at $Y = -319$, so that is also good Y coordinate for f's below-anchor.

I asked Copilot/GPT-5.1 to calculate a good X coordinate for f's below-anchor in italic and semibold italic, taking into account the U-pocket of f's descender and the below-anchor of `dotbelowcomb`. I decided that this U-pocket strategy was appropriate after testing other strategies that accounted for the slant of onf the major stroke. Here is a reasonbly faithful summary of the calculations and reasoning by Copilot/GPT-5.1.

> For italic, the descender's U-pocket is defined by the control points at $(14, -238)$ and $(33, -205)$, giving a U-center around $x \approx 24$. Using the (symmetric) italic `dotbelowcomb` anchor as the dot's optical center, the dot should sit under this U-pocket, not under the ball or the stem. A balanced, U-centered, slightly inner-biased choice is $X \approx 30$.
>
> For semibold italic, the U-pocket of the semibold italic descender is wider and more asymmetric, with a geometric U-center of $X = -33.5$. The semibold italic `dotbelowcomb` itself is also skewed, with its anchor already shifted relative to its mass. In practice, the visually correct position for the dot below f is under the open side of the U-pocket, clearly right of the ball and closer the right of the stem. The anchor candidate is $X \approx 62$.

Here are examples with the patched fonts, with f's below-anchor in italic at $(30, -319)$ and semibold italic at $(62, -319)$. This does not solve the problem of inconsistent dot size below letters, and I did not touch p, q, or y.

Original	Patched italics (f only)
tiɣə ɾfa	tiɣə ɾfa
tiɣə ɾfa	*tiɣə ɾfa*
tiɣə ɾfa	**tiɣə ɾfa**
tiɣə ɾfa	***tiɣə ɾfa***

Original (repeated from above)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

Patched italics (f only)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

Ashkun

`a\char"0304\char"0303` ã̄ *ã̄* **ã̄** ***ã̄***

`\char"0101\char"0303` ã̄ *ã̄* **ã̄** ***ã̄***

`u\char"0304\char"0303` ũ̄ *ũ̄* **ũ̄** ***ũ̄***

`\char"016B\char"0303` ũ̄ *ũ̄* **ũ̄** ***ũ̄***

Patched

`u\char"0304\char"0303` ũ̄ *ũ̄* **ũ̄** ***ũ̄***

`u\cgj\char"0304\char"0303` ũ̄ *ũ̄* **ũ̄** ***ũ̄***

`\char"016B\cgj\char"0303` ũ̄ *ũ̄* **ũ̄** ***ũ̄***

`\char"016B\char"0303` ũ̄ *ũ̄* **ũ̄** ***ũ̄***

Move box without patching font

`\char"016B\kern-4.132pt\raisebox{0.88pt}{\char"02DC}` ũ̄ *ũ̄* **ũ̄** ***ũ̄***

Tacks

a̰ḛḭo̰ṵa̰ḛḭo̰ṵ

a̰ḛḭo̰ṵa̰ḛḭo̰ṵ

Upsilon and grave

Original, then patched, versions of upsilon and combining grave, after setting upsilon's above-anchor.

ὺ ύ̀ **ὺ** **ύ̀**

ὺ ύ̀ **ὺ** **ύ̀**

New upsilon above-anchor: regular $(253, 645)$; italic $(358, 645)$; semibold $(267, 645)$; semibold italic $(369, 645)$

Revised combining grave above-anchor: semibold italic $(0, 683)$. Moved from $(-142, 683)$, which is far left of the optical center of the mark, which has the effect of attaching the mark too far right.

Spoof to move modifier grave (which has the exact same outline as combining grave) to an anchored position, if upsilon had an above-anchor at $(358, 645)$ (keeping combining grave's above-anchor as is). Note upsilon is U+028A and modifier grave is U+02CB.

```
\textit{\char"028A\kern-4.66pt\raisebox{-0.38pt}{\char"02CB}}
```

ὺ

Patch superscript consonant anchors

Libertinus serif regular

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ʶ ˢ ˣ ˠ ᵇ ᵏ ᵐ ᵖ ᵗ ᵚ ᵛ ᵛ̟ ᵝ ᵞ ᵟ

ʰ̟ ʱ̟ ʲ̟ ʳ̟ ʴ̟ ʵ̟ ʷ̟ ʸ̟ ʶ̟ ˢ̟ ˣ̟ ˠ̟ ᵇ̟ ᵏ̟ ᵐ̟ ᵖ̟ ᵗ̟ ᵚ̟ ᵛ̟ ᵛ̟̟ ᵝ̟ ᵞ̟ ᵟ̟

ʰ̠ ʱ̠ ʲ̠ ʳ̠ ʴ̠ ʵ̠ ʷ̠ ʸ̠ ʶ̠ ˢ̠ ˣ̠ ˠ̠ ᵇ̠ ᵏ̠ ᵐ̠ ᵖ̠ ᵗ̠ ᵚ̠ ᵛ̠ ᵛ̟̠ ᵝ̠ ᵞ̠ ᵟ̠

ʰ̃ ʱ̃ ʲ̃ ʳ̃ ʴ̃ ʵ̃ ʷ̃ ʸ̃ ʶ̃ ˢ̃ ˣ̃ ˠ̃ ᵇ̃ ᵏ̃ ᵐ̃ ᵖ̃ ᵗ̃ ᵚ̃ ᵛ̃ ᵛ̟̃ ᵝ̃ ᵞ̃ ᵟ̃

ʰ̀ ʱ̀ ʲ̀ ʳ̀ ʴ̀ ʵ̀ ʷ̀ ʸ̀ ʶ̀ ˢ̀ ˣ̀ ˠ̀ ᵇ̀ ᵏ̀ ᵐ̀ ᵖ̀ ᵗ̀ ᵚ̀ ᵛ̀ ᵛ̟̀ ᵝ̀ ᵞ̀ ᵟ̀

ʰ́ ʱ́ ʲ́ ʳ́ ʴ́ ʵ́ ʷ́ ʸ́ ʶ́ ˢ́ ˣ́ ˠ́ ᵇ́ ᵏ́ ᵐ́ ᵖ́ ᵗ́ ᵚ́ ᵛ́ ᵛ̟́ ᵝ́ ᵞ́ ᵟ́

ʰ̂ ʱ̂ ʲ̂ ʳ̂ ʴ̂ ʵ̂ ʷ̂ ʸ̂ ʶ̂ ˢ̂ ˣ̂ ˠ̂ ᵇ̂ ᵏ̂ ᵐ̂ ᵖ̂ ᵗ̂ ᵚ̂ ᵛ̂ ᵛ̟̂ ᵝ̂ ᵞ̂ ᵟ̂

Libertinus serif regular, patched

ʰ’ ʱ’ ʲ’ ʳ’ ʴ’ ʵ’ ʷ’ ʸ’ ʶ’ ˢ’ ˣ’ ˠ’ ᵇ’ ᵏ’ ᵐ’ ᵖ’ ᵗ’ ᵚ’ ᵛ’ ᵛ̟’ ᵝ’ ᵞ’ ᵟ’

ʰ̋’ ʱ̋’ ʲ̋’ ʳ̋’ ʴ̋’ ʵ̋’ ʷ̋’ ʸ̋’ ʶ̋’ ˢ̋’ ˣ̋’ ˠ̋’ ᵇ̋’ ᵏ̋’ ᵐ̋’ ᵖ̋’ ᵗ̋’ ᵚ̋’ ᵛ̋’ ᵛ̟̋’ ᵝ̋’ ᵞ̋’ ᵟ̋’

h ̏’ ɦ ̏’ j ̏’ r ̏’ ɹ ̏’ ɻ ̏’ ʁ ̏’ w ̏’ y ̏’ ʏ ̏’ l ̏’ s ̏’ x ̏’ ɣ ̏’ b ̏’ k ̏’ m ̏’ p ̏’ t ̏’ ɯ ̏’ v ̏’ ʋ ̏’ β ̏’ γ ̏’ ð ̏’

h ̏ ɦ ̏ j ̏ r ̏ ɹ ̏ ɻ ̏ ʁ ̏ w ̏ y ̏ ʏ ̏ l ̏ s ̏ x ̏ ɣ ̏ b ̏ k ̏ m ̏ p ̏ t ̏ ɯ ̏ v ̏ ʋ ̏ β ̏ γ ̏ ð ̏

h ̏ ɦ ̏ j ̏ r ̏ ɹ ̏ ɻ ̏ ʁ ̏ w ̏ y ̏ ʏ ̏ l ̏ s ̏ x ̏ ɣ ̏ b ̏ k ̏ m ̏ p ̏ t ̏ ɯ ̏ v ̏ ʋ ̏ β ̏ γ ̏ ð ̏

h ̀ ɦ ̀ j ̀ r ̀ ɹ ̀ ɻ ̀ ʁ ̀ w ̀ y ̀ ʏ ̀ l ̀ s ̀ x ̀ ɣ ̀ b ̀ k ̀ m ̀ p ̀ t ̀ ɯ ̀ v ̀ ʋ ̀ β ̀ γ ̀ ð ̀

h ́ ɦ ́ j ́ r ́ ɹ ́ ɻ ́ ʁ ́ w ́ y ́ ʏ ́ l ́ s ́ x ́ ɣ ́ b ́ k ́ m ́ p ́ t ́ ɯ ́ v ́ ʋ ́ β ́ γ ́ ð ́

h ̂ ɦ ̂ j ̂ r ̂ ɹ ̂ ɻ ̂ ʁ ̂ w ̂ y ̂ ʏ ̂ l ̂ s ̂ x ̂ ɣ ̂ b ̂ k ̂ m ̂ p ̂ t ̂ ɯ ̂ v ̂ ʋ ̂ β ̂ γ ̂ ð ̂

Gentium Plus

ʰ’ ʱ’ ʲ’ ʳ’ ʴ’ ʷ’ ʸ’ ˠ’ ᵖ’ ˢ’ ˣ’ ˤ’ ᵇ’ ᵏ’ ᵐ’ ᵖ’ ᵗ’ ɯ’ ᵛ’ ʋ’ β’ γ’ ð’

ʰ ʱ ʲ ʳ ʴ ʷ ʸ ˠ ᵖ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ɯ ᵛ ʋ β γ ð

ʰ¯ ʱ¯ ʲ¯ ʳ¯ ʴ¯ ʷ¯ ʸ¯ ˠ¯ ᵖ¯ ˢ¯ ˣ¯ ˤ¯ ᵇ¯ ᵏ¯ ᵐ¯ ᵖ¯ ᵗ¯ ɯ¯ ᵛ¯ ʋ¯ β¯ γ¯ ð¯

h ̀ ɦ ̀ j ̀ r ̀ ɹ ̀ ɻ ̀ ʁ ̀ w ̀ y ̀ ʏ ̀ l ̀ s ̀ x ̀ ɣ ̀ b ̀ k ̀ m ̀ p ̀ t ̀ ɯ ̀ v ̀ ʋ ̀ β ̀ γ ̀ ð ̀

h ́ ɦ ́ j ́ r ́ ɹ ́ ɻ ́ ʁ ́ w ́ y ́ ʏ ́ l ́ s ́ x ́ ɣ ́ b ́ k ́ m ́ p ́ t ́ ɯ ́ v ́ ʋ ́ β ́ γ ́ ð ́

h ̂ ɦ ̂ j ̂ r ̂ ɹ ̂ ɻ ̂ ʁ ̂ w ̂ y ̂ ʏ ̂ l ̂ s ̂ x ̂ ɣ ̂ b ̂ k ̂ m ̂ p ̂ t ̂ ɯ ̂ v ̂ ʋ ̂ β ̂ γ ̂ ð ̂

A ̄ B ̄ C ̄ D ̄ E ̄ F ̄ G ̄ H ̄ I ̄ J ̄ K ̄ L ̄ M ̄ N ̄ O ̄ P ̄ Q ̄ R ̄ S ̄ T ̄ U ̄ V ̄ W ̄ X ̄ Y ̄ Z ̄ a ̄ b ̄ c ̄ d ̄ e ̄ f ̄ g ̄ h ̄ i ̄ j ̄ k ̄ l ̄ m ̄ n ̄ o ̄ p ̄ q ̄ r ̄ s ̄ t ̄ u ̄ v ̄ w ̄ x ̄ y ̄ z ̄ Æ ̄ æ ̄ Œ ̄ œ ̄ Ø ̄ ø ̄ Đ ̄ ð ̄ Þ ̄ þ ̄ ß ̄ ſ ̄ Ð ̄ đ ̄ Ħ ̄ ħ ̄ ı ̄ ȷ ̄ Ł ̄ ł ̄ Ƀ ̄ Ƃ ̄
À B̀ C̀ D̀ È F̀ G̀ H̀ Ì J̀ K̀ L̀ M̀ Ǹ Ò P̀ Q̀ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Z̀ à b̀ c̀ d̀ è f̀ g̀ h̀ ì j̀ k̀ l̀ m̀ ǹ ò p̀ q̀ r̀ s̀ t̀ ù v̀ ẁ x̀ ỳ z̀ Æ̀ æ̀ Œ̀ œ̀ Ø̀ ø̀ Đ̀ ð̀ Þ̀ þ̀ ß̀ ſ̀ Đ̀ đ̀ Ħ̀ ħ̀ ì j̀ Ł̀ ł̀ B̀ Ƃ̀

Dot removal

To do. Check GSUB to see if dot removal for i and j is complete.

Precomposed characters

To do. Check GSUB if an available precomposed character is encoded by a doublet of the base and a combining mark, or a triplet of the base and the two combining marks, in either mark order, then the precomposed character should be be produced by the font’s GSUB table.

Marks

Here is a catalog of combining marks in Unicode, grouped by the seven GPOS anchor class indices in Libertinus (0–6).

Above marks – GPOS anchor 0

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̚ x̛ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̚ x̛ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̚ x̛ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̚ x̛ x̽ x̾ x̿

Below marks – GPOS anchor 2

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳

Left angle – GPOS anchor 3

x̕ x᾽x̄ x̓

Below-right marks – GPOS anchor 4

x̹ x̺ x̻
x̹ x̺ x̻
x̹ x̺ x̻
x̹ x̺ x̻

Cedilla – GPOS anchor 5

x̡ x̧ x̨ x̩

Overlay marks – GPOS anchor 6

o̶ o̷ o̸ o̵ o̴
o̶ o̷ o̸ o̵ o̴
o̶ o̷ o̸ o̵ o̴
o̶ o̷ o̸ o̵ o̴

Double marks

x͡x x͡x x͞x x͞x x̃x x̃x x͡x x͡x
x͡x x͡x x͞x x͞x x̃x x̃x x͡x x͡x
x̋x x̋x x̏x x̂x x̌x x̆x ẍx
x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x

Greek marks

t̀ t́ t̃ t̄ t̆ ẗ t̩
t̀ t́ t̃ t̩
t̂ t̂ t̀ t̀ t̋ t̋ t̏ ẗ t̩
t̀ t́ t̃ t̄ t̆ ẗ t̩
t̀ t́ t̃ t̩
t̂ t̂ t̀ t̀ t̋ t̋ t̏ t̏ t̩
t̀ t́ t̃ t̄ t̆ ẗ t̩
t̀ t́ t̃ t▯
t̂ t̂ t̀ t̀ t̋ t̃ t̏ ẗ t̩▯
t̀ t́ t̃ t̄ t̆ ẗ t̩
t̀ t́ t̃ t▯
t̂ t̂ t̀ t̀ t̋ t̃ t̏ ẗ t̩▯

Adding missing marks

Semibold and semibold italic lack many of the combining marks of regular and italic. The patch adds them, with mark anchors, in the range U+0332–U+036F. No mark is drawn from scratch: each is an outline that the designer drew in another style. A semibold mark is the regular outline (an above-mark is moved right by 12 units, with its anchor, as the designer's semibold above-marks are). A semibold italic mark is the italic outline moved right by 12 units, with the anchor of regular, as in the semibold italic anchor model. In semibold, U+0358–U+0363 were copies of the capital accents, and they are replaced. The patch also adds `dotaccent.cap` to italic and semibold italic: the dot of the precomposed İ, at the height it has in the precomposed dotted capitals.

The designer does not thicken below-marks between weights, but thickens above-marks by about 12 percent (mean stroke thickness, measured as twice the area of the outline divided by its perimeter: tilde 40 to 45, caron 36 to 42). So the added above-marks are made bolder. Where a mark is, or contains, another mark, the bold style's own mark is used. The rest of the outline is pushed outward by 7.5 percent of its stroke thickness on each side, about 2 to 3 units. These algorithmically emboldened marks are shown in the table on the next page, on a few bases that have above-anchors in all four patched fonts, with the designer's tilde and caron for comparison of weight. The regular and italic columns show the outlines that were copied.

Also added, but not shown in the table: the low line and double low line (U+0332, U+0333), the tilde, stroke and solidus overlays (U+0334–U+0337), the below-marks U+0339–U+033C, U+0347–U+0349, U+034D, U+034E, U+0353–U+0356, U+0359, U+035A, the double overline (U+033F), the dot above right (U+0358), the double marks (U+035C–U+0362) and the combining letters (U+0363–U+036F). These are copied without emboldening, since the designer's lines, below-marks and modifier letters are not thicker in the bold styles; the dot above right is the bold style's own dot. The double marks have no anchors, as in regular. In semibold, the overlays and the dot above right have no anchors, because the semibold lookup table has no anchor class for them. The half rings above (U+0351, U+0357) already exist in semibold, drawn to the right of the origin, so they sit beside the base; they are left as they are. Not added: the custom-width low lines and overlines of the next section, U+0344, U+0345 and U+0483.

		regular	semibold	italic	semibold italic	
U+0303	tilde (designer's, for comparison)	ãñũÕ	**ãñũÕ**	*ãñũÕ*	***ãñũÕ***	
U+030C	caron (designer's, for comparison)	ǎňǔǑ	**ǎňǔǑ**	*ǎňǔǑ*	***ǎňǔǑ***	
U+033D	x above	x̌ňǔǑ	**x̌ňǔǑ**	*x̌ňǔǑ*	***x̌ňǔǑ***	emboldened
U+033E	vertical tilde	a̾n̾u̾O̾	**a̾n̾u̾O̾**	*a̾n̾u̾O̾*	***a̾n̾u̾O̾***	emboldened
U+034A	not tilde above	ãn̋ũÕ̋	**ãn̋ũÕ̋**	*ãn̋ũÕ̋*	***ãn̋ũÕ̋***	emboldened
U+034B	homothetic above	ãn̑ũÕ̑	**ãn̑ũÕ̑**	*ãn̑ũÕ̑*	***ãn̑ũÕ̑***	own tilde; dots emboldened
U+034C	almost equal to above	ãn̆̃ũÕ̆̃	**ãn̆̃ũÕ̆̃**	*ãn̆̃ũÕ̆̃*	***ãn̆̃ũÕ̆̃***	emboldened
U+0350	right arrowhead above	ãn̓ũÕ̓	**ãn̓ũÕ̓**	*ãn̓ũÕ̓*	***ãn̓ũÕ̓***	emboldened
U+0351	left half ring above	ãn̔ũÕ̔	**ãn̔ũÕ̔**	*ãn̔ũÕ̔*	***ãn̔ũÕ̔***	emboldened (semibold italic only)
U+0352	fermata	ãn̗ũÕ̗	**ãn̗ũÕ̗**	*ãn̗ũÕ̗*	***ãn̗ũÕ̗***	own inverted breve; dot emboldened
U+0357	right half ring above	a̾n̾u̾O̾	**a̾n̾u̾O̾**	*a̾n̾u̾O̾*	***a̾n̾u̾O̾***	emboldened (semibold italic only)
U+035B	zigzag above	ãn̛ũỠ	**ãn̛ũỠ**	*ãn̛ũỠ*	***ãn̛ũỠ***	emboldened
U+0340	grave tone mark	àǹùÒ	**àǹùÒ**	*àǹùÒ*	***àǹùÒ***	own grave
U+0341	acute tone mark	áńúÓ	**áńúÓ**	*áńúÓ*	***áńúÓ***	own acute
U+0346	bridge above	ãn̪ũÕ̪	**ãn̪ũÕ̪**	*ãn̪ũÕ̪*	***ãn̪ũÕ̪***	own bridge below, raised

Overline and low line

U+0305 combining overline and U+0332 combining low line are applied by GSUB lookups that substitute a custom-width line mark according to width classes. GSUB lookup 51 effectively classifies the glyphs by width, and GSUB lookups 52–82 provide the appropriate custom-width mark for the base width class. Semibold and semibold italic do not have any combining low lines; the patch adds U+0332 and U+0333 with below-anchors, but not the custom-width variants. Custom-width line marks do not have anchors. They are positioned by HarfBuzz fallback shaping. The overlines are above the ascender line, so they are above all bases. The low lines are below the baseline, but above the descender line, and so they cut though any descenders. If a new glyph is added, and you want to support custom-width line marks for it, then remember to classify the new glyph in GSUB lookup 51.



Above-right marks – GPOS anchor 1

Libertinus has three combining marks with GPOS anchor 1 – U+031B horn, U+0315 comma above right, and U+0358 dot above right.

A few bases in Libertinus have this anchor (anchor 1), but none that possibly need it.

Horn

The Vietnamese alphabet has two horn vowels – Ơơ Ưư– and these vowels may take tonal marks. All these combinations are Unicode characters (with code points).

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

There is a Unicode character for the combining horn mark: ̛ `\char"031B` U+031B, but this is not used in practice. The Vietnamese vowels are usually entered in a LaTeX document as Unicode characters, also with `\usepackage[vietnamese]{babel}` for hyphenation and spacing.

Comma above right

There is a Unicode character for the combining comma above right mark: ̕ `\char"0315` U+0315.

U+0315 (comma above right) is used in rare cases when another comma-like mark does not fit.

In IPA, the glottalization diacritic mark uses U+0313 (comma above) as a combining mark. But when a glottalization mark is needed for a superscript consonant (1) or superscript consonant cluster (2), then the combining mark U+0315 (comma above right) is used instead, because its positioning is hopefully better than that of U+0313.

(1) *k̕wkt* [k̕^{wkx}t] ‘to scrape’ (Nuxalk)

(2) *p’ak* [p’^{kx}ak] ‘he/she cut it’ (Upper Chehalis)

In (1), the superscript consonant cluster is ^{wkx} and it is glottalized. In (2), the superscript consonant ^{x} is glottalized.

Because U+0315 is being used in superscript position on a smaller sized glyph, U+0315 should probably be smaller than U+0313, but it is not so in Libertinus.

Dot above right

There is a Unicode character for the combining dot above right mark: ̙ `\char"0358` U+0358 dot above right

U+0358 should be a small, above-right dot, used to mark palatalization or similar secondary articulations on superscript consonants, created for the same typographic reason U+0315 exists – the standard palatalization diacritic (U+0307, dot above) is too big and too high.

In Libertinus, U+0358 is a 104-diameter Bezier circle, and U+0307 is a smaller Bezier oval, inverting the desired size difference.

There are only a few IPA superscript letters that may serve as the possible host of U+0315 or U+0358 on superscript consonants or on superscript consonant clusters. In Libertinus, regrettably none of these characters has the anchor (anchor 1), so the mark (U+0315 and U+0358) is positioned by HarfBuzz fallback shaping, which seems to place the mark at the right boundary of the glyph at a constant vertical aspect (effectively at the superscript x-height).

These IPA marks U+0315 and U+0358 should only be used in regular (not italic and not semibold). Regrettably, in Libertinus serif semibold, U+0358 looks like a grave, not a dot, and there is no U+0358 glyph in Libertinus serif semibold italic.

There are three comma/apostrophe glyphs that could possibly mark glottalization: U+0313 combining comma above, U+0315 combining comma above right, and U+02BC modifier letter apostrophe. In Libertinus, U+0315 has the worst appearance, even though U+0315 is intended for superscript letters. In Gentium Plus (SIL Global), U+0315 is tightly above-right the superscript letter, as expected.

Libertinus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [kʷˣt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʷˣt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [kʷˣʼt]

Gentium Plus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [kʷˣt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʷˣt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [kʷˣʼt]

I patched Libertinus to add anchors to those IPA superscript letters to support U+0315.

Patched Libertinus, selected IPA superscript letters with U+0315 and U+0358

ʰ’ʱ’ʲ’ʳ’ʴ’ʵ’ʷ’ʸ’ˠ’ˡ’ˢ’ˣ’ˤ’ᵇ’ᵏ’ᵐ’ᵖ’ᵗ’ᵚ’ᵛ’ᵜ’ᵝ’ᵞ’ᵟ’

ʰ‘ʱ‘ʲ‘ʳ‘ʴ‘ʵ‘ʷ‘ʸ‘ˠ‘ˡ‘ˢ‘ˣ‘ˤ‘ᵇ‘ᵏ‘ᵐ‘ᵖ‘ᵗ‘ᵚ‘ᵛ‘ᵜ‘ᵝ‘ᵞ‘ᵟ‘

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʷˣt]

Gentium Plus, selected IPA superscript letters with U+0315 and U+0358

ʰ’ʱ’ʲ’ʳ’ʴ’ʵ’ʷ’ʸ’ˠ’ˡ’ˢ’ˣ’ˤ’ᵇ’ᵏ’ᵐ’ᵖ’ᵗ’ᵚ’ᵛ’ᵜ’ᵝ’ᵞ’ᵟ’

ʰ‘ʱ‘ʲ‘ʳ‘ʴ‘ʵ‘ʷ‘ʸ‘ˠ‘ˡ‘ˢ‘ˣ‘ˤ‘ᵇ‘ᵏ‘ᵐ‘ᵖ‘ᵗ‘ᵚ‘ᵛ‘ᵜ‘ᵝ‘ᵞ‘ᵟ‘

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʷˣt]

Here are the IPA superscript letters considered and their new anchor (anchor 1).

U+02B0: (347, 648), U+02B1: (339, 648), U+02B2: (193, 648), U+02B3: (284, 648), U+02B4: (274, 648), U+02B5: (319, 648), U+02B6: (308, 648), U+02B7: (497, 648), U+02B8: (368, 648), U+02E0: (341, 648), U+02E1: (193, 648), U+02E2: (270, 648), U+02E3: (344, 648), U+02E4: (282, 648), U+1D47: (319, 648), U+1D4F: (353, 648), U+1D50: (533, 648), U+1D56: (334, 648), U+1D57: (241, 648), U+1D5A: (525, 648), U+1D5B: (354, 648), U+1D5C: (508, 648), U+1D5D: (335, 648), U+1D5E: (372, 648), U+1D5F: (353, 648)

I did not change the anchor-1 of U+315 and U+0358. U+0358 has its anchor at the xMin and yMin of its bounding box. But U+0315 already has some clearance built in to its anchor: its anchor is 62 units below its yMin, but 25 units right of it xMin. So, I decided to provide x-clearance of 12 units and y-clearance of 80 units from the bases. All these IPA superscript letters already have a superscript meanline at $Y = 630$. So their anchor will be at $Y = 630 - 62 + 80 = 648$. The X coordinate depends on their bounding box and its xMax. To provide the desired clearance of 12 units, the anchor is at $X = \text{xMax} + 26 + 12$. These anchor coordinate values were optimized for U+0315 but they work fine for U+0358.

Bases

Here are some subgroups of bases.

Latin

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ

IPA

ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞʟɯɲɵɭʒ2ǀǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɢɣɤɥɦɧɨɪɪɺɬɮɭɰɯɱɲɳɴɵœɷɸɹɻɽɾʀʁɕʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀǃ
ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞʟɯɲɵɭʒ2ǀǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɢɣɤɥɦɧɨɪɪɺɬɮɭɰɯɱɲɳɴɵœɷɸɹɻɽɾʀʁɕʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀǃ
ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞʟɯɲɵɭʒ2ǀǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɢɣɤɥɦɧɨɪɪɺɬɮɭɰɯɱɲɳɴɵœɷɸɹɻɽɾʀʁɕʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀǃ
ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞʟɯɲɵɭʒ2ǀǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɢɣɤɥɦɧɨɪɪɺɬɮɭɰɯɱɲɳɴɵœɷɸɹɻɽɾʀʁɕʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀǃ

Superscript consonants

ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛ◌ᵝᵞᵟ
ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛ◌ᵝᵞᵟ
ʰʱ ʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛ◌ᵝᵞᵟ
ʰʱ ʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛ◌ᵝᵞᵟ

Greek

ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω

Cyrillic

АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺҥI
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺҥI
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺҥI
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺҥI

Precomposed capital vowels

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶẸẺẼỀỀỂỄỆỈỊỌỎỐỒỔỖỘỚỜỞỠỢỤỦỨỪỬỮỰ
ỮŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ
ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶẸẺẼỀỀỂỄỆỈỊỌỎỐỒỔỖỘỚỜỞỠỢỤỦỨỪỬỮỰ
ỮŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ
ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶẸẺẼỀỀỂỄỆỈỊỌỎỐỒỔỖỘỚỜỞỠỢỤỦỨ
ỪỬỮŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ
ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶẸẺẼỀỀỂỄỆỈỊỌỎỐỒỔỖỘỚỜỞỠỢỤỦỨ
ỪỬỮŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ

Precomposed small vowels

àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặẹẻẽềềểễệỉịọỏốồổỗộớờởỡợụủứừửữự
àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặẹẻẽềềểễệỉịọỏốồổỗộớờởỡợụủứừửữự
àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặẹẻẽềềểễệỉịọỏốồổỗộớờởỡợụủứừửữự
àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặẹẻẽềềểễệỉịọỏốồổỗộớờởỡợụủứừửữự

Precomposed capital consonants

ḂḄÇḈĈĊČḊĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴƘĹḶĽḺĻḼḾṀṂŃṆŇṊŅṄƝƠṔṖṕŖŘŗŕŖŚŜŞŠṠṢṤṦṨȘŢŤṪṰṬṮȚƬƮŨṾỤỦẂẀẄẂẊẌÝỲỸỶỴỾỸỾȲŹŻŽẐẔẒ
ḂḄÇḈĈĊČḊĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴƘĹḶĽḺĻḼḾṀṂŃṆŇṊŅṄƝƠṔṖṕŖŘŗŕŖŚŜŞŠṠṢṤṦṨȘŢŤṪṰṬṮȚƬƮŨṾỤỦẂẀẄẂẊẌÝỲỸỶỴỾỸỾȲŹŻŽẐẔẒ
ḂḄÇḈĈĊČḊĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴƘĹḶĽḺĻḼḾṀṂŃṆŇṊŅṄƝƠṔṖṕŖŘŗŕŖŚŜŞŠṠṢṤṦṨȘŢŤṪṰṬṮȚƬƮŨṾỤỦẂẀẄẂẊẌÝỲỸỶỴỾỸỾȲŹŻŽẐẔẒ
ḂḄÇḈĈĊČḊĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴƘĹḶĽḺĻḼḾṀṂŃṆŇṊŅṄƝƠṔṖṕŖŘŗŕŖŚŜŞŠṠṢṤṦṨȘŢŤṪṰṬṮȚƬƮŨṾỤỦẂẀẄẂẊẌÝỲỸỶỴỾỸỾȲŹŻŽẐẔẒ

Precomposed small consonants

ƀƅɓƃbçćĉċčɕƈďđḑḓḋḍḏđƌḟĝğġģḡǧǥķĥħḫḥḧḣḩȟĵǰḱǩķƙḳĸĺļľŀłḽḷḻḹŀḿṁṃṅṇňŉṋņṉɲŋơɋɋṕṗƥɍŗřŕȓȑṛṟśŝşšṡṣṥṧșßſŧťẗṫţṭṯŧƭʈũṽỵŵẁẅẘẍẋýÿŷỳẏỷẙƴźżžẑẓẕȝƹƽƿ
ƀƅɓƃbçćĉċčɕƈďđḑḓḋḍḏđƌḟĝğġģḡǧǥķĥħḫḥḧḣḩȟĵǰḱǩķƙḳĸĺļľŀłḽḷḻḹŀḿṁṃṅṇňŉṋņṉɲŋơɋɋṕṗƥɍŗřŕȓȑṛṟśŝşšṡṣṥṧșßſŧťẗṫţṭṯŧƭʈũṽỵŵẁẅẘẍẋýÿŷỳẏỷẙƴźżžẑẓẕȝƹƽƿ
ƀƅɓƃbçćĉċčɕƈďđḑḓḋḍḏđƌḟĝğġģḡǧǥķĥħḫḥḧḣḩȟĵǰḱǩķƙḳĸĺļľŀłḽḷḻḹŀḿṁṃṅṇňŉṋņṉɲŋơɋɋṕṗƥɍŗřŕȓȑṛṟśŝşšṡṣṥṧșßſŧťẗṫţṭṯŧƭʈũṽỵŵẁẅẘẍẋýÿŷỳẏỷẙƴźżžẑẓẕȝƹƽƿ

ƀƅɓƃbçćĉċčɕƈďđḑḓḋḍḏđƌḟĝğġģḡǧǥķĥħḫḥḧḣḩȟĵǰḱǩķƙḳĸĺļľŀłḽḷḻḹŀḿṁṃṅṇňŉṋņṉɲŋơɋɋṕṗƥɍŗřŕȓȑṛṟśŝşšṡṣṥṧșßſŧťẗṫţṭṯŧƭʈũṽỵŵẁẅẘẍẋýÿŷỳẏỷẙƴźżžẑẓẕȝƹƽƿ

No marks: rare, digraph etc.

ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj

Small capitals

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ

Precomposed characters of anchor-relevant bases and marks

àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ

Base and mark shaping – Linguistics notation

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic.

The grids use Latin in IPA bases and combining marks that are used in linguistics notation.

A black color means GPOS anchor positioning.

A blue color means a precomposed glyph.

A red color means HarfBuzz fallback shaping (no anchors, no precomposed glyph, no GSUB substitution).

A faded color – gray, light blue, or pink – means that the base and mark combination is not reasonably expected in notation in linguistics.

A gray color can also mean that mark glyph is missing in the font. In this case, just the gray base is printed. This only happens in Libertinus Serif Semibold and Semibold Italic for three below marks: o͇ (U+0333 double macron below), o͚ (U+033A inverted bridge below), and o͛ (U+033B square below). Otherwise, all considered base and mark combinations can be drawn in XeLaTeX documents in Libertinus without tofu for missing glyphs.

Regular, Bases requiring anchors for common marks, Common combining marks



Regular patched, Bases requiring anchors for common marks, Common combining marks



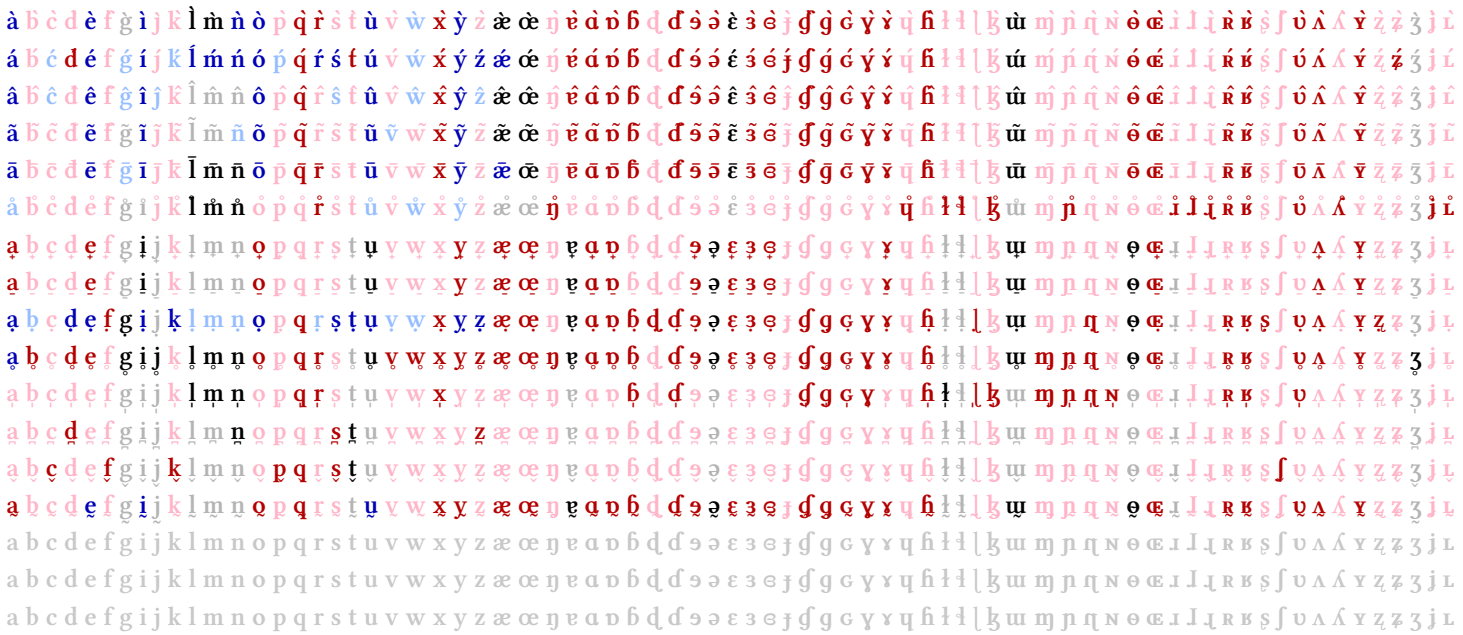
Italic, Bases requiring anchors for common marks, Common combining marks



Italic patched, Bases requiring anchors for common marks, Common combining marks



Semibold, Bases requiring anchors for common marks, Common combining marks



Semibold patched, Bases requiring anchors for common marks, Common combining marks



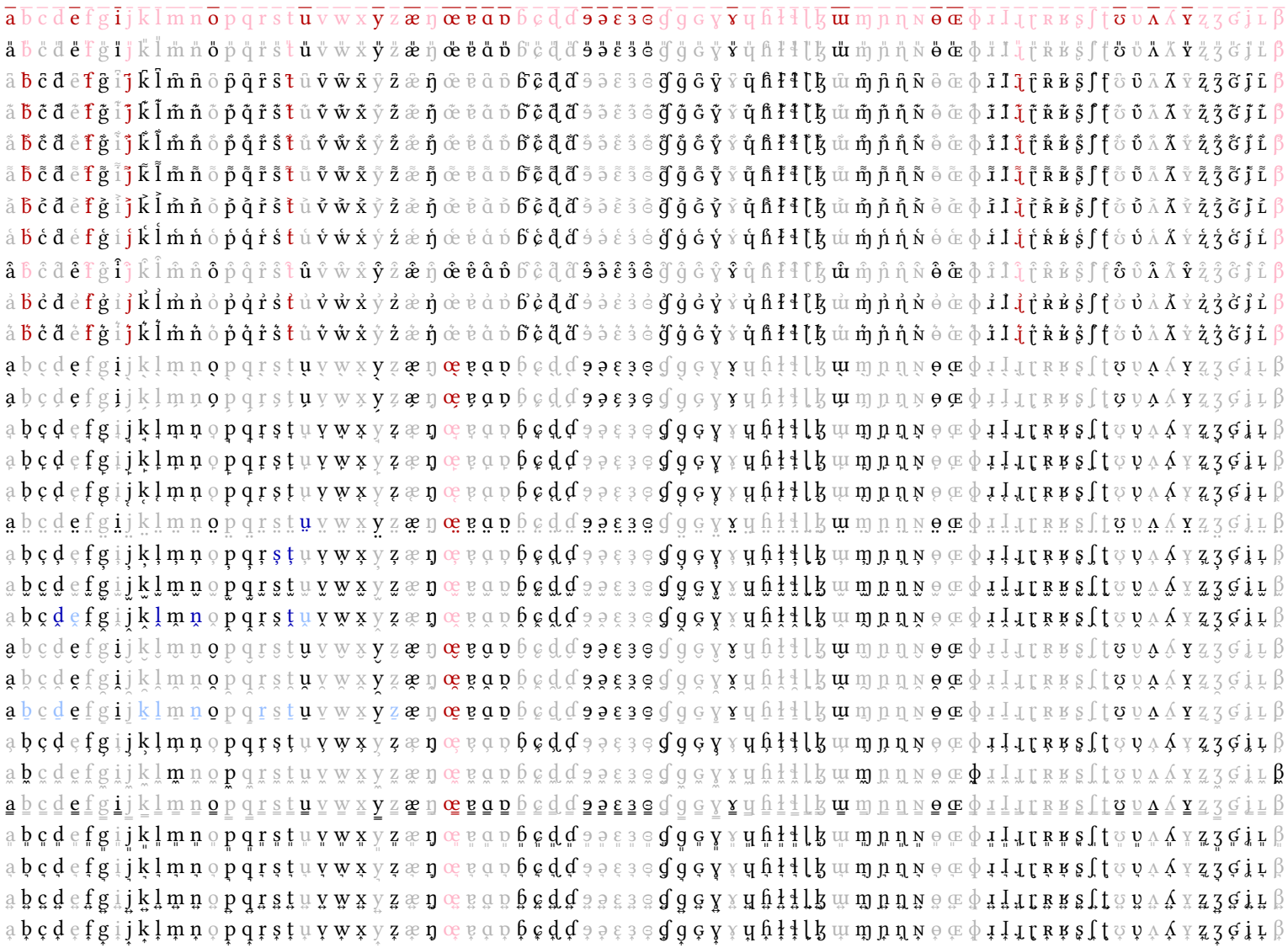
Semibold italic, Bases requiring anchors for common marks, Common combining marks



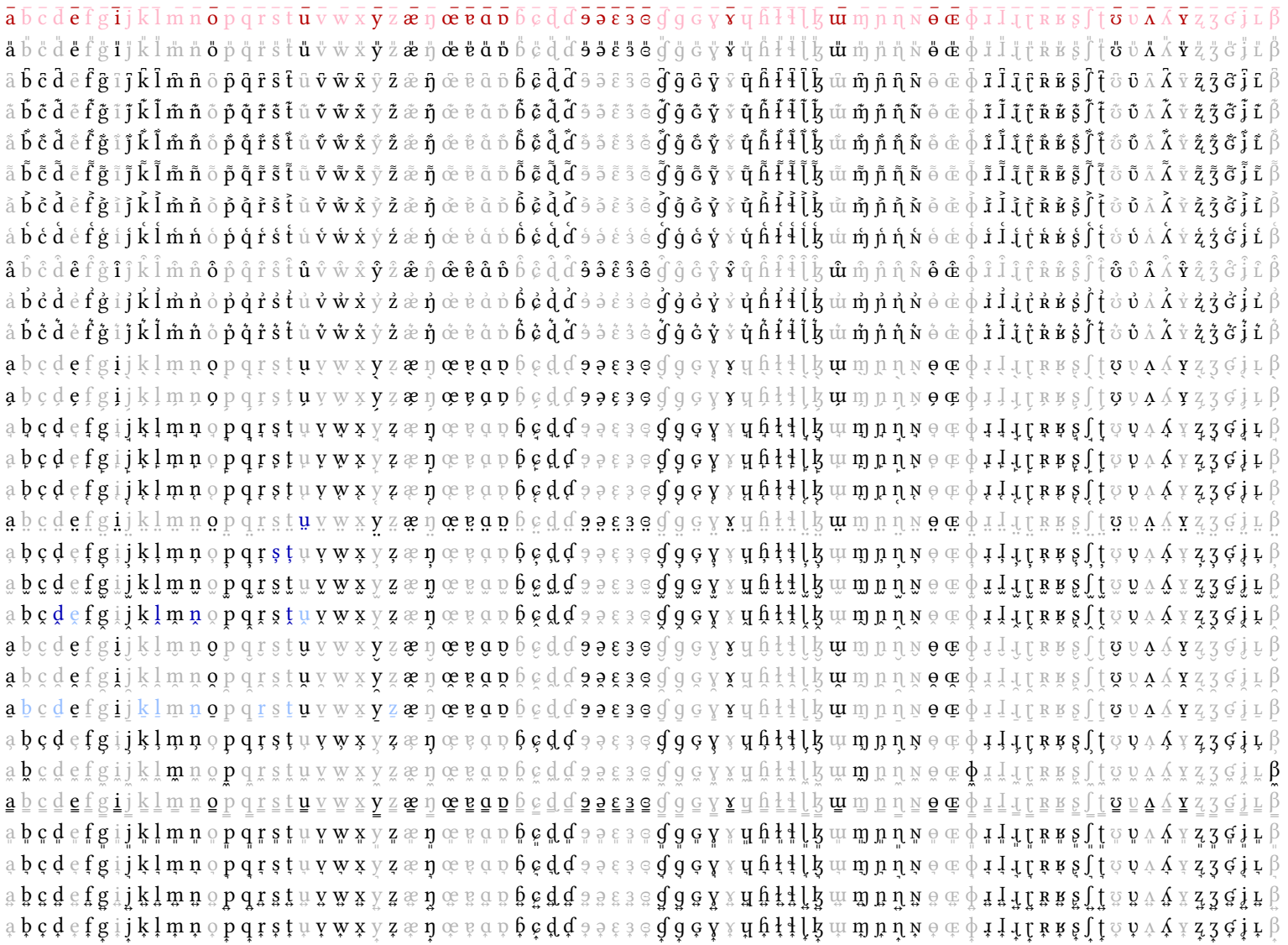
Semibold italic patched, Bases requiring anchors for common marks, Common combining marks



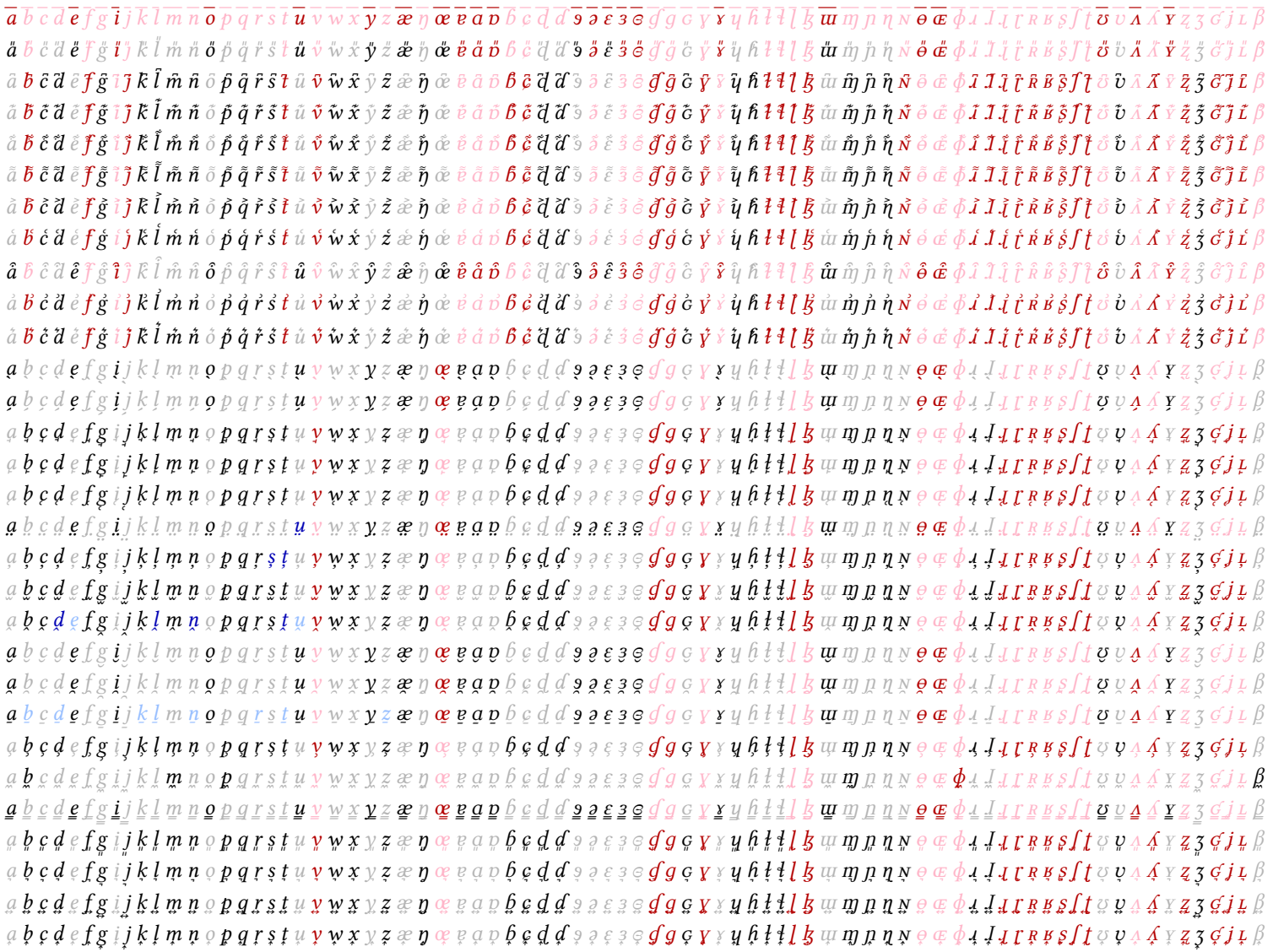
Regular, Bases requiring anchors for rare marks, Rare combining marks



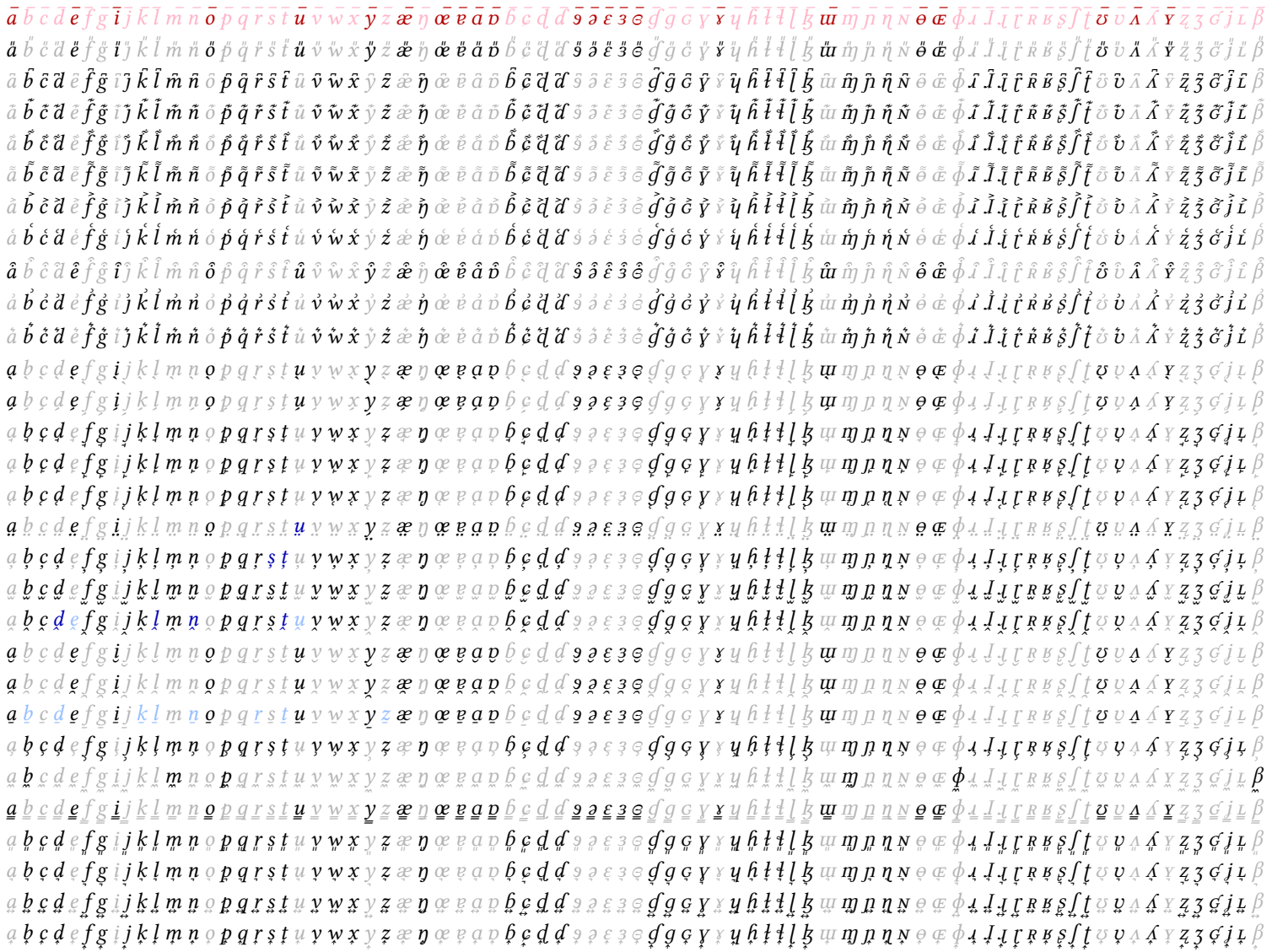
Regular patched, Bases requiring anchors for rare marks, Rare combining marks



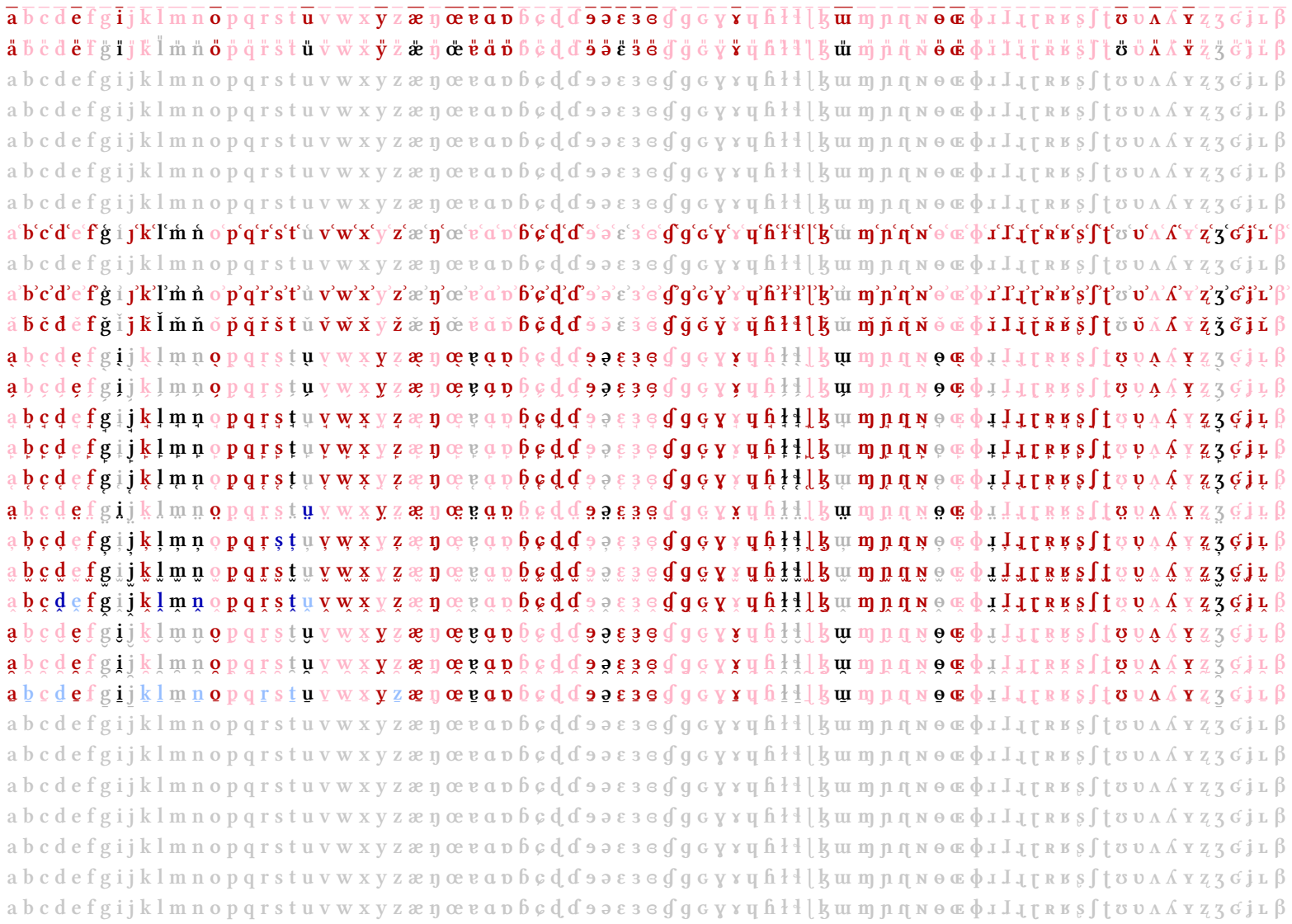
Italic, Bases requiring anchors for rare marks, Rare combining marks



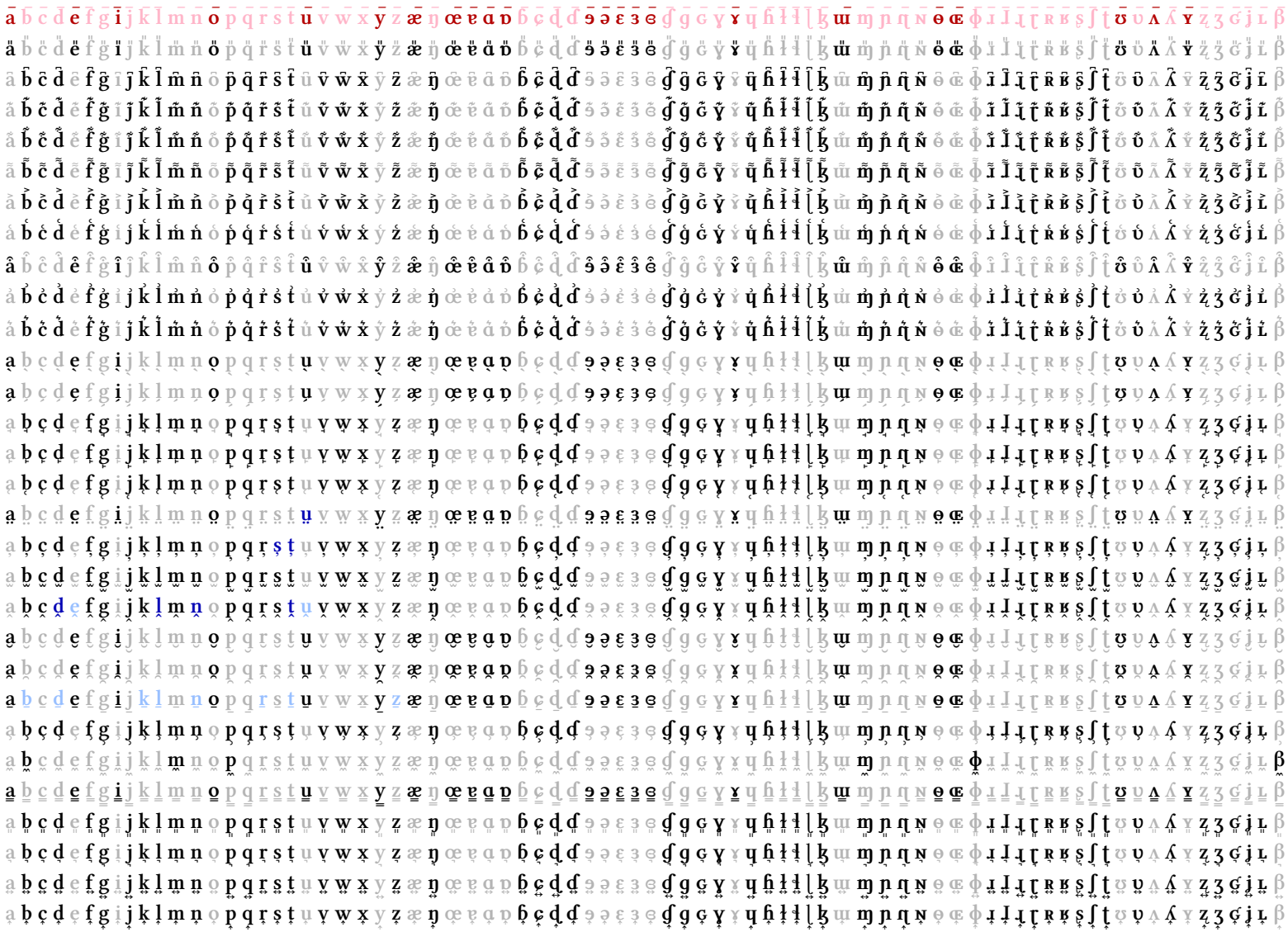
Italic patched, Bases requiring anchors for rare marks, Rare combining marks



Semibold, Bases requiring anchors for rare marks, Rare combining marks



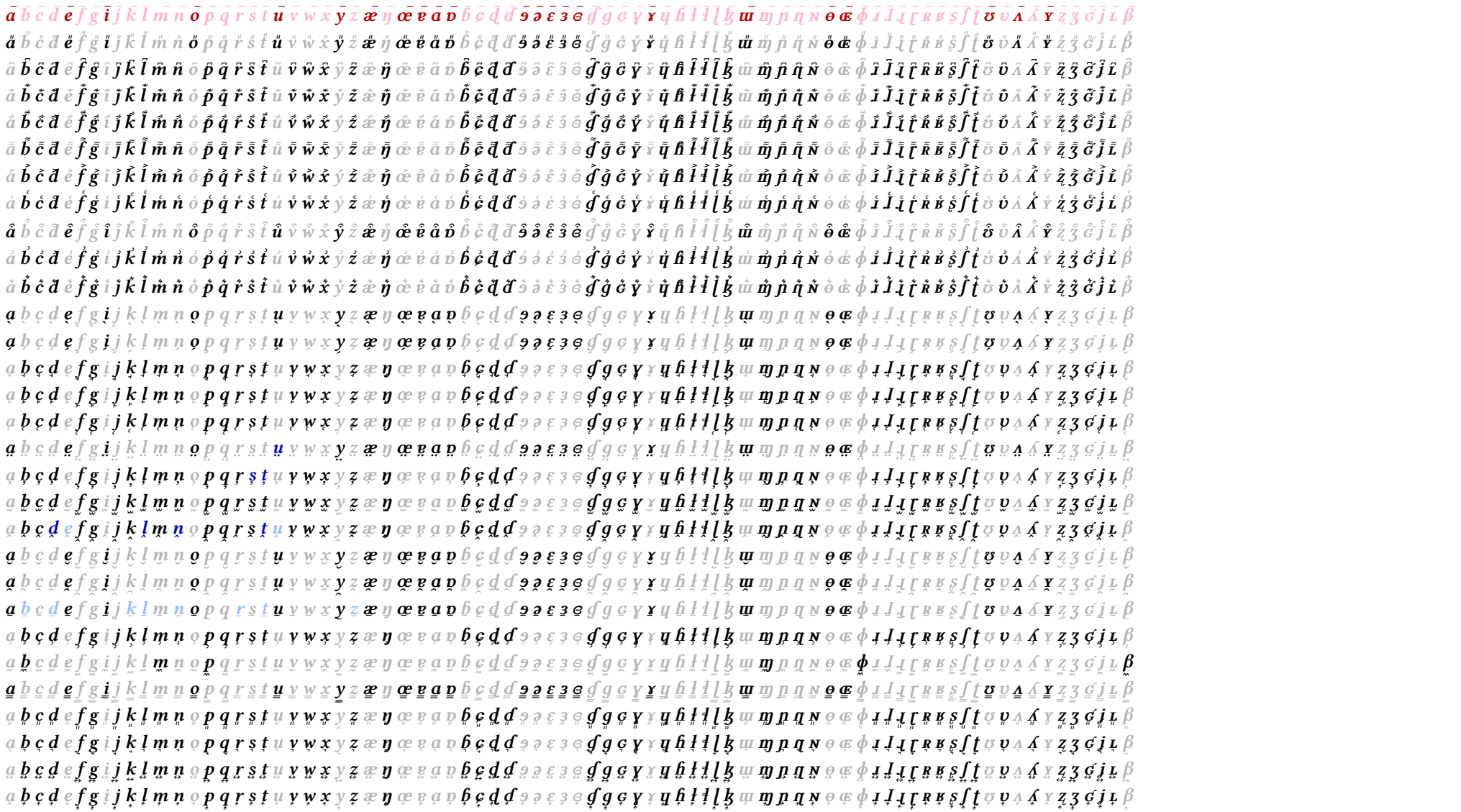
Semibold patched, Bases requiring anchors for rare marks, Rare combining marks



Semibold italic, Bases requiring anchors for rare marks, Rare combining marks



Semibold italic patched, Bases requiring anchors for rare marks, Rare combining marks



Base and mark shaping – Complete

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic. The grids use more complete sets of Latin and IPA bases and above and below marks.

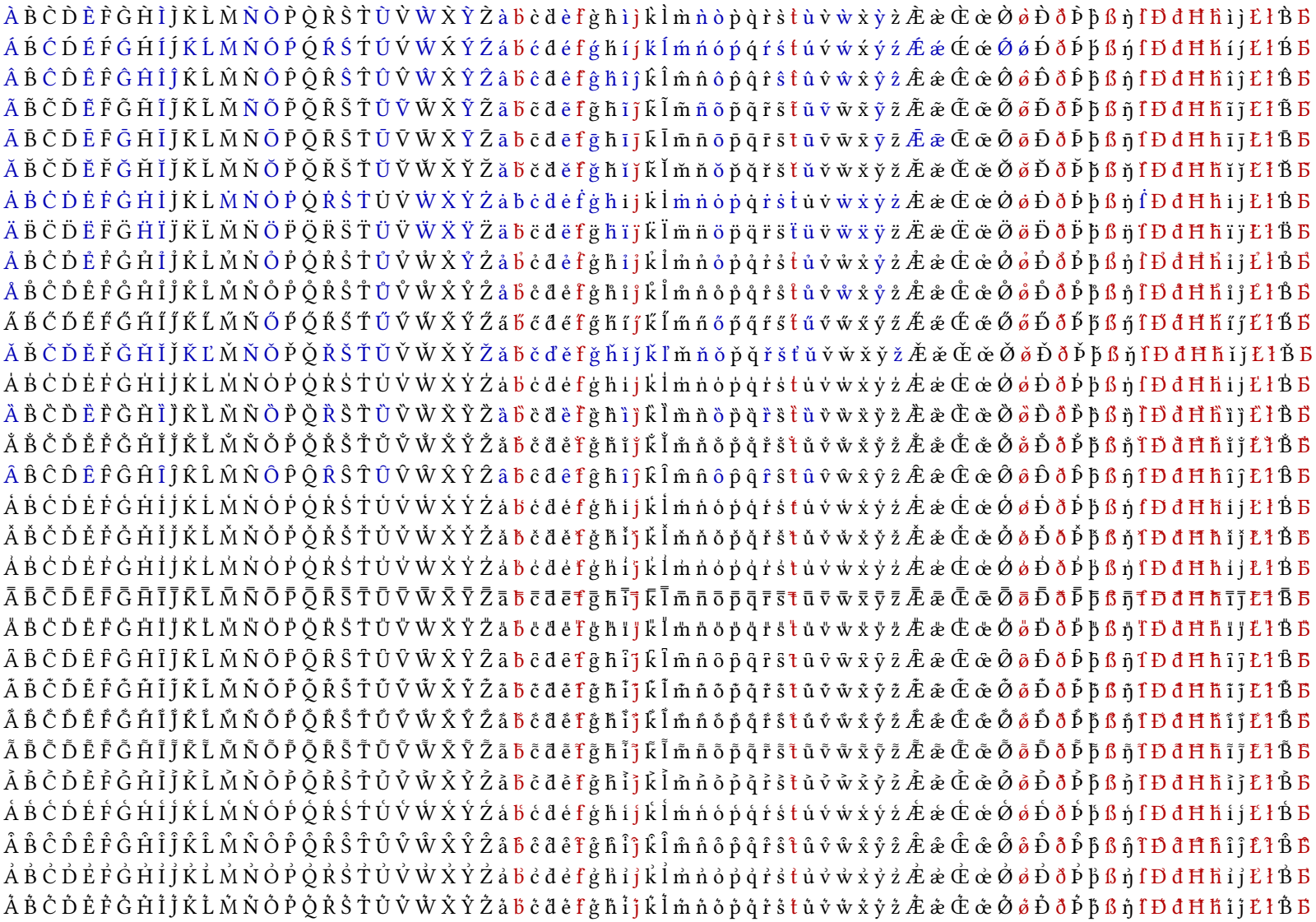
Blue indicates that XeLaTeX uses a precomposed Unicode character glyph.

Black indicates that the mark is positioned using GPOS anchors.

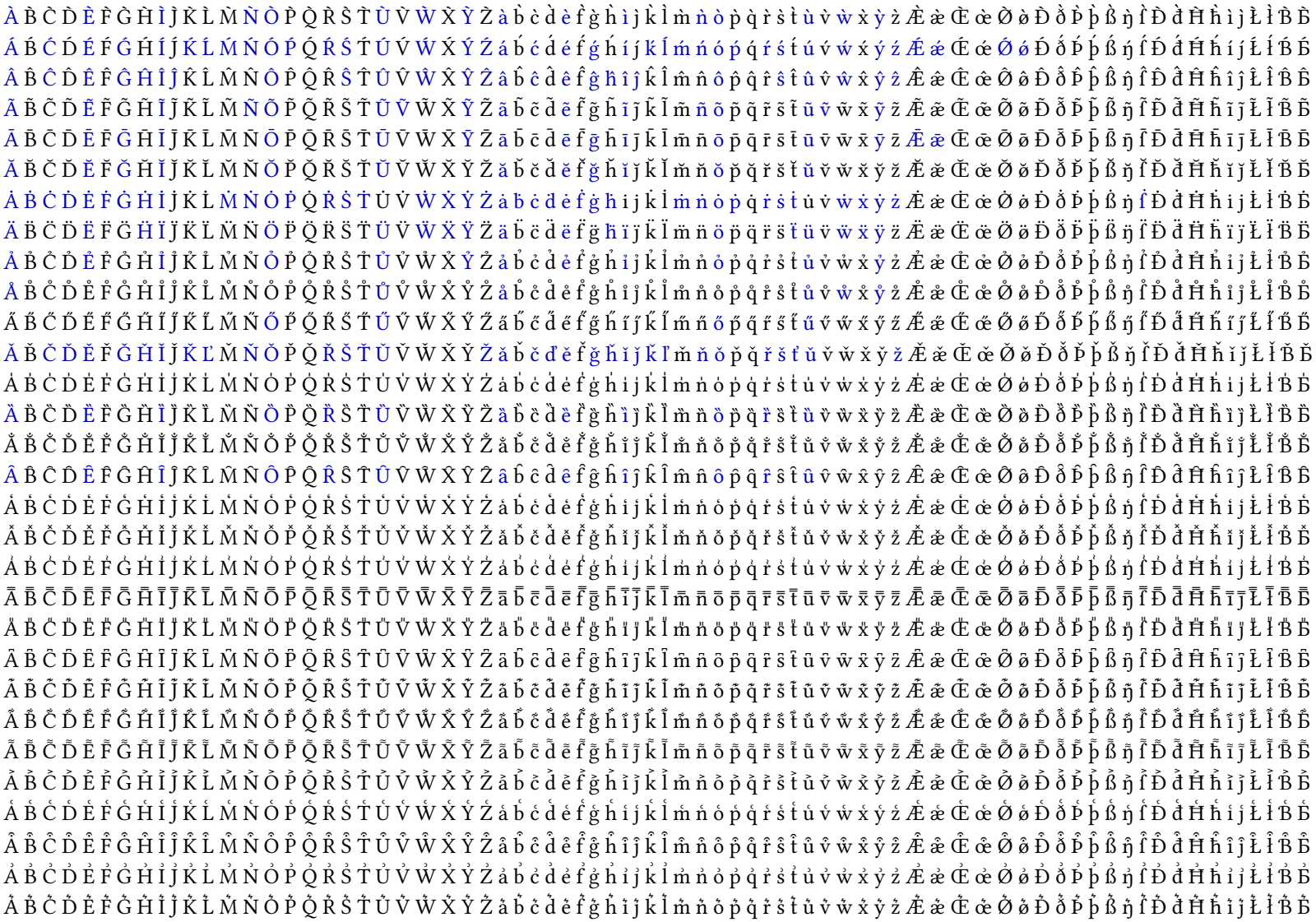
Red indicates HarfBuzz fallback shaping: no precomposed glyph and no GPOS anchors are used.

Light gray indicates that either the base or the mark glyph is missing from the font. In this case only the base glyph is shown.

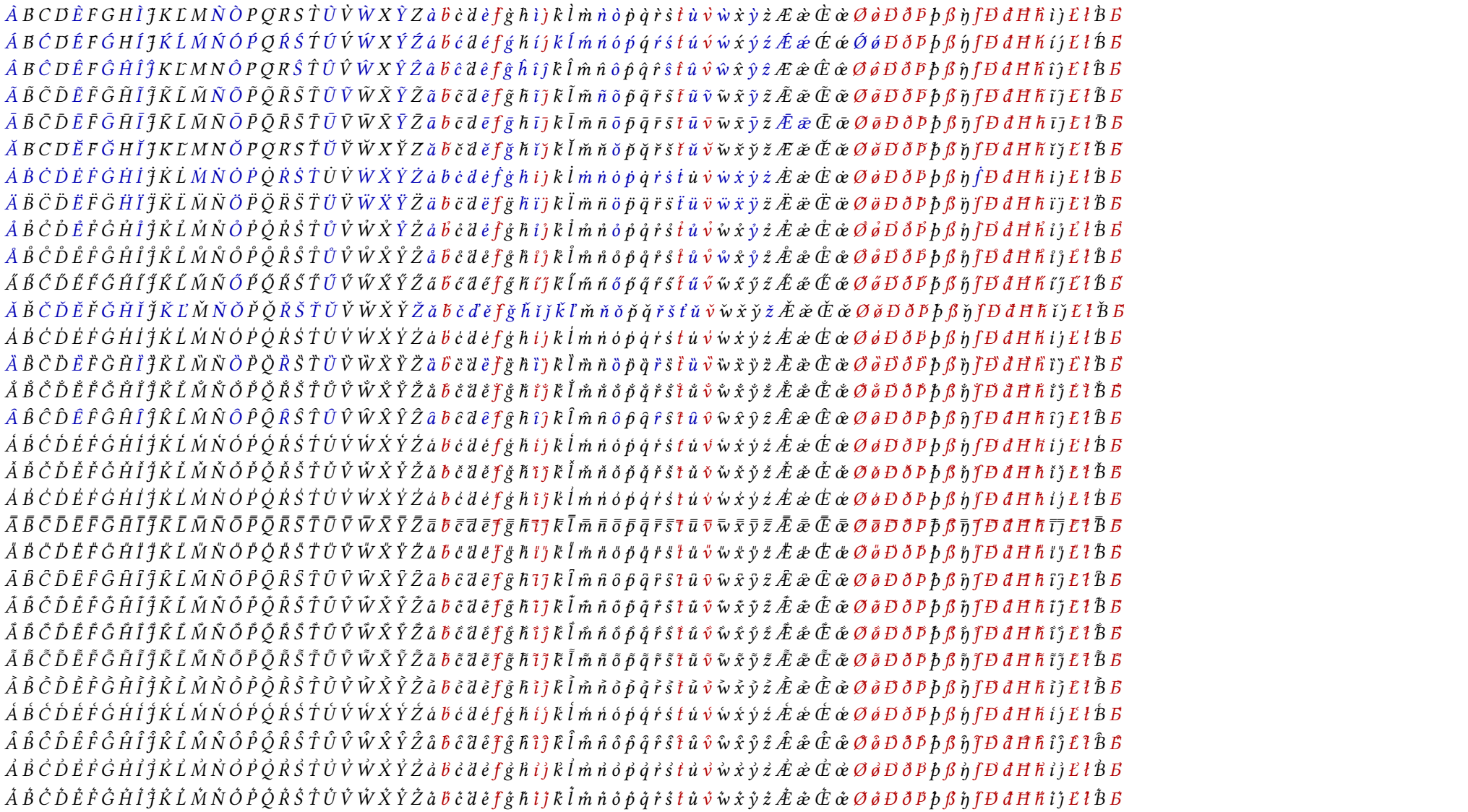
Regular, Latin, Above



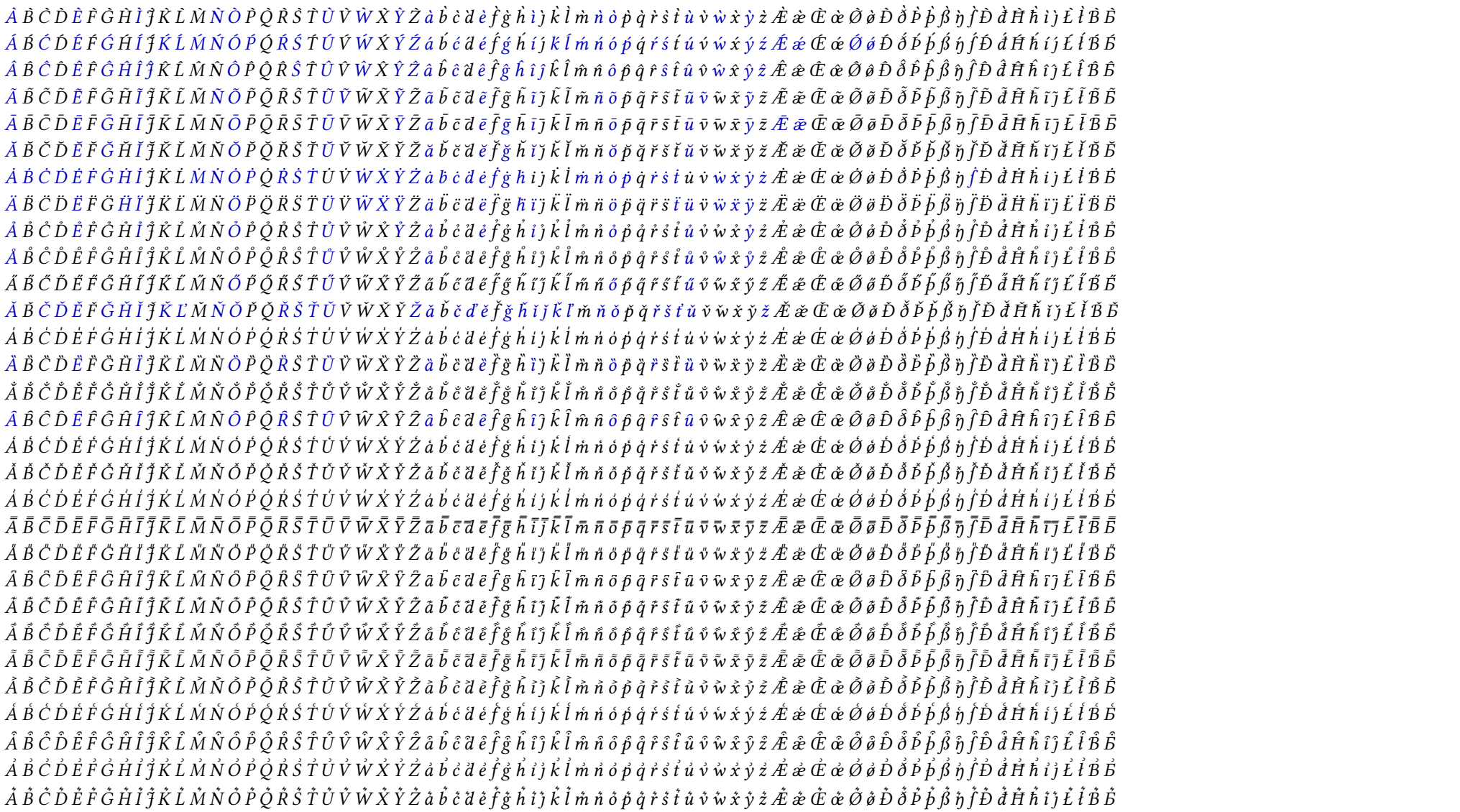
Regular patched, Latin, Above



Italic, Latin, Above



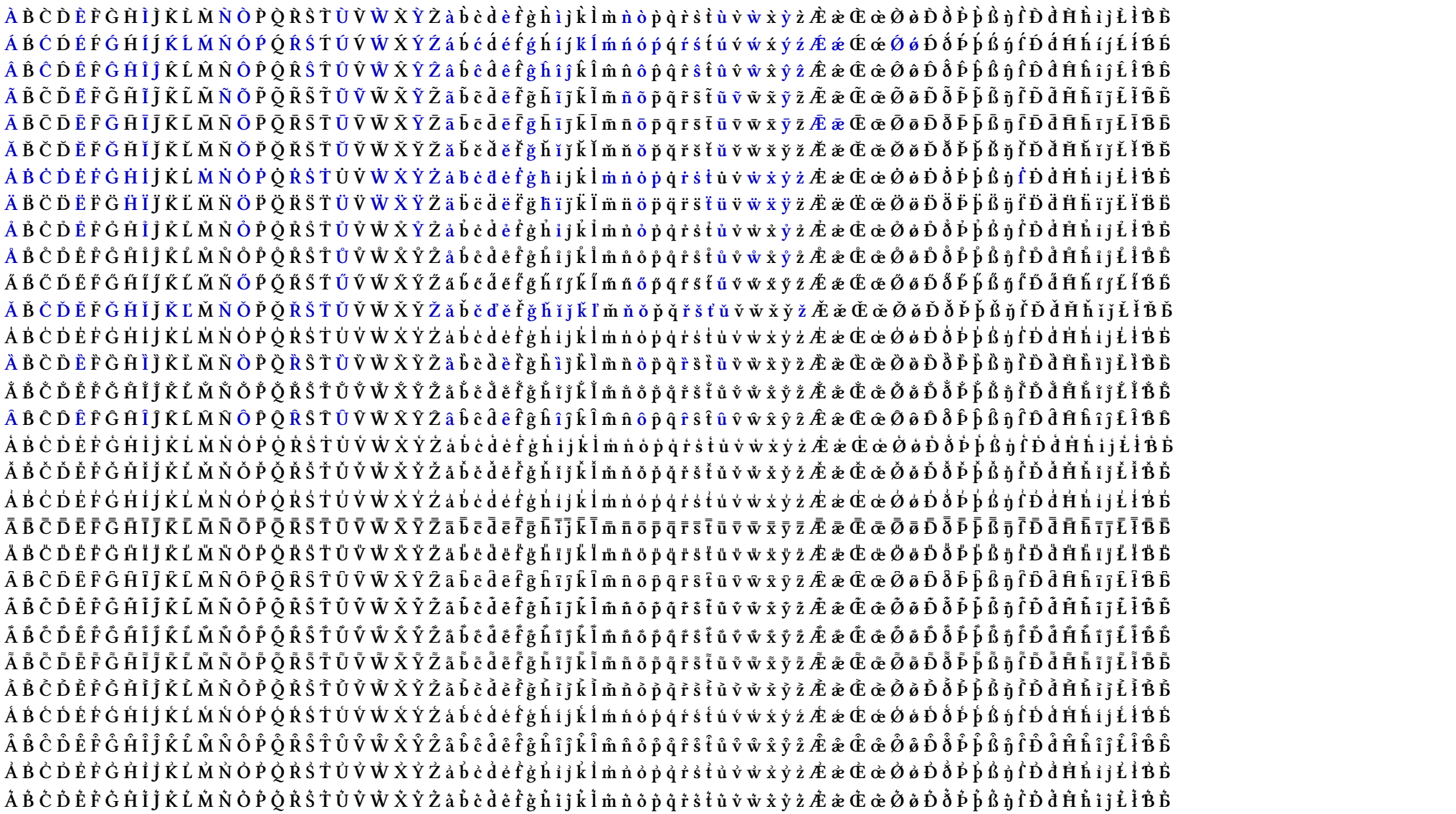
Italic patched, Latin, Above



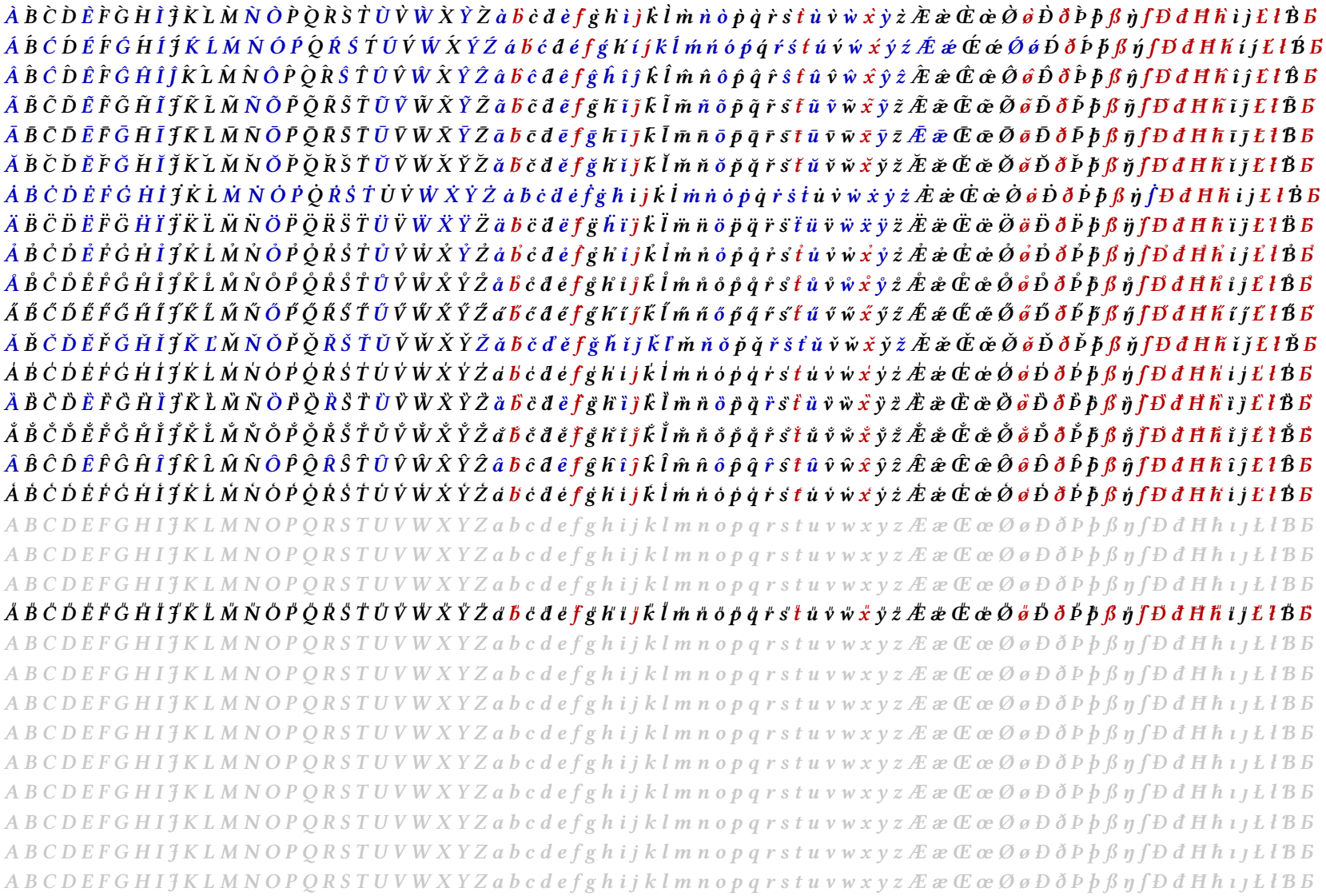
Semibold, Latin, Above



Semibold patched, Latin, Above

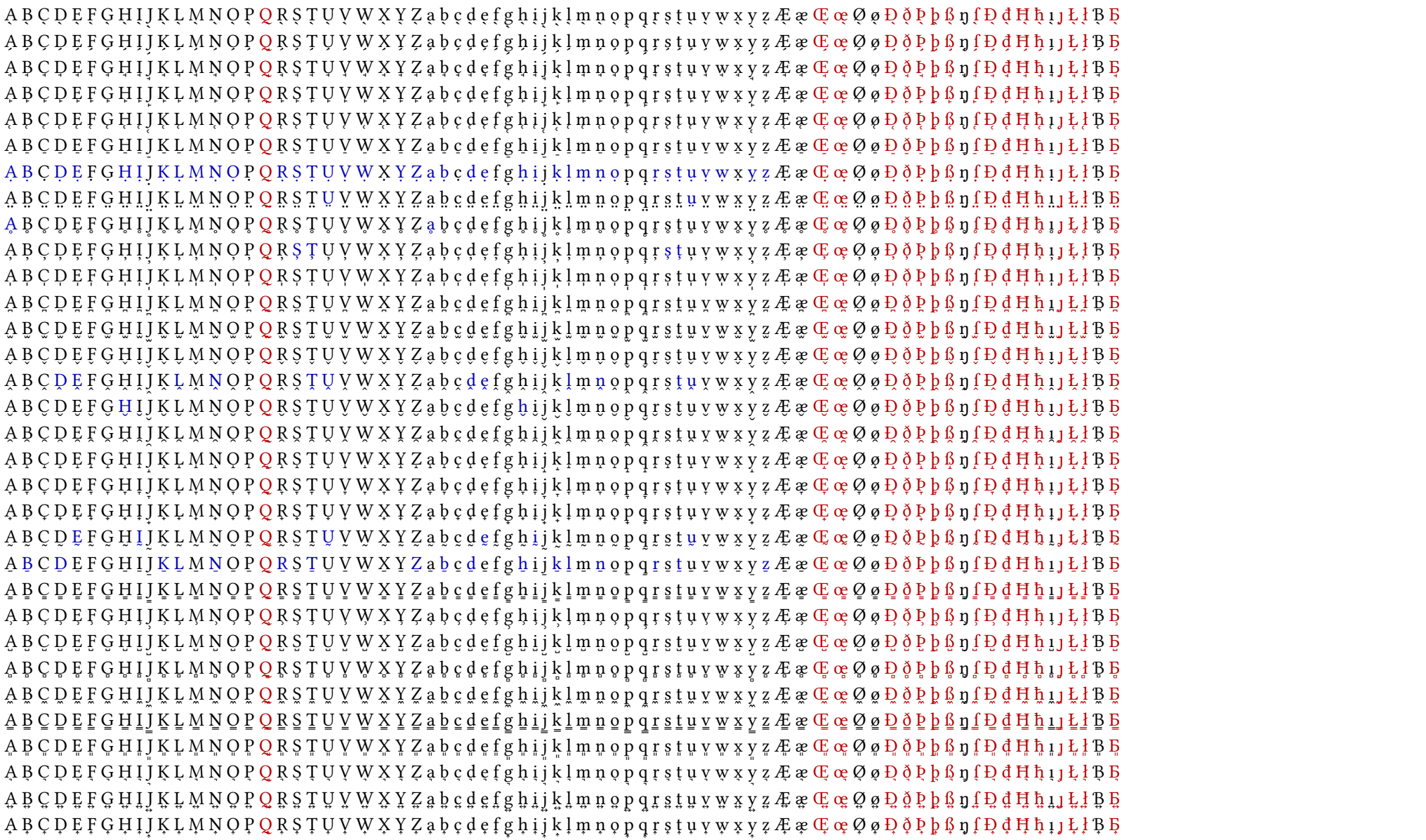


Semibold italic, Latin, Above



Semibold italic patched, Latin, Above

Regular, Latin, Below

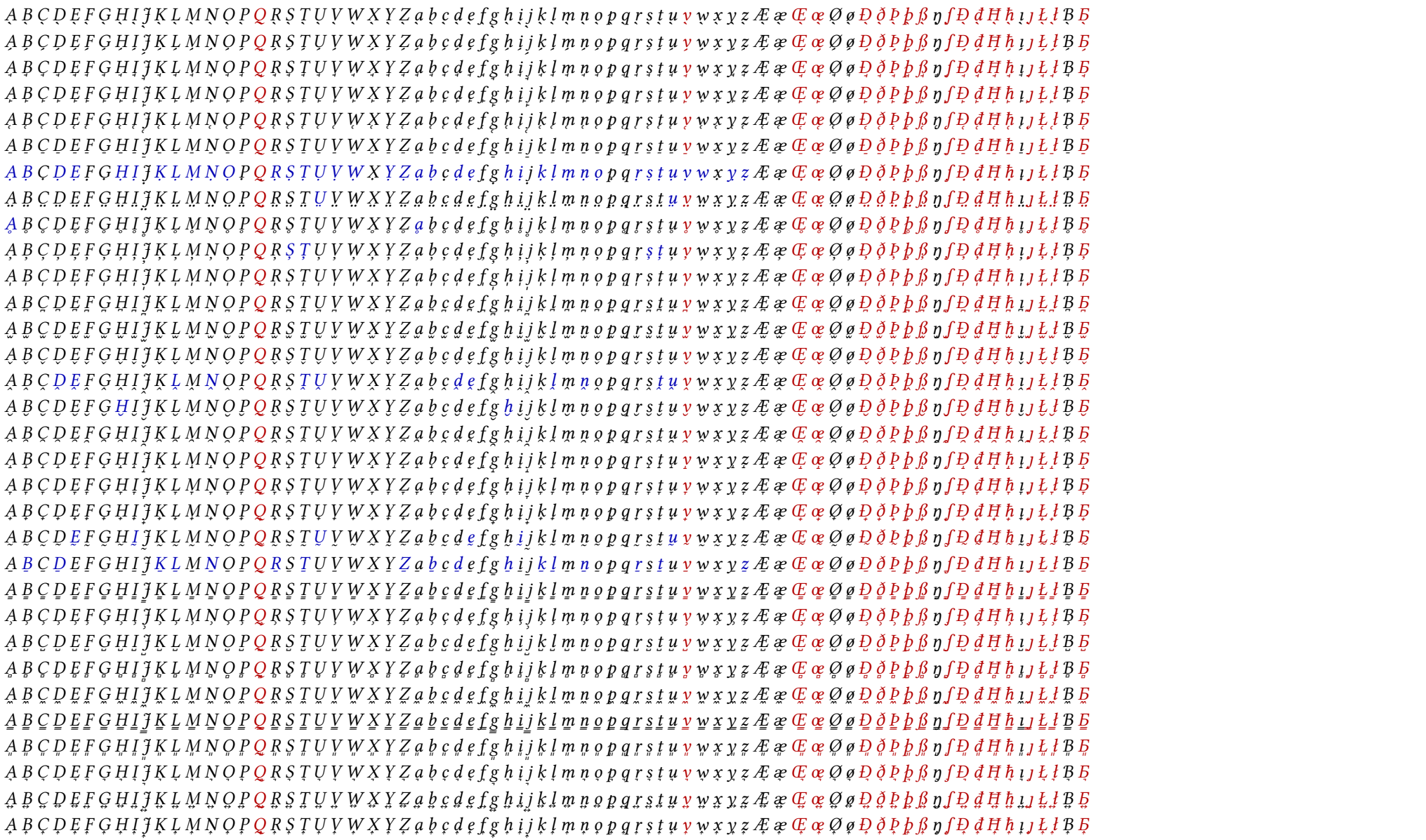


Regular patched, Latin, Below

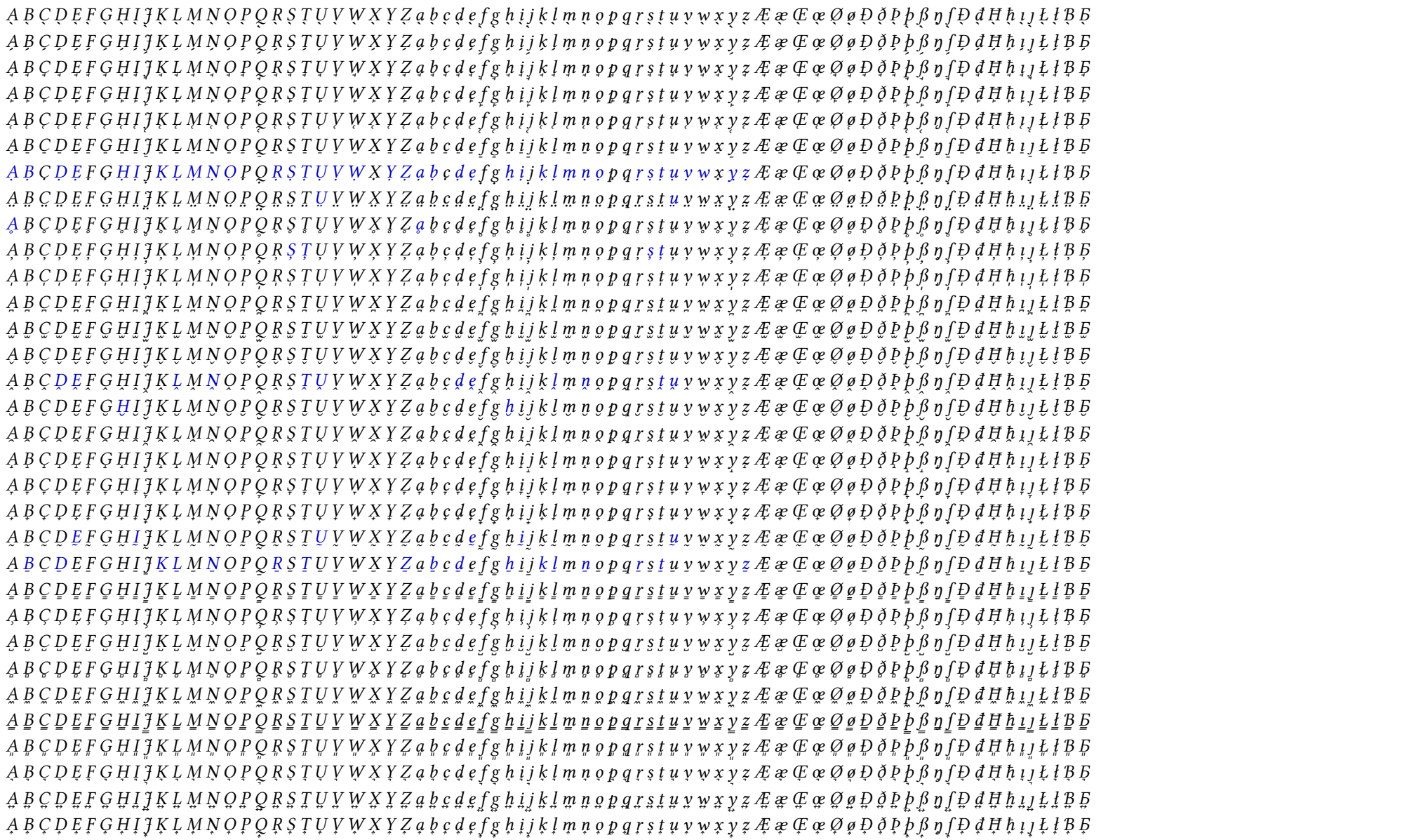
Ạ Ḅ Ç Ḍ Ẹ Ḟ Ḡ Ḥ Ị Ḳ Ḷ Ṃ Ṇ Ọ Ṗ Ǫ Ṛ Ṣ Ṭ Ụ Ṿ Ẉ Ẋ Ỵ Ẓ ạ ḅ ç ḍ ẹ ḟ ḡ ḥ ị ḳ ḷ ṃ ṇ ọ ṗ ǫ ṛ ṣ ṭ ụ ṿ ẉ ẋ ỵ ẓ Æ æ Œ œ Ø ø Ð ð Þ þ ß ŋ ſ Đ đ Ħ ħ ı ȷ Ł ł Ƀ Ɓ

Ạ Ḅ Ç Ḍ Ẹ Ḟ Ḡ Ḥ Ị Ḳ Ḷ Ṃ Ṇ Ọ Ṗ Ǫ Ṛ Ṣ Ṭ Ụ Ṿ Ẉ Ẋ Ỵ Ẓ ạ ḅ ç ḍ ẹ ḟ ḡ ḥ ị ḳ ḷ ṃ ṇ ọ ṗ ǫ ṛ ṣ ṭ ụ ṿ ẉ ẋ ỵ ẓ Æ æ Œ œ Ø ø Ð ð Þ þ ß ŋ ſ Đ đ Ħ ħ ı ȷ Ł ł Ƀ Ɓ

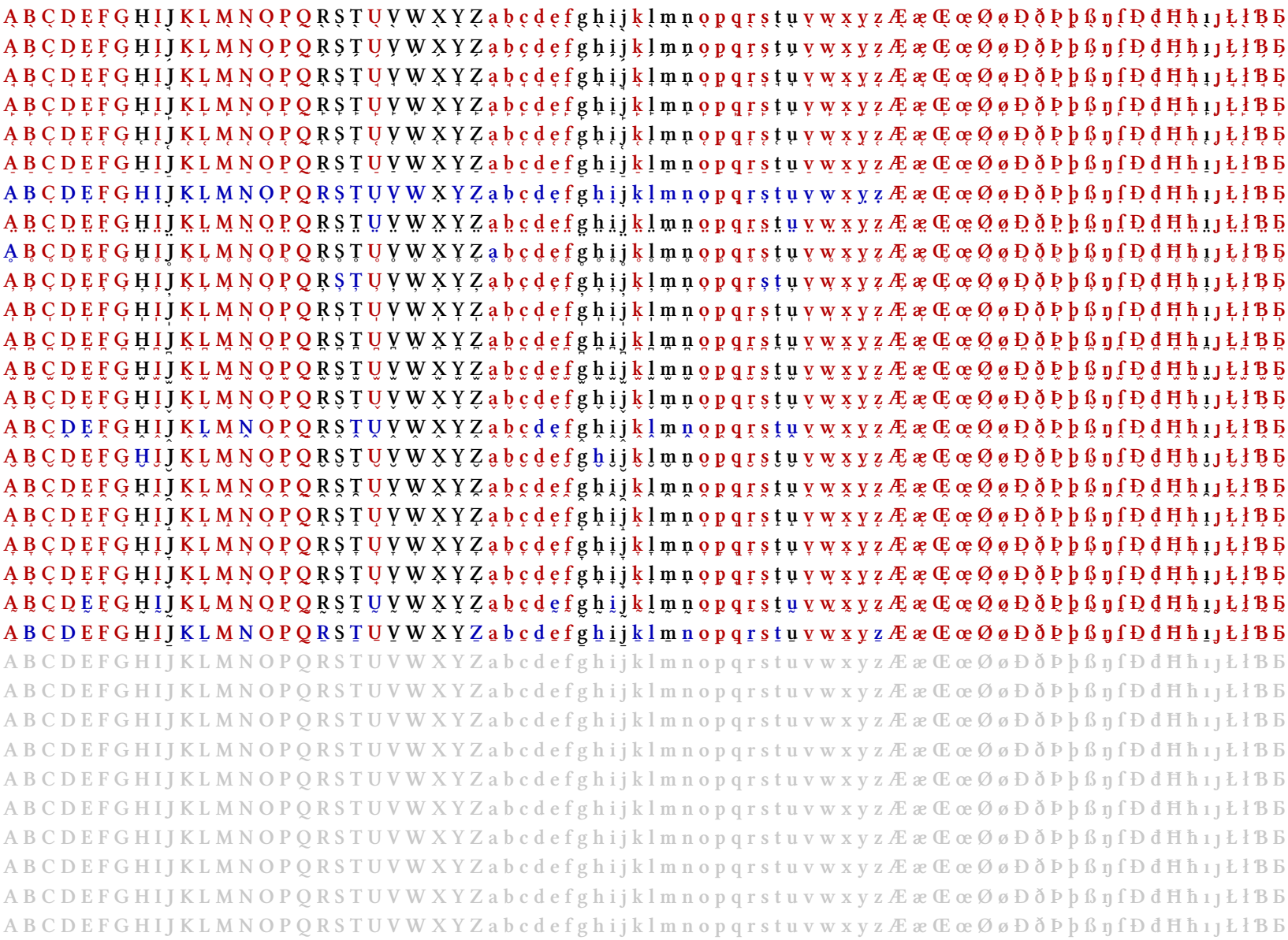
Italic, Latin, Below



Italic patched, Latin, Below



Semibold, Latin, Below

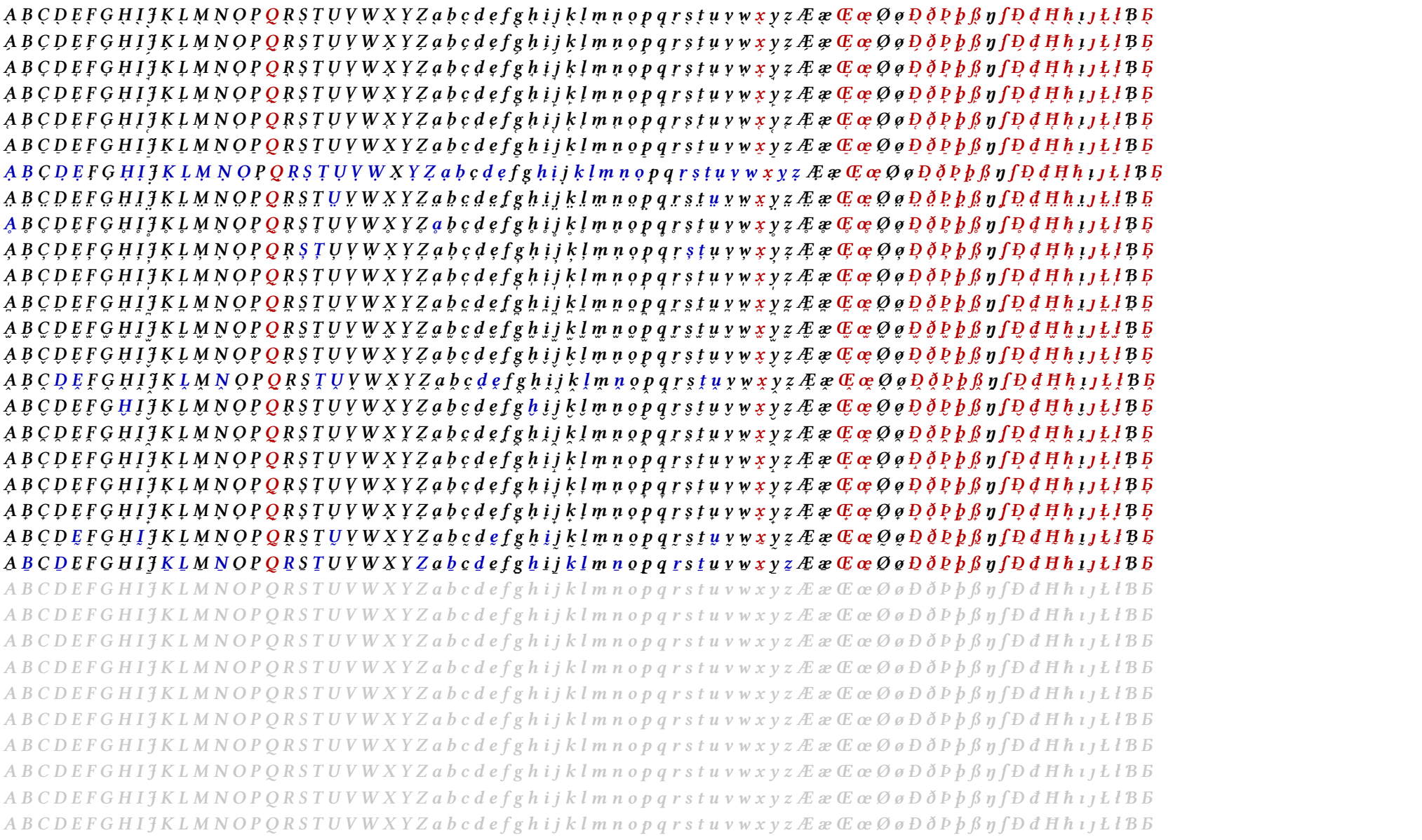


Semibold patched, Latin, Below

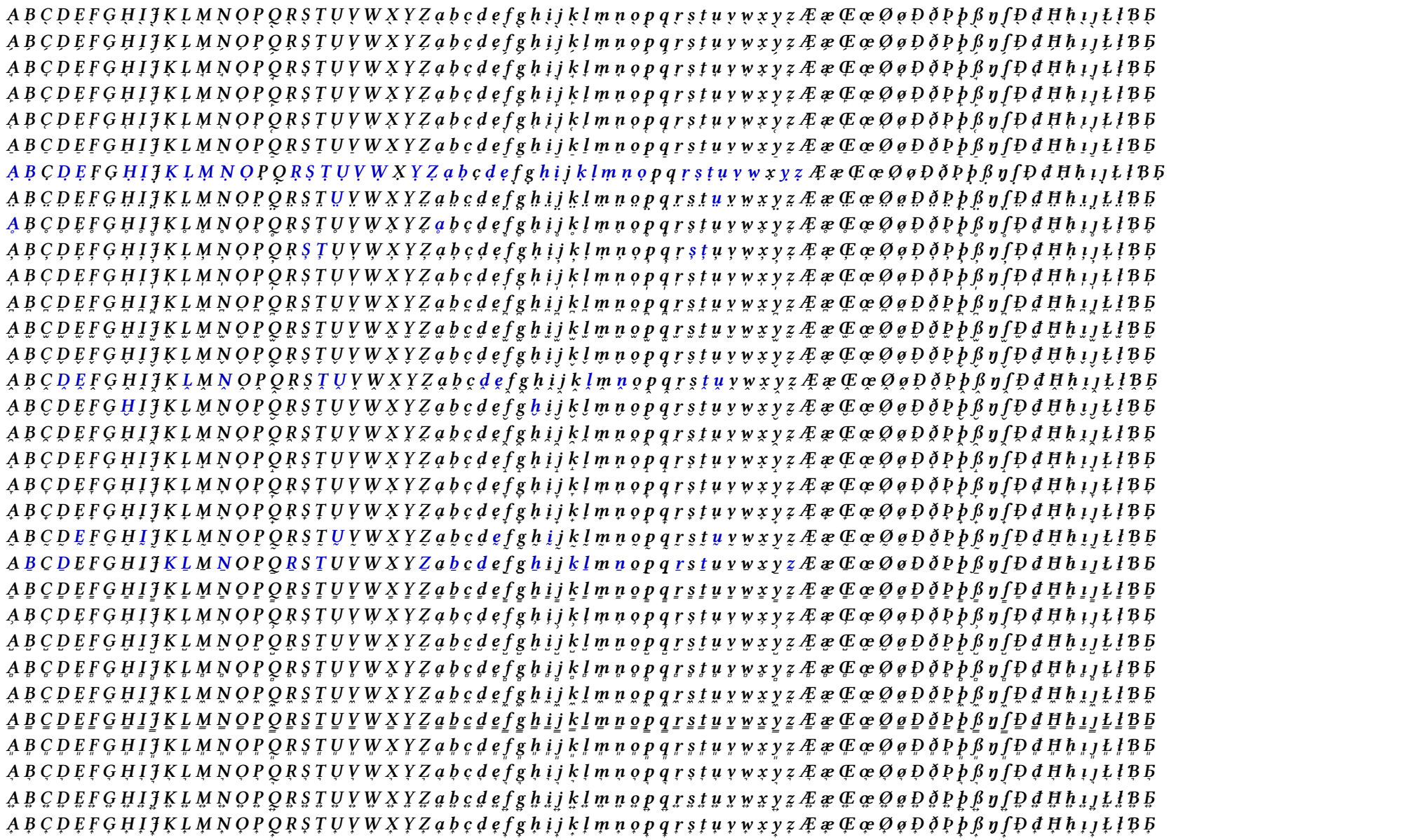
Ạ Ḅ Ḍ Ẹ Ḥ Ị Ḳ Ḷ Ṃ Ṇ Ọ Ṛ Ṣ Ṭ Ụ Ṿ Ẉ Ỵ Ẓ ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ Æ æ Œ œ Ø ø Ð ð Þ þ ß ŋ ʃ Đ đ Ħ ħ ı ȷ Ł ł Ƀ Ƶ

Ạ Ḅ Ḍ Ẹ Ḥ Ị Ḳ Ḷ Ṃ Ṇ Ọ Ṛ Ṣ Ṭ Ụ Ṿ Ẉ Ỵ Ẓ ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ Æ æ Œ œ Ø ø Ð ð Þ þ ß ŋ ʃ Đ đ Ħ ħ ı ȷ Ł ł Ƀ Ƶ

Semibold italic, Latin, Below



Semibold italic patched, Latin, Below



Regular, IPA, Above



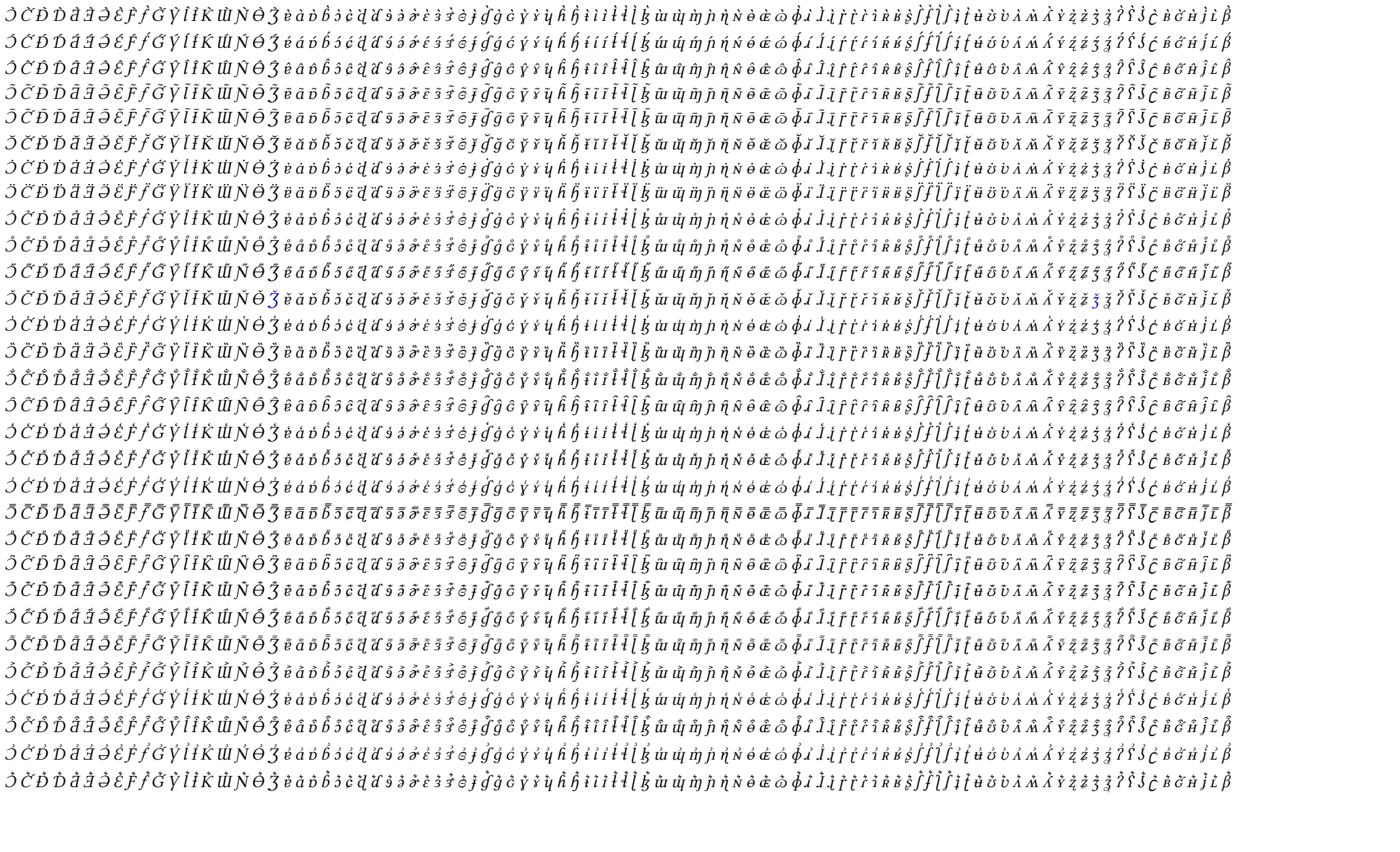
Regular patched, IPA, Above



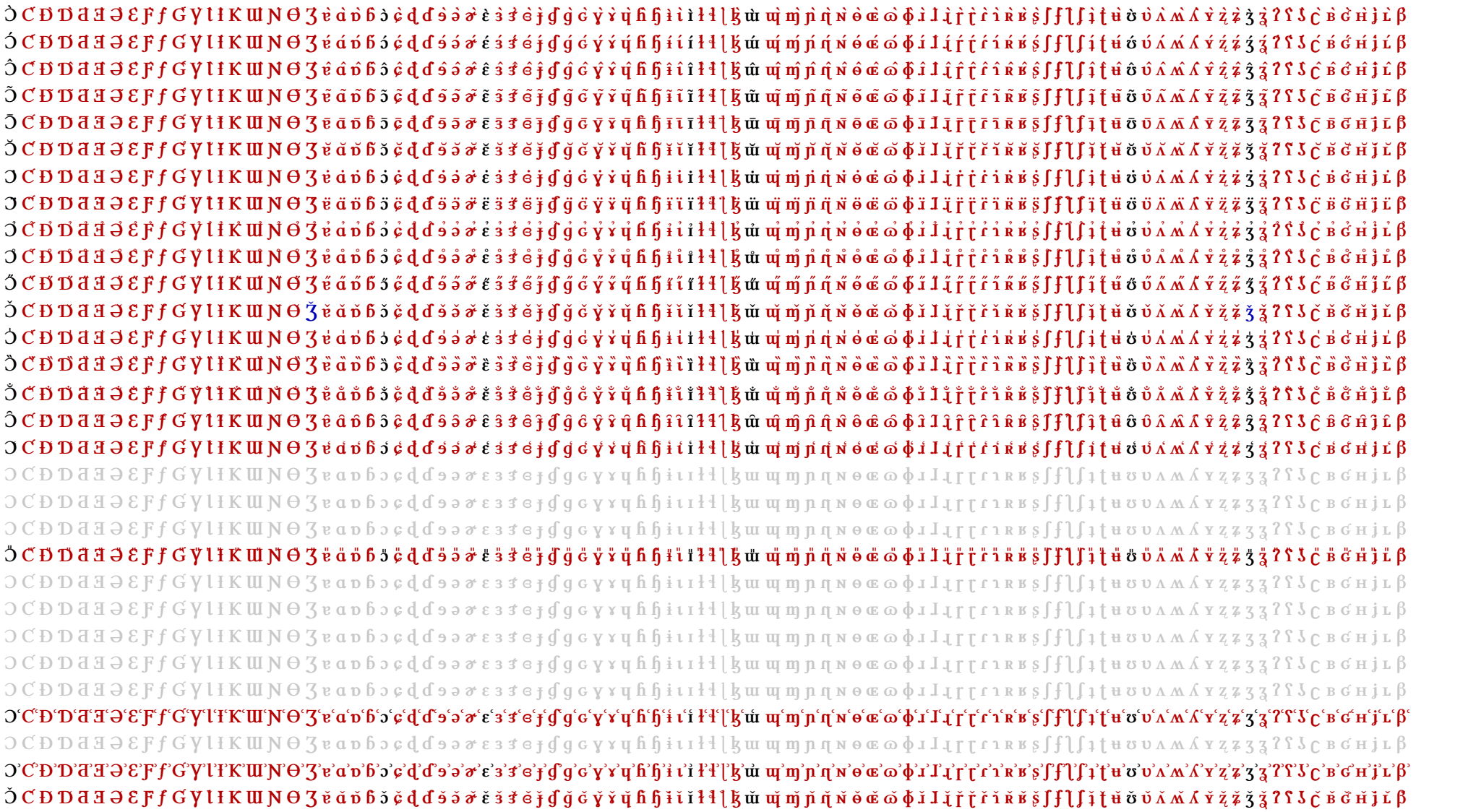
Italic, IPA, Above



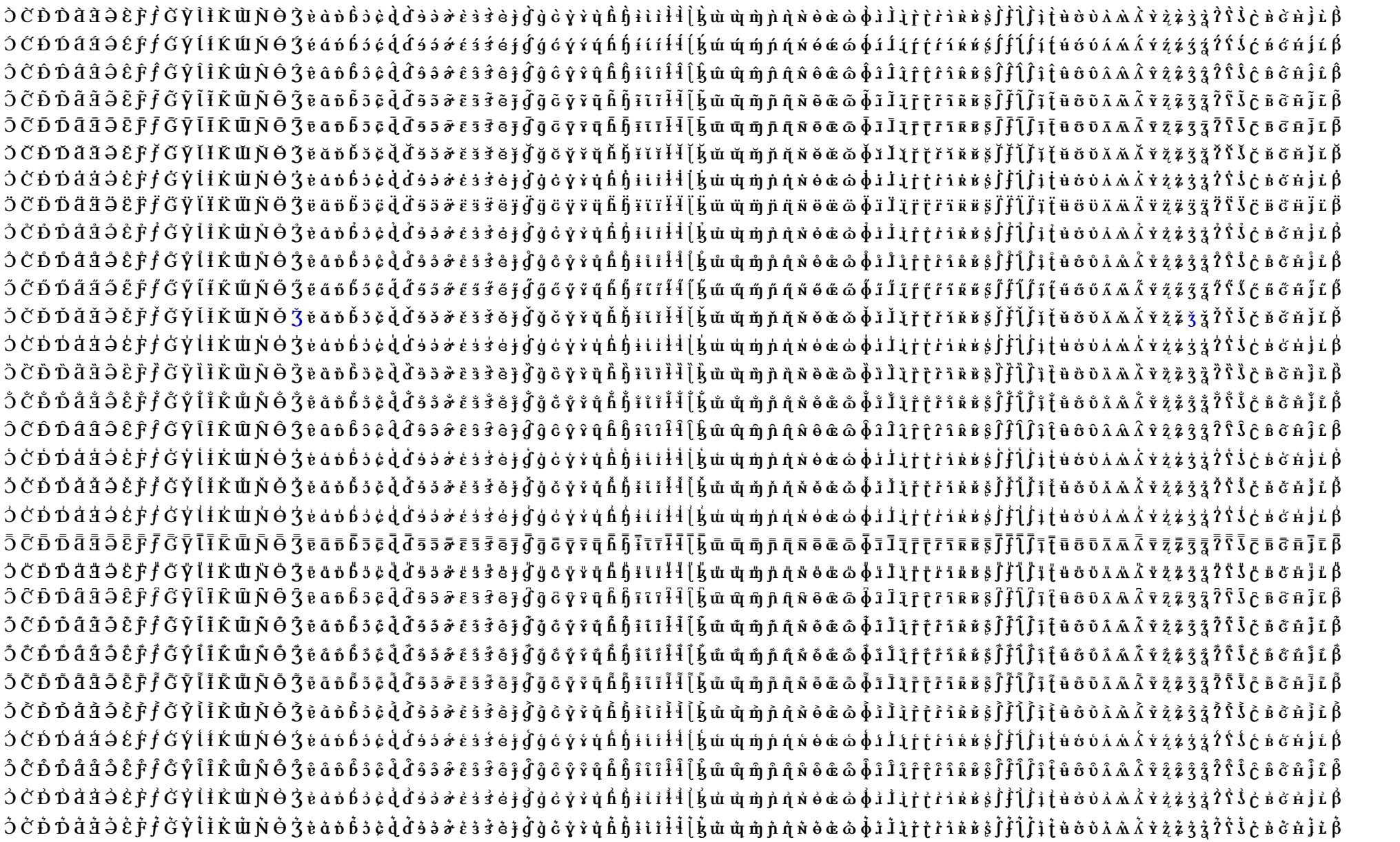
Italic patched, IPA, Above



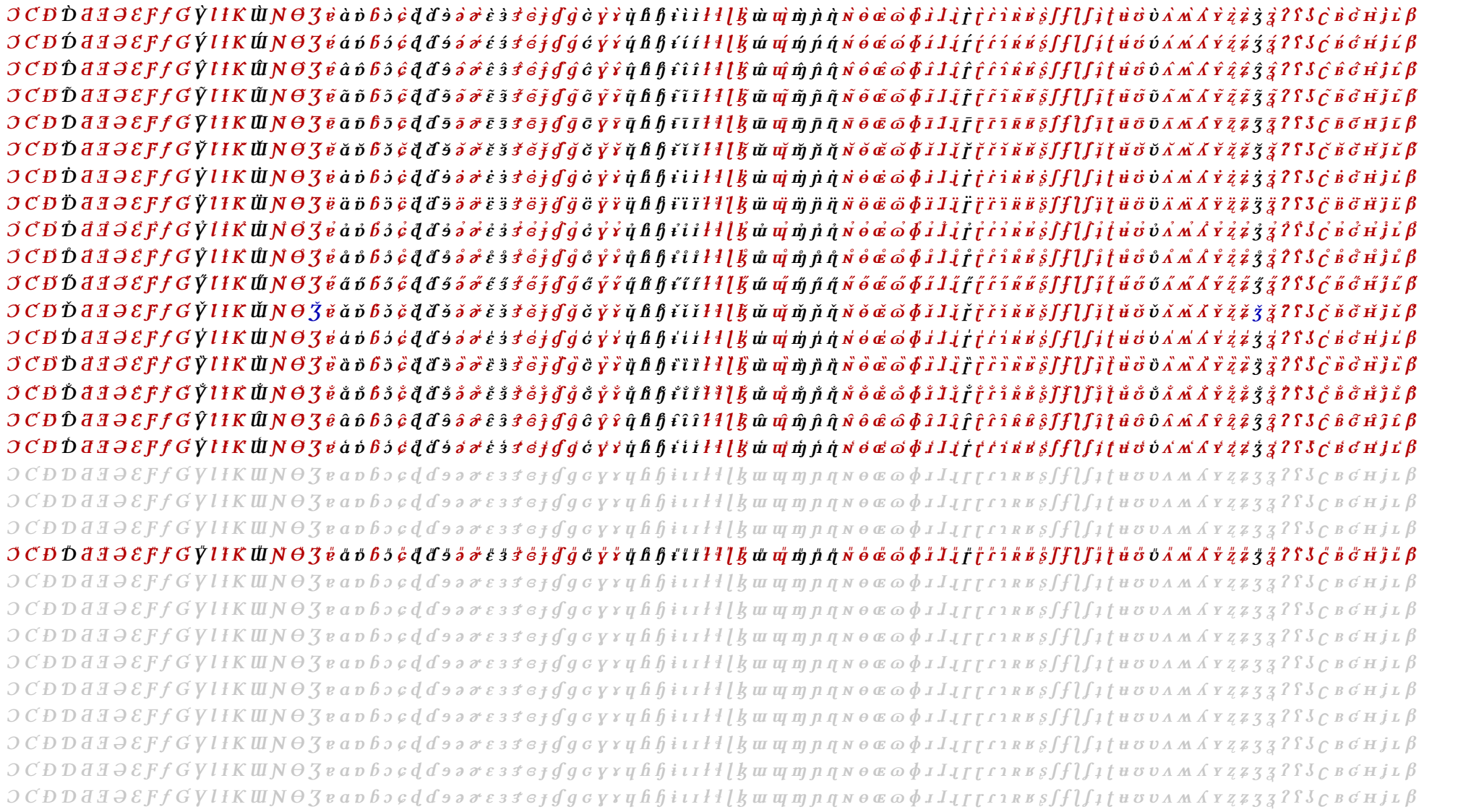
Semibold, IPA, Above



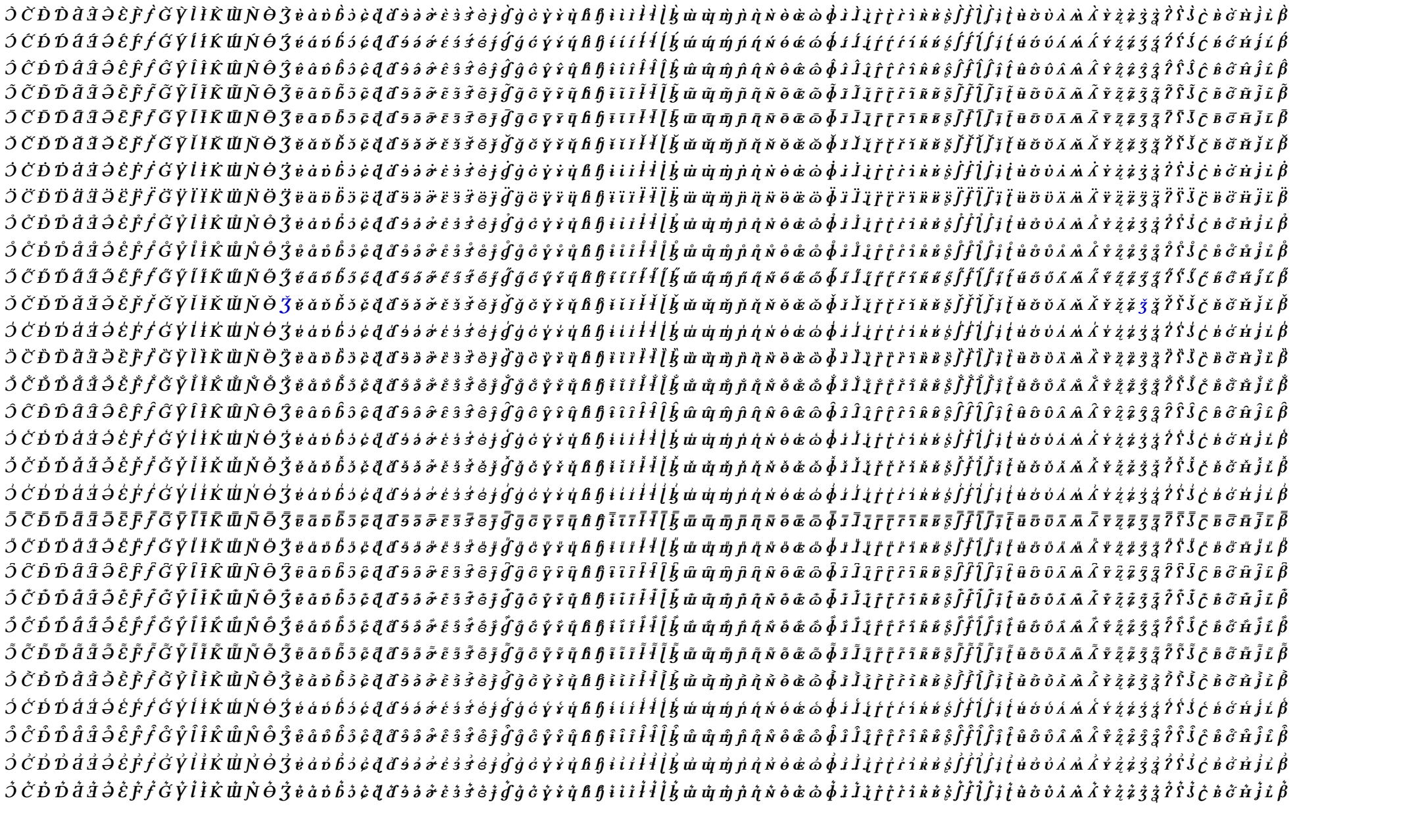
Semibold patched, IPA, Above



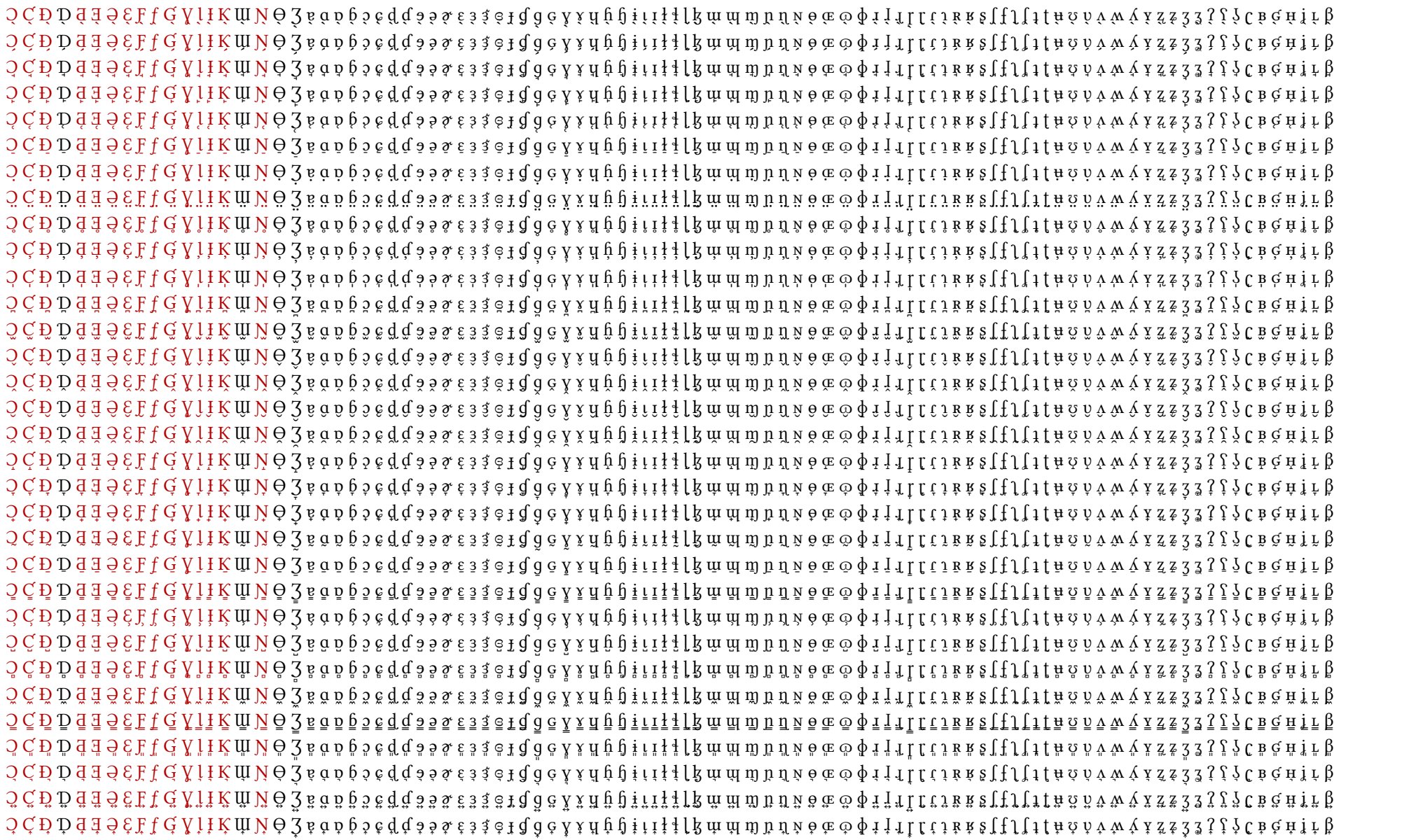
Semibold italic, IPA, Above



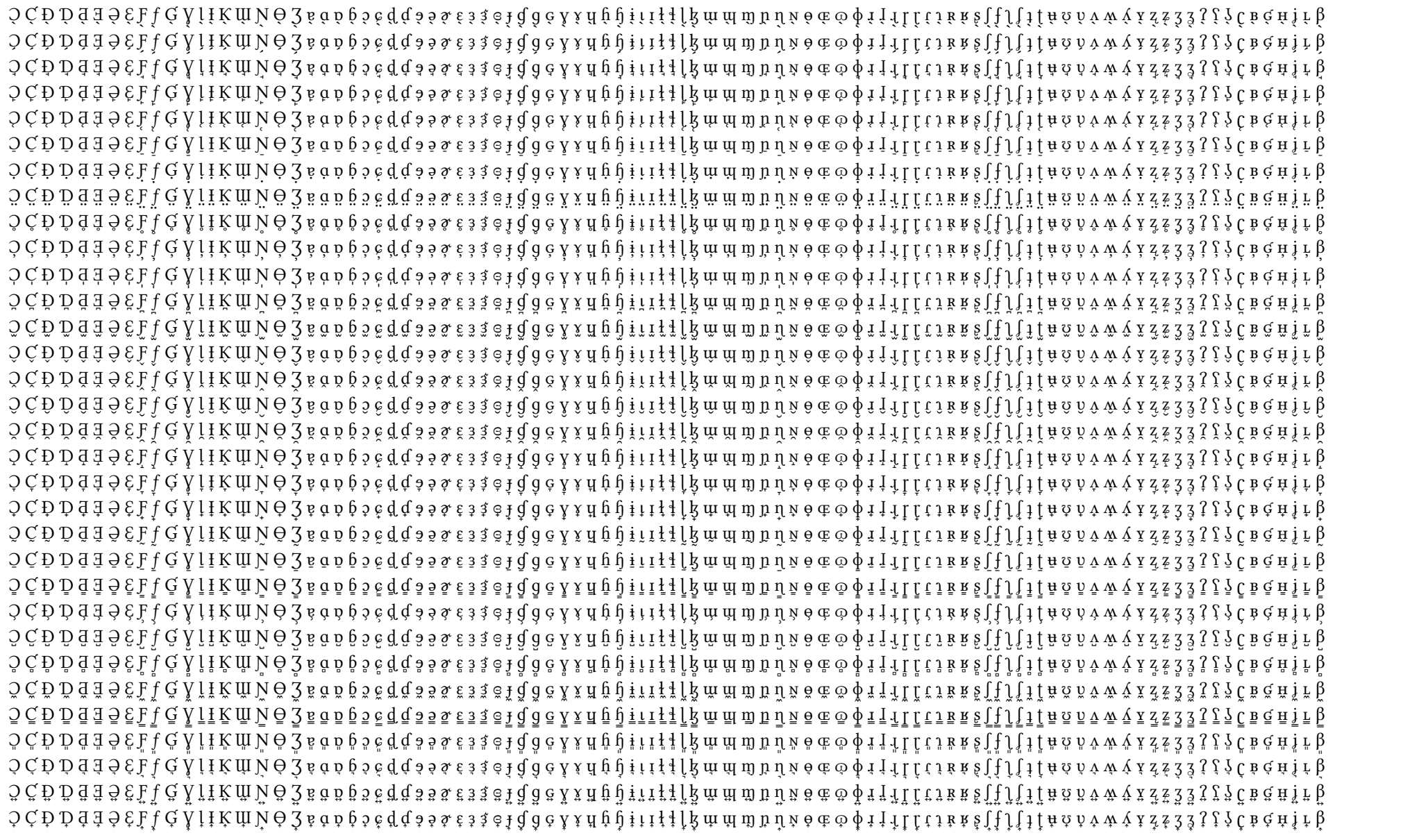
Semibold italic patched, IPA, Above



Regular, IPA, Below



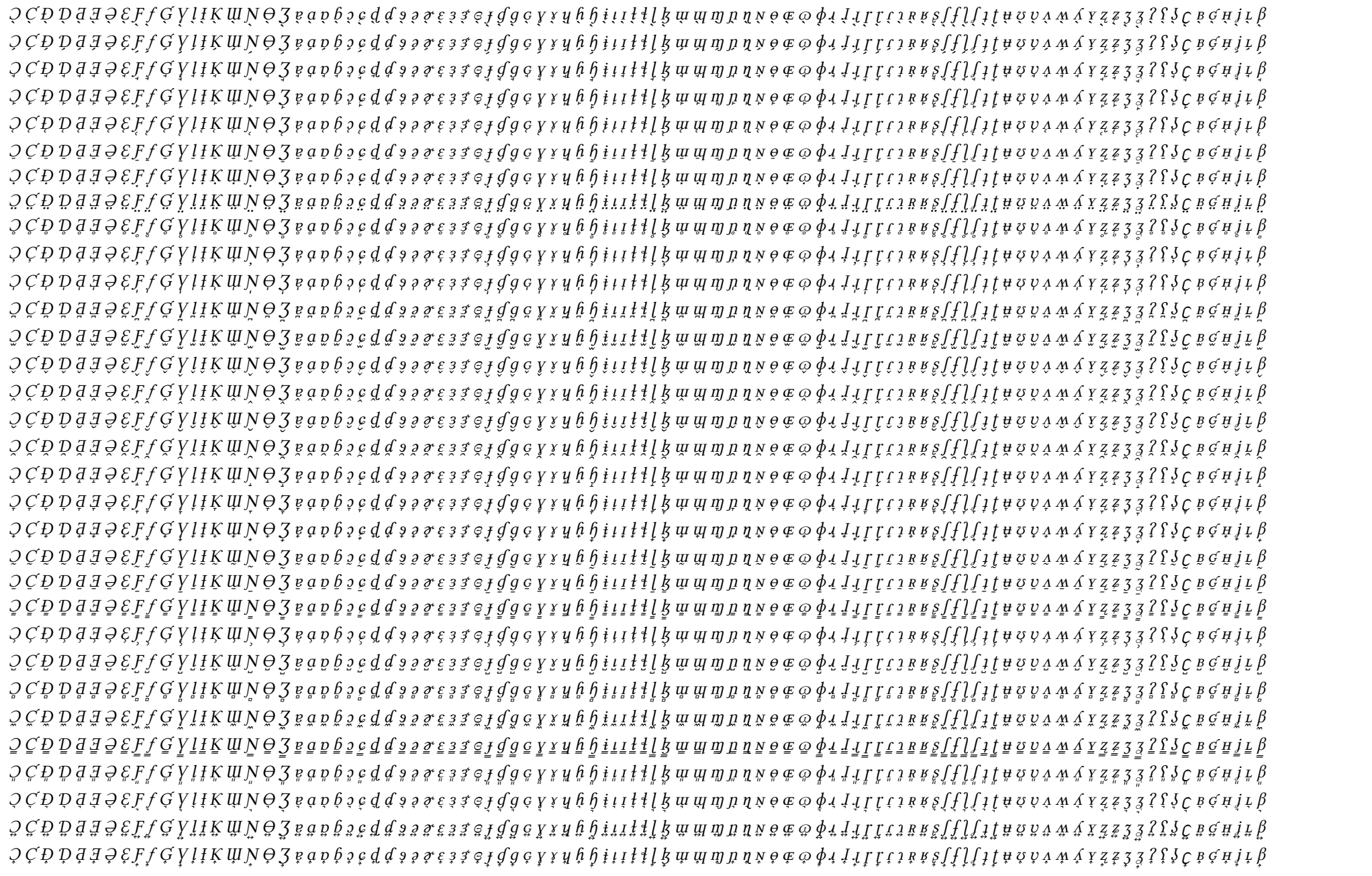
Regular patched, IPA, Below



Italic, IPA, Below

𝛽 𝑡 𝑓 𝐻 𝐵 𝐵 𝐶 𝐶 𝑆 𝑆 𝑍 𝑍 𝑍 𝑍 𝑋 𝑋 𝑌 𝑌 𝑊 𝑊 𝑉 𝑉 𝑂 𝑂 𝐻 𝐻 𝑇 𝑇 𝐽 𝐽 𝑆 𝑆 𝑅 𝑅 𝑈 𝑈 𝐽 𝐽 𝑇 𝑇 𝑃 𝑃 𝜙 𝜙 𝜔 𝜔 𝜖 𝜖 𝑁 𝑁 𝑢 𝑢 𝑈 𝑈 𝑚 𝑚 𝑚 𝑚 𝑚 𝑚 𝑚 𝑚 𝑘 𝑘 𝜆 𝜆 𝜒 𝜒 𝜂 𝜂 𝑦 𝑦 𝑔 𝑔 𝑖 𝑖 𝑖 𝑖 𝑖 𝑖 𝑖 𝑖 𝑙 𝑙 𝑙 𝑙 𝑙 𝑙 𝑙 𝑙 𝑚 𝑚 𝑚 𝑚 𝑚 𝑚 𝑚 𝑚 𝑛 𝑛 𝑛 𝑛 𝑛 𝑛 𝑛 𝑛 𝑜 𝑜 𝜔 𝜔 𝜖 𝜖 𝑝 𝑝 𝑝 𝑝 𝑝 𝑝 𝑝 𝑝 𝑞 𝑞 𝑞 𝑞 𝑞 𝑞 𝑞 𝑞 𝑟 𝑟 𝑟 𝑟 𝑟 𝑟 𝑟 𝑟 𝑠 𝑠 𝑠 𝑠 𝑠 𝑠 𝑠 𝑠 𝑡 𝑡 𝑡 𝑡 𝑡 𝑡 𝑡 𝑡 𝑢 𝑢 𝑢 𝑢 𝑢 𝑢 𝑢 𝑢 𝑣 𝑣 𝑣 𝑣 𝑣 𝑣 𝑣 𝑣 𝑤 𝑤 𝑤 𝑤 𝑤 𝑤 𝑤 𝑤 𝑥 𝑥 𝑥 𝑥 𝑥 𝑥 𝑥 𝑥 𝑦 𝑦 𝑦 𝑦 𝑦 𝑦 𝑦 𝑦 𝑧 𝑧 𝑧

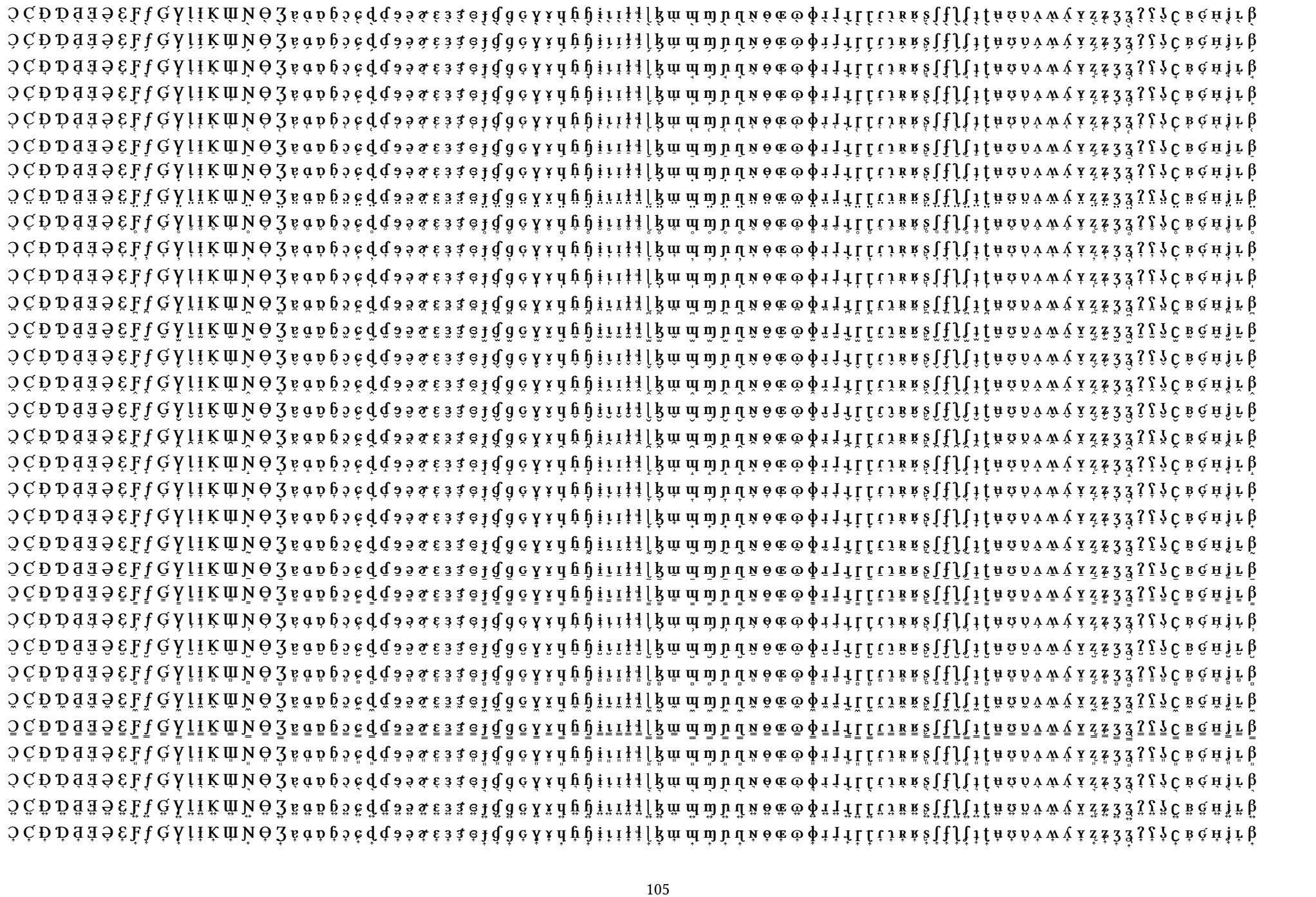
Italic patched, IPA, Below



Semibold, IPA, Below



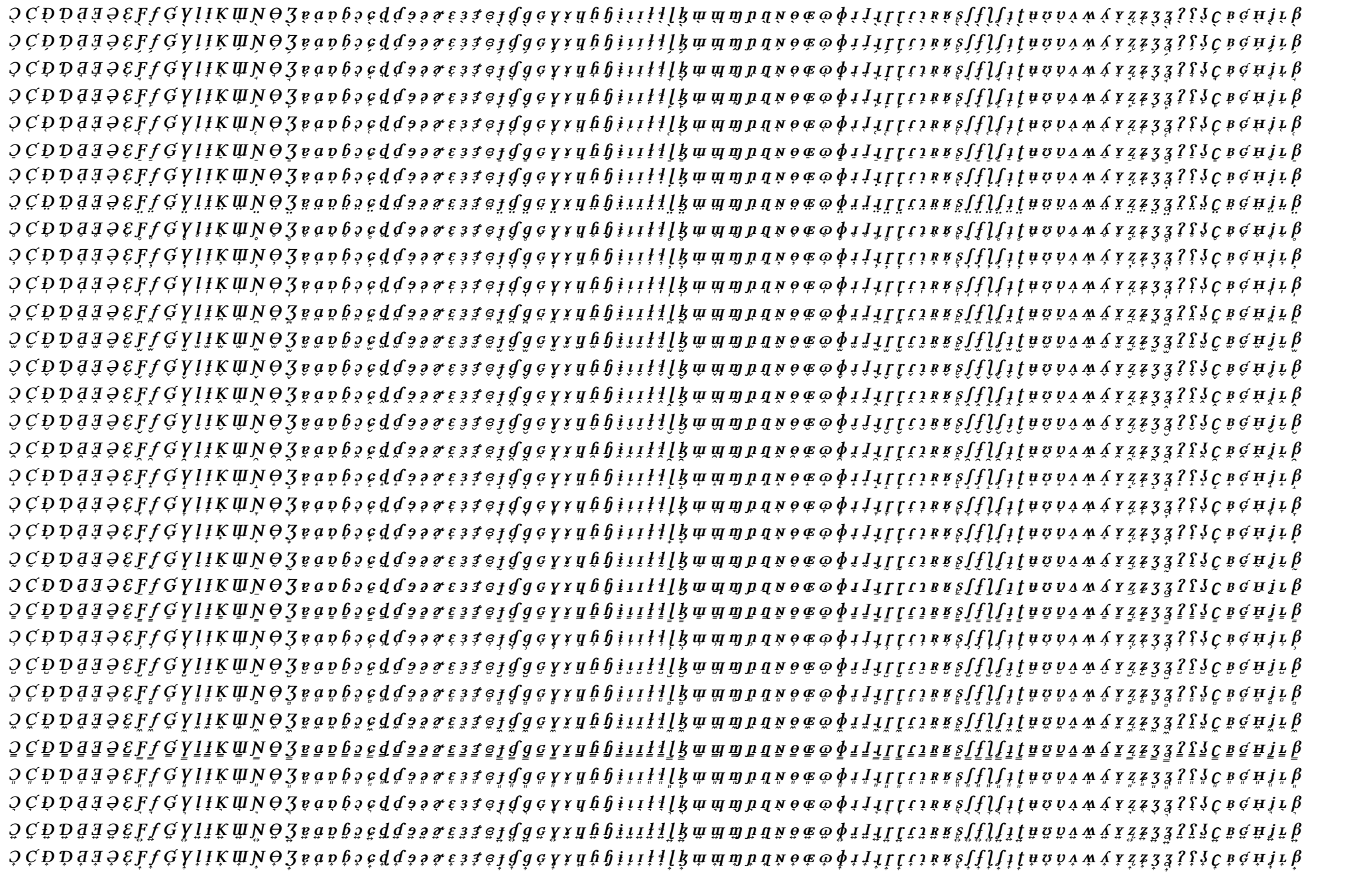
Semibold patched, IPA, Below



Semibold italic, IPA, Below



Semibold italic patched, IPA, Below



Capital alternates

These grids display capitals with alternate marks.

In the patched fonts, the mark anchor of each capital alternate is set by script (`font_patching_capmarks.py`) from the precomposed capitals A–Z. The Y coordinate is the capital anchor reference (850) less the height by which the mark is raised in the precomposed glyphs. The X coordinate of a symmetric mark is the center of its outline on its own axis; for the grave, acute, double acute and double grave it is the median position of the mark in the precomposed glyphs, relative to the midline of the base. In semibold italic both are shifted left by 150 units, as in the semibold italic anchor model. A human-curated anchor overrules the scripted one. Measured over the capitals that have an above-anchor, the median distance of an alternate mark from the midline falls from 142 to 17 units in italic and from 171 to 42 units in semibold italic; regular (16) and semibold (11) are unchanged.

Check these by eye:

- Italic. All ten curated anchors are kept and overrule the script. Their X coordinates differ from the scripted ones by 6 to 62 units: grave -45 (script 17), acute -65 (-37), caron -50 (-29), inverted breve -23 (-2), double grave -47 (-34), breve -10 (-1), circumflex 16 (24), hook -27 (-19), dieresis -75 (-69), double acute -109 (-103). The curated values fit the designer's capital anchors, which are left of the midline; the scripted values assume an anchor on the midline. If the capital anchors are moved to the midline, the curated values should be dropped.
- Regular. The curated grave $(-175, 723)$ and acute $(-210, 723)$ are kept; the script gives -162 and -218. The other eight curated anchors are within 4 units of the script and are commented out in `fontanchors_human`.
- Semibold italic. All anchors are scripted. The marks are still about 35 units left of the midline, because the designer's capital anchors are left of the candidates of the semibold italic anchor model; J and L are far off (their base anchors do not conform).
- Semibold. Only J, O, Q, R, S, T, V, W, X, Y, Z have an above-anchor; on the other capitals the alternates are placed by fallback. The hook above is raised by 89 units, and the double acute and double grave are lowered by 30 and 23 units, relative to the designer's anchors.
- The acute, grave and their doubles. The precomposed glyphs place these by eye, letter by letter, so the implied X varies by up to 190 units over the bases (most for the acute, because of L), and one anchor is a compromise.
- `dotaccent.cap` is new in italic and semibold italic, and in regular its anchor is raised by 50 units to match the precomposed dotted capitals.
- Not used as evidence, because the mark outline differs in the precomposed glyph: Ÿ and Ẅ (dieresis), Ľ (comma-style caron), and Ĵ in semibold italic.

Regular, Capitals, Marks with capital alternates

À B̀ C̀ D̀ È F̀ G̀ H̀ Ì J̀ K̀ L̀ M̀ Ǹ Ò P̀ Q̀ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Z̀ Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̀ Ḥ Ị̀ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̀ Ẹ̀ Ị̀ Ọ̀ Ụ̀
Á B́ Ć D́ É F́ Ǵ H́ Í J́ Ḱ Ĺ Ḿ Ń Ó Ṕ Q́ Ŕ Ś T́ Ú V́ Ẃ X́ Ý Ź Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę́ Ḥ Í Ṛ Ṣ Ṭ Ṿ Ẓ Ạ́ Ẹ́ Í Ọ́ Ụ́
Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̂ Ḥ Ị̂ Ṛ Ṣ Ṭ Ṿ Ẓ Ậ Ệ Ị̂ Ộ Ụ̂
Ă B̆ Č D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ Ǩ L̆ M̆ N̆ Ŏ P̆ Q̆ Ř Š T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̆ Ḥ Ị̆ Ṛ Ṣ Ṭ Ṿ Ẓ Ặ Ệ Ǐ Ọ̆ Ụ̆
Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J̇ K̇ L̇ Ṁ Ṅ Ȯ Ṗ Q̇ Ṙ Ṡ Ṫ U̇ V̇ Ẇ Ẋ Ẏ Ż Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̇ Ḥ Ị̇ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̇ Ẹ̇ İ Ọ̇ Ụ̇
Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̈ Ḥ Ị̈ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̈ Ẹ̈ Ï Ọ̈ Ụ̈
Å B̊ C̊ D̊ E̊ F̊ G̊ Ḧ I̊ J̊ K̊ L̊ M̊ N̊ O̊ P̊ Q̊ R̊ S̊ T̊ Ů V̊ W̊ Ẍ Y̊ Ż Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̊ Ḥ Ị̊ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̊ Ẹ̊ İ̊ Ọ̊ Ụ̊
Ő B̋ C̋ D̋ Ű F̋ G̋ H̋ Ő J̋ K̋ L̋ M̋ N̋ Ő P̋ Q̋ R̋ S̋ T̋ Ű V̋ W̋ X̋ Ÿ Ż Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̋ Ḥ Ị̋ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̋ Ẹ̋ Ő Ọ̋ Ụ̋
Ǎ B̌ Č Ď Ě F̌ Ǧ Ȟ Ǐ J̌ Ǩ Ľ M̌ Ň Ǒ P̌ Q̌ Ř Š Ť Ǔ V̌ W̌ X̌ Y̌ Ž Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̌ Ḥ Ị̌ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̌ Ẹ̌ Ǐ Ọ̌ Ụ̌
Ȁ B̏ C̏ D̏ Ȅ F̏ G̏ Ȟ Ȉ J̏ K̏ L̏ M̏ N̏ Ȍ P̏ Q̏ Ȑ S̏ T̏ Ȕ V̏ W̏ X̏ Ÿ Ż Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̏ Ḥ Ị̏ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̏ Ẹ̏ Ȉ Ọ̏ Ụ̏
Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̂ Ḥ Ị̂ Ṛ Ṣ Ṭ Ṿ Ẓ Ậ Ệ Î Ộ Ụ̂

Regular patched, Capitals, Marks with capital alternates

À B̀ C̀ D̀ È F̀ G̀ H̀ Ì J̀ K̀ L̀ M̀ Ǹ Ò P̀ Q̀ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Z̀ Ɔ̀ ə̀ ɛ̀ ɵ̀ Ɛ̀ ʊ̀ Ù Ỳ Ƶ̀ Ḛ̀ Ḥ̀ Į̀ Ḛ̀ Ṣ̀ T̀ V̀ Ƶ̀ Ạ̀ Ḛ̀ Į̀ Ọ̀ Ụ̀
Á B́ Ć D́ É F́ Ǵ H́ Í J́ Ḱ Ĺ Ḿ Ń Ó Ṕ Q́ Ŕ Ś T́ Ú V́ Ẃ X́ Ý Ź Ɔ́ ə́ ɛ́ ɵ́ Ɛ́ ʊ́ Ú Ý Ƶ́ Ḛ́ Ḥ́ Į́ Ḛ́ Ṣ́ T́ V́ Ƶ́ Ạ́ Ḛ́ Į́ Ọ́ Ụ́
Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ ə̂ ɛ̂ ɵ̂ Ɛ̂ ʊ̂ Û Ŷ Ƶ̂ Ḛ̂ Ḥ̂ Į̂ Ḛ̂ Ṣ̂ T̂ V̂ Ƶ̂ Ậ Ḛ̂ Į̂ Ộ Ụ̂
Ă B̆ Č D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ Ǩ L̆ M̆ N̆ Ŏ P̆ Q̆ Ř Š T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Ɔ̆ ə̆ ɛ̆ ɵ̆ Ɛ̆ ʊ̆ Ŭ Y̆ Ƶ̆ Ḛ̆ Ḥ̆ Į̆ Ḛ̆ Ṣ̆ T̆ V̆ Ƶ̆ Ặ Ḛ̆ Į̆ Ọ̆ Ụ̆
Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J̇ K̇ L̇ Ṁ Ṅ Ȯ Ṗ Q̇ Ṙ Ṡ Ṫ U̇ V̇ Ẇ Ẋ Ẏ Ż Ɔ̇ ə̇ ɛ̇ ɵ̇ Ɛ̇ ʊ̇ U̇ Ẏ Ƶ̇ Ḛ̇ Ḥ̇ Į̇ Ḛ̇ Ṩ Ṫ V̇ Ƶ̇ Ạ̇ Ḛ̇ Į̇ Ọ̇ Ụ̇
Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ̈ ə̈ ɛ̈ ɵ̈ Ɛ̈ ʊ̈ Ü Ÿ Ƶ̈ Ḛ̈ Ḥ̈ Į̈ Ḛ̈ Ṣ̈ T̈ V̈ Ƶ̈ Ạ̈ Ḛ̈ Į̈ Ọ̈ Ụ̈
Å B̊ C̊ D̊ E̊ F̊ G̊ H̊ I̊ J̊ K̊ L̊ M̊ N̊ O̊ P̊ Q̊ R̊ S̊ T̊ Ů V̊ Ẇ Ẋ Ẏ Ż Ɔ̊ ə̊ ɛ̊ ɵ̊ Ɛ̊ ʊ̊ Ů Ẏ Ƶ̊ Ḛ̊ Ḥ̊ Į̊ Ḛ̊ Ṣ̊ T̊ V̊ Ƶ̊ Ạ̊ Ḛ̊ Į̊ Ọ̊ Ụ̊
Ő B̋ Ć D̋ Ő F̋ Ǵ H̋ Í J̋ K̋ L̋ M̋ Ń Ő P̋ Q̋ R̋ S̋ T̋ Ű V̋ W̋ X̋ Ÿ Ź Ɔ̋ ə̋ ɛ̋ ɵ̋ Ɛ̋ ʊ̋ Ű Ÿ Ƶ̋ Ḛ̋ Ḥ̋ Į̋ Ḛ̋ Ṣ̋ T̋ V̋ Ƶ̋ Ạ̋ Ḛ̋ Į̋ Ọ̋ Ụ̋
Ă B̆ Č D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ Ǩ Ľ M̆ N̆ Ŏ P̆ Q̆ Ř Š T̆ Ŭ V̆ W̆ X̆ Ÿ Ž Ɔ̆ ə̆ ɛ̆ ɵ̆ Ɛ̆ ʊ̆ Ŭ Ÿ Ƶ̆ Ḛ̆ Ḥ̆ Į̆ Ḛ̆ Ṣ̆ T̆ V̆ Ƶ̆ Ặ Ḛ̆ Į̆ Ọ̆ Ụ̆
Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ̈ ə̈ ɛ̈ ɵ̈ Ɛ̈ ʊ̈ Ü Ÿ Ƶ̈ Ḛ̈ Ḥ̈ Į̈ Ḛ̈ Ṣ̈ T̈ V̈ Ƶ̈ Ạ̈ Ḛ̈ Į̈ Ọ̈ Ụ̈
Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ ə̂ ɛ̂ ɵ̂ Ɛ̂ ʊ̂ Û Ŷ Ƶ̂ Ḛ̂ Ḥ̂ Į̂ Ḛ̂ Ṣ̂ T̂ V̂ Ƶ̂ Ậ Ḛ̂ Į̂ Ộ Ụ̂

Italic, Capitals, Marks with capital alternates

Â B C D È F G H Ì J K L M Ǹ Ò P Q R S T̀ Ù V̀ Ẁ X Ỳ Z Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą Ḛ Ị Ǫ Ṵ

Á B Ć D É F Ǵ H Í J Ḱ Ĺ Ḿ Ń Ó Ṕ Q Ŕ Ś T́ Ú V́ Ẃ X Ý Ź Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą Ḛ Ị Ǫ Ṵ

Â B Ĉ D Ê F Ĝ Ĥ Î J̃ K L̂ M N̂ Ô P Q R Ŝ T̂ Û V̂ Ŵ X Ŷ Ẑ Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą̂ Ḛ̂ Ị Ǫ̂ Ṵ

Ă B C D Ĕ F Ğ H Ǐ J K L M N Ŏ P Q R S Ť Ŭ V̆ W̆ X Y̆ Z Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą̆ Ḛ̆ Ị Ǫ̆ Ṵ

Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J̇ K̇ L̇ Ṁ Ṅ Ȯ Ṗ Q̇ Ṙ Ṡ Ṫ U̇ V̇ Ẇ Ẋ Ẏ Ż Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą Ḛ Ị Ǫ Ṵ

Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą Ḛ Ị Ǫ Ṵ

Ả Ḃ Ċ Ḋ Ẻ Ḟ Ġ Ḥ Ỉ J̇ K̇ Ḷ Ṁ Ṅ Ỏ Ṗ Q̇ Ṙ Ṡ Ṫ Ủ V̇ Ẇ Ẋ Ỷ Ż Ɔ Ə Ɛ Θ Σ Ʊ Ủ Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą Ḛ Ị Ǫ Ṵ

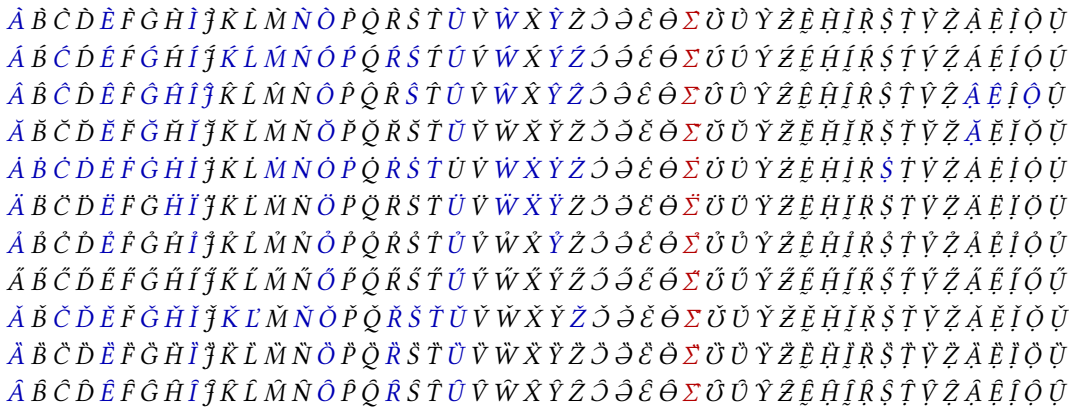
Á B̋ C̋ D̋ É F̋ G̋ H̋ Í J̋ K̋ L̋ M̋ N̋ Ő P̋ Q̋ R̋ S̋ T̋ Ű V̋ W̋ X̋ Ӳ Ż Ɔ Ə Ɛ Θ Σ Ʊ Ű Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą Ḛ Ị Ǫ Ṵ

Ă B̌ C̆ D̆ Ě F̌ Ǧ H̆ Ǐ J̌ K̆ Ľ M̆ N̆ Ǒ P̌ Q̆ Ř Š Ť Ǔ V̆ W̆ X̆ Y̆ Ž Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą Ḛ Ị Ǫ Ṵ

Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą Ḛ Ị Ǫ Ṵ

Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î J̃ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ę Ḥ Į Ḳ Ṣ Ṭ Ṿ Ẓ Ą Ḛ Ị Ǫ Ṵ

Italic patched, Capitals, Marks with capital alternates



Semibold, Capitals, Marks with capital alternates

À B C D È F G H Ì J K L M N Ò P Q R S T Ù V W X Y Z Ɔ Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Á B Ć D É F G H Í J Ḱ Ĺ M Ń Ó P Q Ŕ Ś T Ú V Ẃ X Ý Ź Ɔ Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Â B Ĉ D Ê F Ĝ Ĥ Î Ĵ K L M N Ô P Q R Ŝ T Û V Ŵ X Ŷ Ẑ Ɔ Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ậ Ệ Ị Ộ Ụ
Ă B C D Ĕ F Ğ H Ĭ J K L M N Ŏ P Q Ř Š T Ŭ V W X Y Ž Ɔ Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ặ Ẹ Ị Ǒ Ŭ
Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J K L Ṁ Ṅ Ȯ Ṗ Q Ṙ Ṡ Ṫ U V Ẇ Ẋ Ẏ Ż Ɔ Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Ä B C D Ë F Ḧ Ï J K L M N Ö P Q R S T Ü V Ẅ Ẍ Ÿ Ż Ɔ Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Å Ḃ Ċ Ḋ Ḛ Ḟ Ġ Ḣ Ỉ J Ḳ Ḷ Ṁ Ṅ Ỏ Ṗ Q Ṛ Ṡ Ṫ Ů V W̊ Ẍ Y̊ Ż Ɔ Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Ấ Ḃ Ć Ḍ Ế Ḟ Ǵ Ḥ Í J Ḱ Ḽ Ḿ Ń Ố Ṕ Q Ŕ Ś Ṭ Ű V Ẃ X Ý Ź Ő Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Ă B Č Ď Ě F Ğ Ȟ Ǐ J Ǩ Ľ M Ň Ǒ P Q Ř Š Ť Ŭ V W X Y Ž Ɔ Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ǫ Ǔ
Ä Ḃ Ć Ḏ Ë Ḟ Ǵ Ḧ Ì J Ḳ Ḷ Ṁ Ṅ Ö Ṗ Q Ṛ Ṡ Ṫ Ü V Ẅ Ẍ Ÿ Ż Ɔ Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Â B C D Ê F G H Î J K L M N Ô P Q R Ŝ T Û V Ŵ X Ŷ Ẑ Ɔ Ə Ɛ Ɵ Σ Ʊ U Y Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ

Semibold patched, Capitals, Marks with capital alternates

À B̀ C̀ D̀ È F̀ G̀ H̀ Ì J̀ K̀ L̀ M̀ Ǹ Ò P̀ Q̀ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Z̀ Ɔ̀ ə̀ ɛ̀ ɵ̀ Σ̀ ʊ̀ ɔ̀ ɤ̀ Ƶ̀ Ę̀ Ḥ̀ Į̀ Ṛ̀ S̀ T̀ V̀ Ƶ̀ Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀

Á B́ Ć D́ É F́ Ǵ H́ Í J́ Ḱ Ĺ Ḿ Ń Ó Ṕ Q́ Ŕ Ś T́ Ú V́ Ẃ X́ Ý Ź Ɔ́ ə́ ɛ́ ɵ́ Σ́ ʊ́ ɔ́ ɤ́ Ƶ́ Ę́ Ḥ́ Į́ Ṛ́ Ś T́ V́ Ƶ́ Ą́ Ę́ Į́ Ǫ́ Ų́

Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ ə̂ ɛ̂ ɵ̂ Σ̂ ʊ̂ ɔ̂ ɤ̂ Ƶ̂ Ę̂ Ḥ̂ Į̂ Ṛ̂ Ŝ T̂ V̂ Ƶ̂ Ą̂ Ę̂ Į̂ Ǫ̂ Ų̂

Ă B̆ Č D̆ Ĕ F̆ Ğ Ĥ Ĭ J̆ Ǩ L̆ M̆ N̆ Ŏ P̆ Q̆ Ř S̆ T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Ɔ̆ ə̆ ɛ̆ ɵ̆ Σ̆ ʊ̆ ɔ̆ ɤ̆ Ƶ̆ Ę̆ Ḥ̆ Į̆ Ṛ̆ S̆ T̆ V̆ Ƶ̆ Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆

Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J̇ K̇ L̇ Ṁ Ṅ Ȯ Ṗ Q̇ Ṙ Ṡ Ṫ U̇ V̇ Ẇ Ẋ Ẏ Ż Ɔ̇ ə̇ ɛ̇ ɵ̇ Σ̇ ʊ̇ ɔ̇ ɤ̇ Ƶ̇ Ę̇ Ḥ̇ Į̇ Ṛ̇ Ṡ Ṫ V̇ Ƶ̇ Ą̇ Ę̇ Į̇ Ǫ̇ Ų̇

Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ̈ ə̈ ɛ̈ ɵ̈ Σ̈ ʊ̈ ɔ̈ ɤ̈ Ƶ̈ Ę̈ Ḥ̈ Į̈ Ṛ̈ S̈ T̈ V̈ Ƶ̈ Ä̈ Ë̈ Ï̈ Ö̈ Ü̈

Å Ḃ Ċ Ḋ Ḛ Ḟ Ġ Ḣ İ J̊ K̊ L̊ Ṁ Ṅ Ȯ Ṗ Q̊ Ṙ Ṡ Ṫ Ů V̊ Ẇ Ẋ Ẏ Ż Ɔ̊ ə̊ ɛ̊ ɵ̊ Σ̊ ʊ̊ ɔ̊ ɤ̊ Ƶ̊ Ę̊ Ḥ̊ Į̊ Ṛ̊ Ṡ Ṫ V̊ Ƶ̊ Ą̊ Ę̊ Į̊ Ǫ̊ Ų̊

Ő B̋ Ć D̋ Ȅ F̋ G̋ H̋ Ȉ J̋ K̋ L̋ M̋ Ń Ó P̋ Q̋ R̋ S̋ T̋ Ű V̋ Ẅ X̋ Ÿ Ź Ɔ̋ ə̋ ɛ̋ ɵ̋ Σ̋ ʊ̋ ɔ̋ ɤ̋ Ƶ̋ Ę̋ Ḥ̋ Į̋ Ṛ̋ S̋ T̋ V̋ Ƶ̋ Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋

Ă B̌ Č Ď Ě F̌ Ǧ Ȟ Ǐ J̌ Ǩ Ľ M̌ Ň Ǒ P̌ Q̌ Ř Š Ť Ǔ V̌ W̌ X̌ Y̌ Ž Ɔ̌ ə̌ ɛ̌ ɵ̌ Σ̌ ʊ̌ ɔ̌ ɤ̌ Ƶ̌ Ę̌ Ḥ̌ Į̌ Ṛ̌ Š Ť V̌ Ƶ̌ Ą̌ Ę̌ Į̌ Ǫ̌ Ų̌

Ȁ B̏ Ȃ D̏ Ȇ F̏ G̏ H̏ Ȋ J̏ K̏ L̏ M̏ N̏ Ȍ P̏ Q̏ Ȓ S̏ T̏ Ȕ V̏ Ẁ X̏ Ỹ Ȝ Ɔ̏ ə̏ ɛ̏ ɵ̏ Σ̏ ʊ̏ ɔ̏ ɤ̏ Ƶ̏ Ę̏ Ḥ̏ Į̏ Ṛ̏ S̏ T̏ V̏ Ƶ̏ Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏

Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ ə̂ ɛ̂ ɵ̂ Σ̂ ʊ̂ ɔ̂ ɤ̂ Ƶ̂ Ę̂ Ḥ̂ Į̂ Ṛ̂ Ŝ T̂ V̂ Ƶ̂ Ą̂ Ę̂ Į̂ Ǫ̂ Ų̂

Semibold italic, Capitals, Marks with capital alternates



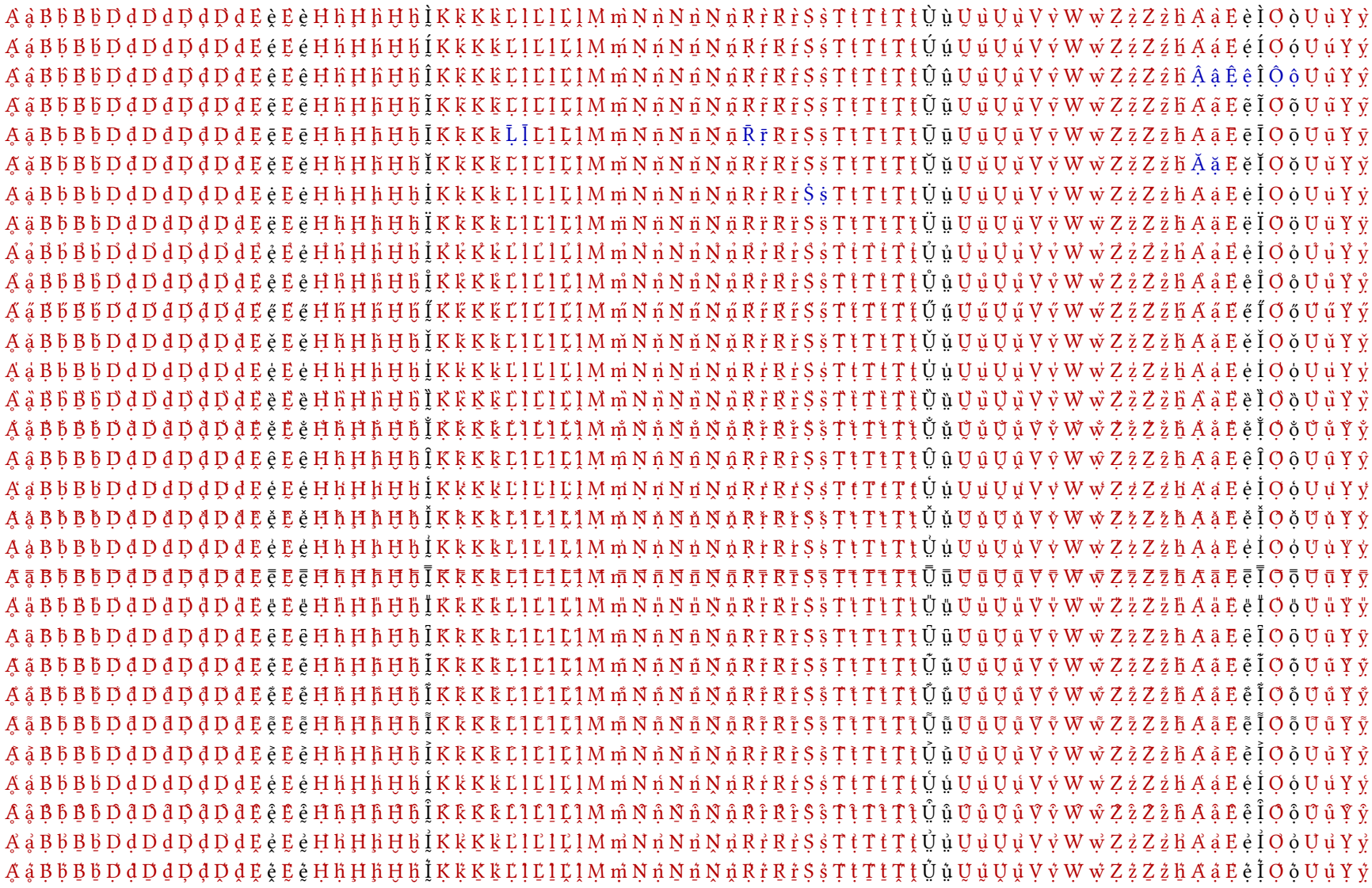
Semibold italic patched, Capitals, Marks with capital alternates

À Ɓ Ć Ď È Ḟ Ġ Ḧ Ì Ĵ Ḱ Ĺ Ṁ Ń Ò Ṕ Q Ŕ Ṡ Ť Ù Ṽ Ẁ Ẋ Ỳ Ż Ɔ Ə Ɛ Ɵ Σ Ʊ Ụ Ỷ Ƶ Ȩ Ḫ Į Ḛ Ṣ Ţ Ṿ Ẓ Ạ Ḝ Ị Ọ Ụ
Á Ɓ Ć Ď É Ḟ Ǵ Ḧ Í Ĵ Ḱ Ĺ Ḿ Ń Ó Ṕ Q Ŕ Ś Ť Ú Ṽ Ẃ Ẋ Ý Ź Ɔ Ə Ɛ Ɵ Σ Ʊ Ụ Ý Ƶ Ȩ Ḫ Í Ḛ Ṣ Ţ Ṿ Ẓ Ạ Ḝ Ị Ọ Ụ
Â Ɓ Ĉ Ď Ê Ḟ Ĝ Ḧ Î Ĵ Ḱ Ĺ Ṁ Ń Ô Ṕ Q Ŕ Ŝ Ť Û Ṽ Ŵ Ẋ Ŷ Ẑ Ɔ Ə Ɛ Ɵ Σ Ʊ Û Ŷ Ẑ Ȩ Ḫ Î Ḛ Ṣ Ţ Ṿ Ẓ Ạ Ḝ Ị Ọ Ụ
Ă Ɓ Č Ď Ĕ Ḟ Ğ Ḧ Ǐ Ĵ Ḱ Ľ Ḿ Ň Ŏ Ṕ Q Ř Š Ť Ŭ Ṽ Ẅ Ẋ Ỹ Ž Ɔ Ə Ɛ Ɵ Σ Ʊ Ŭ Ỹ Ž Ȩ Ḫ Ǐ Ḛ Ṣ Ţ Ṿ Ẓ Ạ Ḝ Ị Ọ Ụ
Ȧ Ɓ Ċ Ḋ Ė Ḟ Ġ Ḣ İ Ĵ Ḱ Ŀ Ṁ Ṅ Ȯ Ṗ Q Ṙ Ṡ Ṫ Ụ Ṿ Ẇ Ẋ Ẏ Ż Ɔ Ə Ɛ Ɵ Σ̇ Ʊ Ụ Ẏ Ż Ȩ Ḣ İ Ḛ Ṣ Ṫ Ṿ Ẓ Ạ Ḝ Ị Ọ Ụ
Ä Ɓ Č Ď Ë Ḟ Ġ Ḧ Ï Ĵ Ḱ Ľ Ḿ Ṅ Ö Ṗ Q Ř Ṡ Ť Ü Ṽ Ẅ Ẍ Ÿ Ž Ɔ Ə Ɛ Ɵ Σ̈ Ʊ Ü Ÿ Ž Ȩ Ḫ Ï Ḛ Ṣ Ṫ Ṿ Ẓ Ä Ḝ Ị Ö Ü
Ả Ɓ Ċ Ď Ẻ Ḟ Ġ Ḣ Ỉ Ĵ Ḱ Ĺ Ṁ Ṅ Ỏ Ṗ Q Ṙ Ṡ Ť Ủ Ṽ Ẃ Ẋ Ỷ Ż Ɔ Ə Ɛ Ɵ Σ̉ Ʊ Ủ Ỷ Ż Ȩ Ḫ Ỉ Ḛ Ṣ Ṫ Ṿ Ẓ Ạ Ḝ Ị Ọ Ụ
Ấ Ɓ Č Ď Ế Ḟ Ǧ Ḧ Ǐ Ĵ Ḱ Ĺ Ḿ Ṅ Ố Ṗ Q Ṙ Ś Ť Ứ Ṽ Ẃ Ẋ Ỹ Ž Ɔ Ə Ɛ Ɵ Σ́ Ʊ Ứ Ỹ Ž Ȩ Ḫ Ǐ Ḛ Ṣ Ṫ Ṿ Ẓ Ặ Ḝ Ị Ọ Ụ
Ắ Ɓ Č Ď Ĕ Ḟ Ğ Ḧ Ǐ Ĵ Ḱ Ľ Ḿ Ň Ŏ Ṕ Q Ř Š Ť Ŭ Ṽ Ẅ Ẋ Ỹ Ž Ɔ Ə Ɛ Ɵ Σ Ʊ Ŭ Ỹ Ž Ȩ Ḫ Ǐ Ḛ Ṣ Ţ Ṿ Ẓ Ặ Ḝ Ị Ọ Ụ
Ầ Ɓ Č Ď Ề Ḟ Ǧ Ḧ Ì Ĵ Ḱ Ľ Ḿ Ň Ồ Ṗ Q Ř Ṡ Ť Ừ Ṽ Ẁ Ẋ Ỳ Ž Ɔ Ə Ɛ Ɵ Σ̀ Ʊ Ừ Ỳ Ž Ȩ Ḫ Ì Ḛ Ṣ Ṫ Ṿ Ẓ Ạ̀ Ḝ Ị Ọ Ụ
Â Ɓ Ĉ Ď Ê Ḟ Ĝ Ḧ Î Ĵ Ḱ Ĺ Ṁ Ń Ô Ṕ Q Ŕ Ṡ Ť Û Ṽ Ŵ Ẋ Ŷ Ẑ Ɔ Ə Ɛ Ɵ Σ Ʊ Û Ŷ Ẑ Ȩ Ḫ Î Ḛ Ṣ Ţ Ṿ Ẓ Ậ Ḝ Ị Ọ Ụ

Precomposed characters

These grids display precomposed characters with referenced glyphs, where a precomposed character can inherit an anchor from the references glyph.

Regular, Precomposed bases that can inherit an above anchor, Above



Regular patched, Precomposed bases that can inherit an above anchor, Above



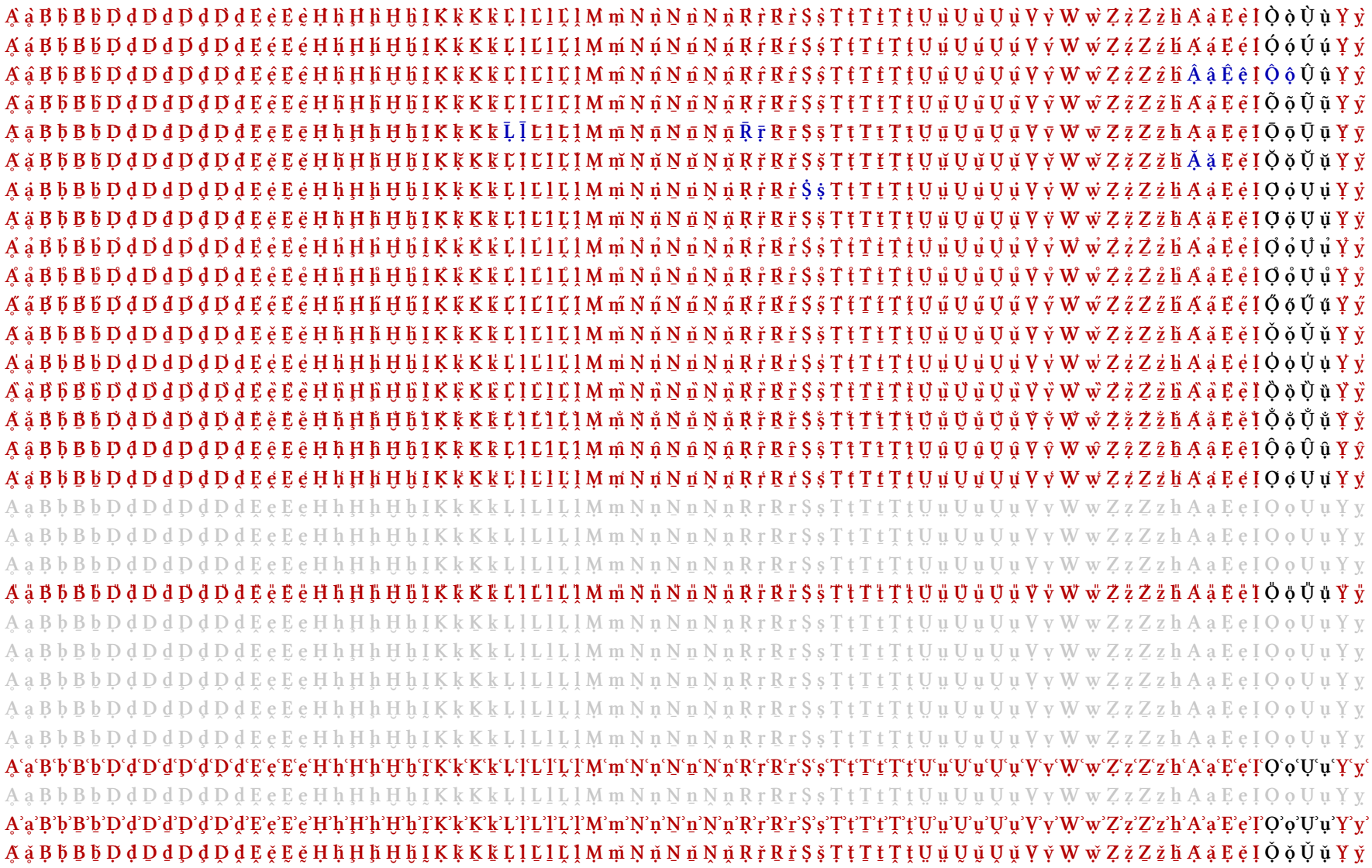
Italic, Precomposed bases that can inherit an above anchor, Above



Italic patched, Precomposed bases that can inherit an above anchor, Above



Semibold, Precomposed bases that can inherit an above anchor, Above



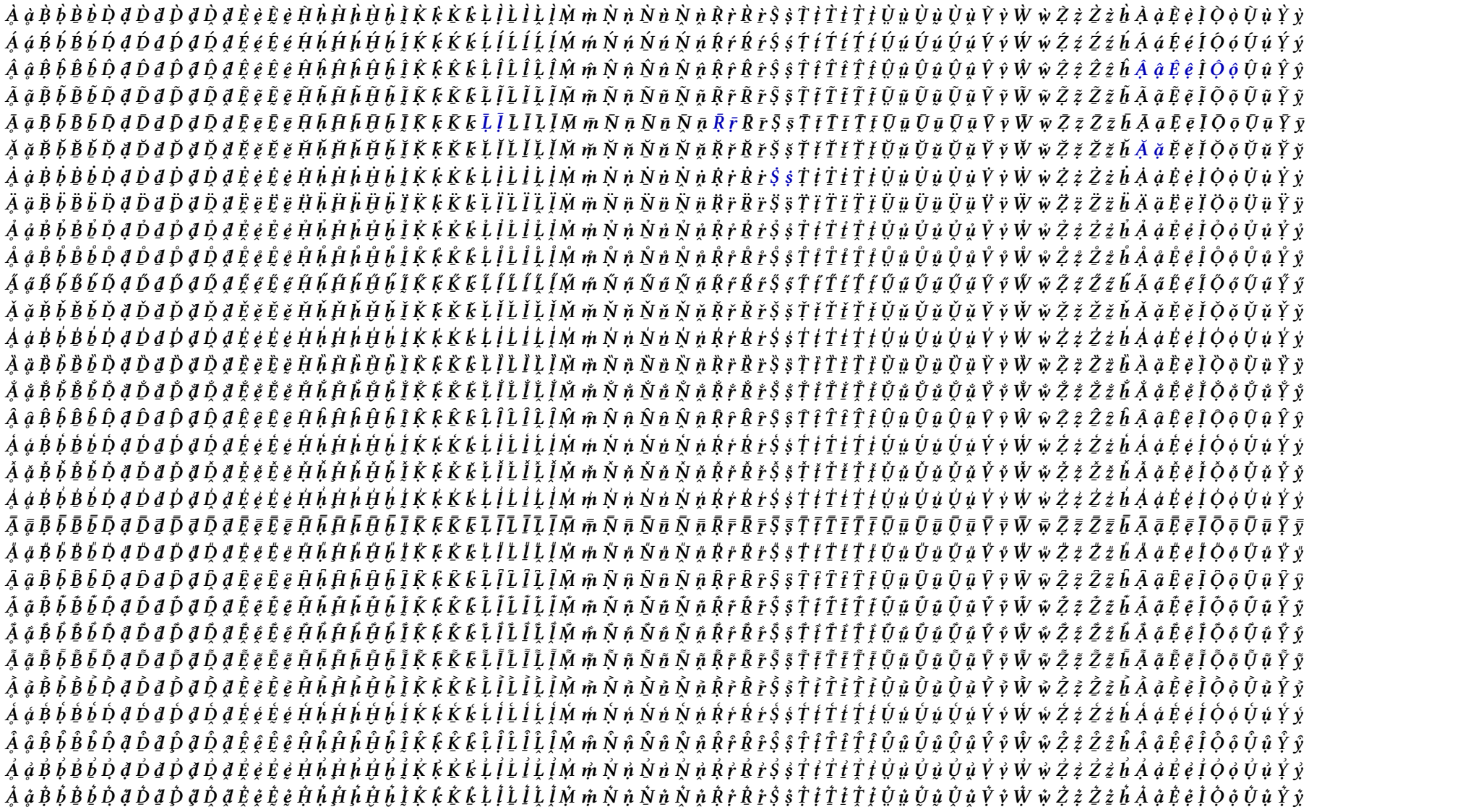
Semibold patched, Precomposed bases that can inherit an above anchor, Above



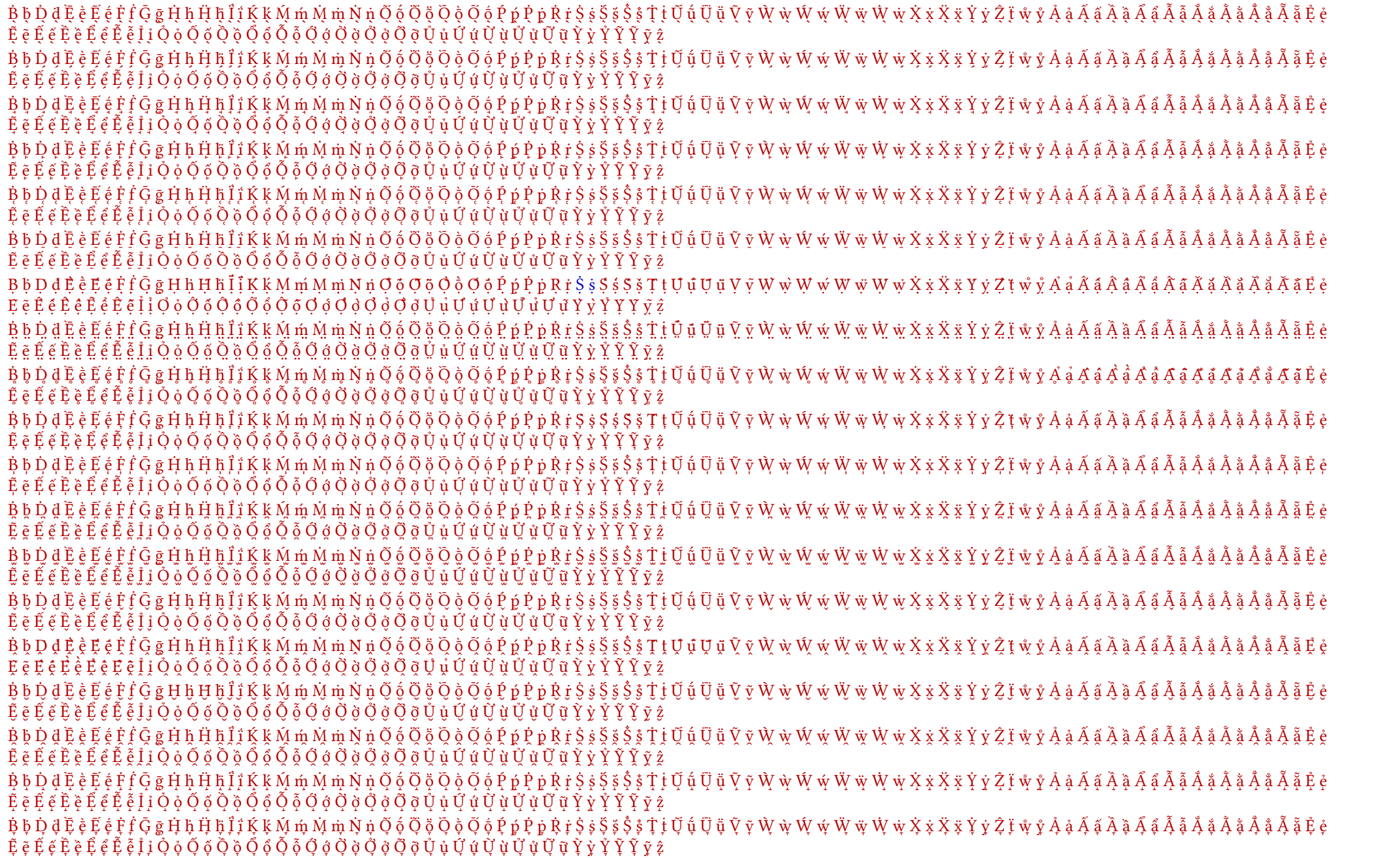
Semibold italic, Precomposed bases that can inherit an above anchor, Above



Semibold italic patched, Precomposed bases that can inherit an above anchor, Above



Regular, Precomposed bases that can inherit a below anchor, Below



Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṧ ṧ Ṭ ṭ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ ἀ
Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṧ ṧ Ṭ ṭ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ḙ ḙ Ḛ ḛ Ḝ ḝ Ḯ ḯ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṥ ṥ Ṫ ṫ Ṳ ṳ Ṵ ṵ Ṽ ṽ Ẅ ẅ Ẍ ẍ T̈ ẗ Ỳ ỳ Ỵ ỵ Ỹ ỹ Ỿ ỿ ẓ
Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṧ ṧ Ṭ ṭ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ḙ ḙ Ḛ ḛ Ḝ ḝ Ḯ ḯ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṧ ṧ Ṭ ṭ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ

Regular patched, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṙ ṙ Ṣ ṣ Ṩ ṩ Ṯ ṯ Ṳ ṳ Ṷ ṷ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṩ ṩ Ṫ ṫ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẁ ẁ Ẅ ẅ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẗ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Italic, Precomposed bases that can inherit a below anchor, Below



Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṫ ṫ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ặ ặ Ẳ ẳ Ẵ ẵ Ẹ ẹ Ẽ ẽ Ế ế
Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

Italic patched, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ḽ ḽ Ṃ ṃ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṙ ṙ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṭ ṭ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ Ẕ ẕ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṭ ṭ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế
Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Semibold, Precomposed bases that can inherit a below anchor, Below



Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṫ ṫ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

Semibold patched, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḫ ḫ Ḭ ḭ Ḱ ḱ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṙ ṙ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṭ ṭ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ Ẕ ẕ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ḛ ḛ Ḝ ḝ Ḙ ḙ Ḛ ḛ Ḝ ḝ Ḙ ḙ Ḭ ḭ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ ỿ ẑ

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḫ ḫ Ḭ ḭ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṣ ṣ Ṩ ṩ Ṯ ṯ Ṳ ṳ Ṷ ṷ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ḕ ḕ Ḗ ḗ Ḝ ḝ Ḙ ḙ Ḛ ḛ Ḭ ḭ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ ẓ

Semibold italic, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ṉ ṉ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṭ ṭ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ṽ ṽ Ẉ ẉ Ẅ ẅ Ẇ ẇ Ẍ ẍ Ẏ ẏ Ẓ ẓ Ẕ ẕ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ặ Ẹ ẹ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ ẑ

Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḣ ḣ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḳ ḳ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṅ ṅ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṫ ṫ Ṭ ṭ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ṿ ṿ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ Ẓ ẓ Ẕ ẕ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḣ ḣ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḳ ḳ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṅ ṅ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṫ ṫ Ṭ ṭ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ṿ ṿ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ Ẓ ẓ Ẕ ẕ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Semibold italic patched, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḵ ḵ Ṁ ṁ Ṃ ṃ Ṇ ṇ Ṓ ṓ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṭ ṭ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḵ ḵ Ṁ ṁ Ṃ ṃ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṫ ṫ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

1. ”i” not losing its dot when there is a tone diacritic AND a tilde below

   i̱ ı́ ı̱

   ı̱́

   ı̱̀

2. troubles with capital I

   I̱ I̱ I̱

   Solution: Use combnining grapheme joiner (CGJ)

   I̱ I̱ I̱

3. double acute critic shifted right in bold face

   á a̋ ą ą́ ą̋ ą̋

   Solution: Patch anchor of double acute

   á a̋ ą ą́ ą̋ ą̋ **á a̋ ą ą́ ą̋ ą̋** ***á a̋ ą ą́ ą̋ ą̋***

4. wrong tilde placement together with caron (and grave? and acute? and double grave? OK, it seems it’s rather about vowels...) with some vowels

   ą̌ ą̌ ą̌ ą̌

   ų̀ ***ų̀ ų ų̀***

   Solution: Use combnining grapheme joiner (CGJ)

   ą̌ ą̌ ą̌ ***ą̌***

   ų̀ ***ų̀ ų̀ ų̀***

5. wrong acute placement in bold face

   ɛ́ *ɛ́* **ɛ́** ***ɛ́***

   á *á* **á** ***á***

   ɔ́ *ɔ́* **ɔ́** ***ɔ́***

   ú *ú* **ú** ***ú***

   ɛ̀ *ɛ̀* **ɛ̀** ***ɛ̀***

   à *à* **à** ***à***

   ɔ̀ *ɔ̀* **ɔ̀** ***ɔ̀***

   ù *ù* **ù** ***ù***

   Solution: Patch above-anchor of acute and grave in (semi)bold.

   ɛ́ *ɛ́* **ɛ́** ***ɛ́*** á *á* **á** ***á***

   ɔ́ *ɔ́* **ɔ́** ***ɔ́*** ú *ú* **ú** ***ú***

   ɛ̀ *ɛ̀* **ɛ̀** ***ɛ̀*** à *à* **à** ***à***

   ɔ̀ *ɔ̀* **ɔ̀** ***ɔ̀*** ù *ù* **ù** ***ù***

6. italic examples with tilde below

Flą̰

+ *Flą̰*

gṵ̀ ą̰

gṵ̀ą̰

+ *gṵ̀ą̰*

lɛ̰̃ ɔ̰

+ *lɛ̰̃ ɔ̰*

mṵ

kṵ̃,

kṵ̃

+ *kṵ̃*

nṵ̃

+ *nṵ̃*

nɔ̰̌

nṵ̀

nṵ̀

N-ą̰

vɔ̰ṵ̃ṵ̃

fɔ̰ṵ̃fɔ̰ṵ̃

lɛ̰̃ ɔ̰

Flą̰

gṵ̀ ą̰

gṵ̀ą̰

lɛ̰̃ ɔ̰

lɛ̰̃ ɔ̰

mṵ

kṵ̃,

kṵ̃

nṵ̃

nɔ̰̌

nɔ̰̌

nù̱

nù̱

N-ą̆

7. Ɛ̏ bold italics

   Ɛ̏

   Ɛ̀ Ɛ̏ **Ɛ̀ Ɛ̏**

   Solution: Set above-anchor for open-E, patch GSUB to use capital alternate mark glyph ”uni”